GENERAL ORGANIC CHEMISTRY - I (GOC-I)

Electronic effect :- Effect due to movements of electrons in a molecule.

- Permanent effect
  1. Inductive (I) effect
  2. Resonance (R) / Mesomeric (M) effect
  3. Hyper conjugation (H.C)
- Temporary effect
  - Inductomeric effect
  - Electromeric effect

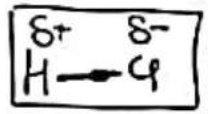
GENERAL ORGANIC CHEMISTRY -I (GOC-I)

Inductive effect (I)

Partial

∧ displacement of σ bond electrons towards more electronegative atom. due to this displacement molecule / bond becomes polar.

Resultant effects due to polarity of bond is called induct effect.



eg $CH_3-\overset{\delta\delta\delta+}{CH_2}-\overset{\delta\delta+}{CH_2}\rightarrow\overset{\delta+}{CH_2}\rightarrow\overset{\delta-}{CH_2}\rightarrow Cl$

No I effect

⇒ I effect ⇒ applicable only on σ bond (not on π bond.

⇒ " " $\propto \frac{1}{\text{distance}}$

⇒ H is Reference atom for I effect

Electronegativity (Pauling)

F = 4	I = 2.6
O = 3.5	S = 2.56
Cl = 3.1	P = 2.19
N = 3.04	C = 2.55
Br = 2.9	H = 2.1

GENERAL ORGANIC CHEMISTRY -I

Types of I Groups $\begin{cases} -I \\ +I \end{cases}$

① **-I Groups:-** These are σ bond electron withdrowing groups.

⇒ Relative Strength of -I groups.

$$-\overset{\oplus}{N}F_3 > -\overset{\oplus}{N}R_3 > -\overset{\oplus}{S}R_2 > -\overset{\oplus}{N}H_3 > -N(=O)_2 > -SO_3H > -CN$$

$$> -C(=O)-H > -C(=O)-R > -C(=O)-OH > -F > -Cl > -Br > -I >$$

$$> -OR > -OH > -\underset{Sp}{C\equiv C-H} > -\underset{SP^3}{NH_2} > -Ph > -CH=CH_2 > H$$

GENERAL ORGANIC CHEMISTRY -I

Note Electronegativity $\propto$ % S Character

" " $C_{sp} > C_{sp^2} > C_{sp^3}$

" " $N_{sp} > N_{sp^2} > N_{sp^3}$

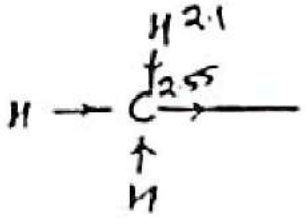
" " $N_{sp} > N_{sp^2} > C_{sp} > N_{sp^3} > C_{sp^2} > C_{sp^3}$

(2) +I groups:- σ Bond donating groups.

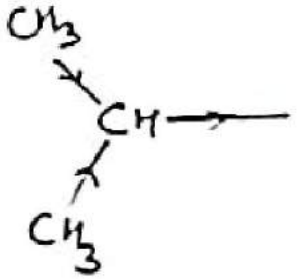
Relative order.

$$\overset{\ominus}{C}H_2- > -NH_2 > -O^{\ominus} > -\overset{O}{\overset{\|}{C}}-O^{\ominus} > -\underset{3^\circ}{\overset{R}{\underset{R}{\overset{|}{\underset{|}{C}}}}}-R > -\underset{2^\circ}{CH}(R)-R > -\underset{1^\circ}{CH_2}-R > -\underset{1^\circ}{CH_3} > D > \boxed{H}$$

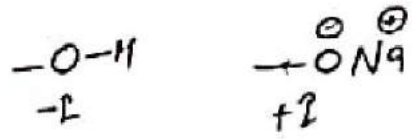
GENERAL ORGANIC CHEMISTRY - I

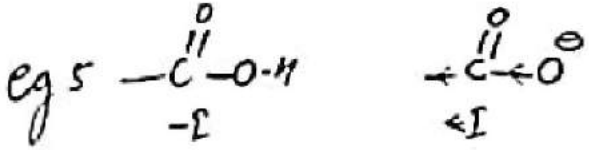
eg 1 

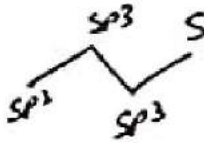
(+I)

eg 2 

eg 3 $CH_3- < C_2H_5-$ +I

eg 4 

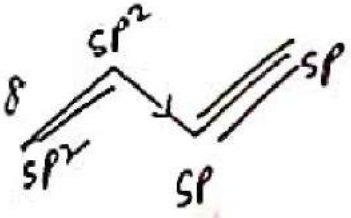
eg 5 

eg 6 

No I effect

eg 7  " " "

For alkyl group +I effect $\propto$ no. of carbon atoms
(Same degree)

eg 8 

GENERAL ORGANIC CHEMISTRY -I

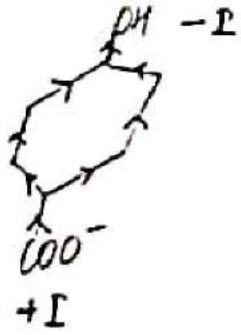
eg 9 $CH_3-CH_2-\overset{\delta-}{C}H_2-MgX$

(organometallic comp.)

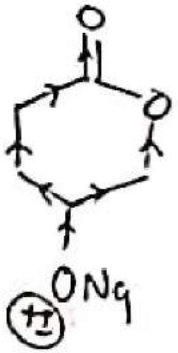
C-Metal

eg10 $\overset{\delta\delta-}{CH_3}-\overset{\delta-}{CH_2}-\overset{\delta+}{Li}$

eg11



eg12



Q

① $-CH_3 < -CH_2-CH_3 < -CH_2-CH_2-CH_3 <$  (+I)

② $NO_2-CH_2- > Cl-CH_2-$ (-I)

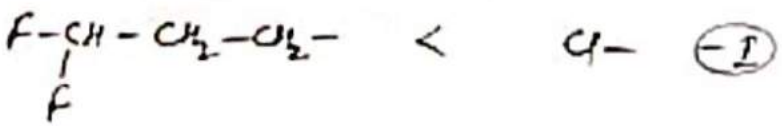
③ $NO_2-CH_2-CH_2- < Cl-$ (-I)

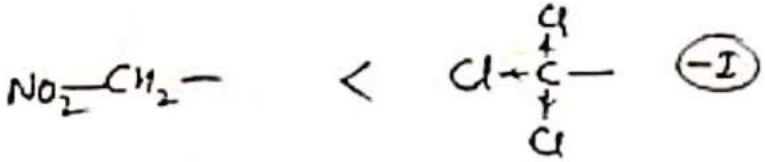
D > N > P

④ $F-\underset{F}{CH}-CH_2-CH_2- < NO_2-$

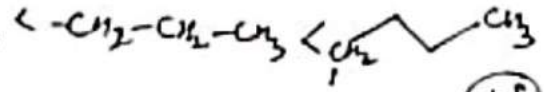
GENERAL ORGANIC CHEMISTRY - I

(5)



eg b 

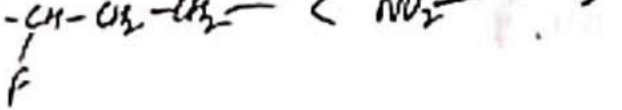
Q.

(1) $-CH_3 < -CH_2-CH_3 < -CH_2-CH_2-CH_3 <$  (+I)

(2) $NO_2-CH_2- > Cl-CH_2-$ (-I)

(3) $NO_2-CH_2-CH_2- < Cl-$ (-I)

D > N > P

(4) 

GENERAL ORGANIC CHEMISTRY - I [GOC-I]

Resonance :- It is phenomenon of delocalisation of conjugated π electrons.

When a single structure of a compound / an ion can not explain all the properties of compound/ion, then we use different Lewis electronic structures for explanation of all properties of comp./ion.

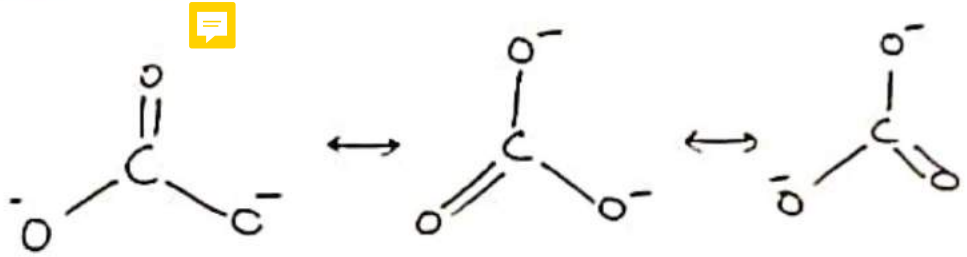
These lewis structures are called Resonating structures (R.S) / Canonical Structures

- These resonating structures are hypothetical structures (Can not be isolated)

⇒ The actual structure of compound/ion is represented by **Resonance hybrid**:

GENERAL ORGANIC CHEMISTRY - I [GOC-I]

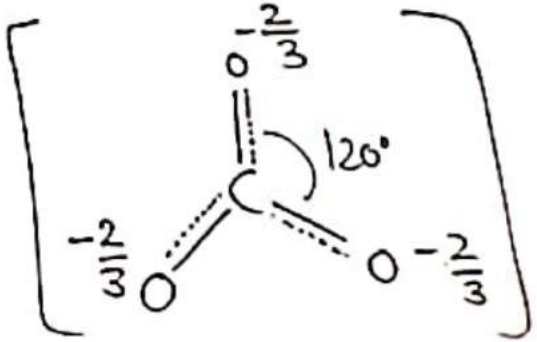
⇒ Carbonate CO_3^{-2}



$1\,A^\circ = 10^{-10}\,M$

$1\,PM = 10^{-12}\,M$

in Carbonate ion all c-o bond length are same (1.28 A°)



Resonance Hybrid

Bond length

① C—C → 1.54 A° → 154 PM

② C=C ⇒ 1.34 A°

③ C≡C → 1.20 A°

④ $C_{SP^3}-H$ → 1.10 A°

⑤ $C_{SP^2}-H$ ⇒ 1.09 A°

⑥ $C_{SP}-H$ ⇒ 1.06 A°

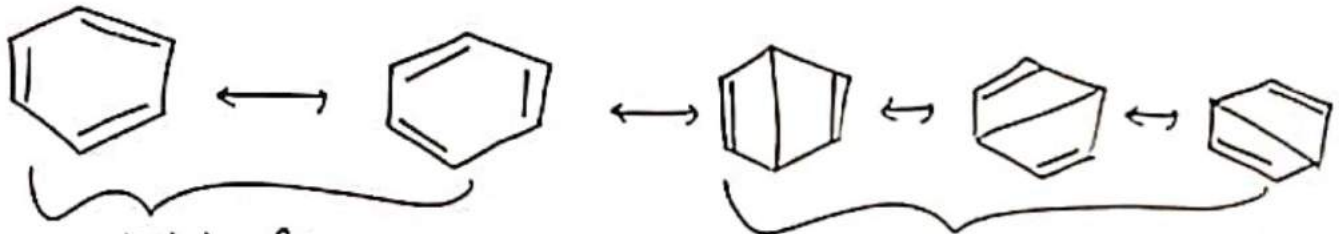
⑦ C—O ⇒ 1.43 A°

⑧ >C=O → 1.23 A°

GENERAL ORGANIC CHEMISTRY - I [GOC-I]

eg Benzene (C_6H_6) ← D.U = 4

- Hydrogenation ($3H_2$, Δ) →
- Ozonolysis (Δ) → 3 (CH=O, CH=O) Glyoxal





Resonance Hybrid of Benzene.



Bond length

1. C—C → 1.54 A° → 154 PM
2. C=C → 1.34 A°
3. C≡C → 1.20 A°
4. $C_{sp^3}-H$ → 1.10 A°
5. $C_{sp^2}-H$ → 1.09 A°
6. $C_{sp}-H$ → 1.06 A°
7. C—O → 1.43 A°
8. >C=O → 1.23 A°

GENERAL ORGANIC CHEMISTRY - I

Resonance hybrid:- It is actual structure of compounds or ion.

- Resonance hybrid has all the properties of Resonating structures (R.S)

⇒ All R.S. give contribution in R.H.

⇒ More stable R.S. give greater contribution in resonance hybride

⇒ Compounds gain stability by delocalisation of electrons (Resonance)

⇒ Stability of comp./ion α no. of Resonating structures.

⇒ Resonance hybrid has least potential energy (w.r.t. Resonating structures)

GENERAL ORGANIC CHEMISTRY - I

=1 Resonance energy (R.E) :-

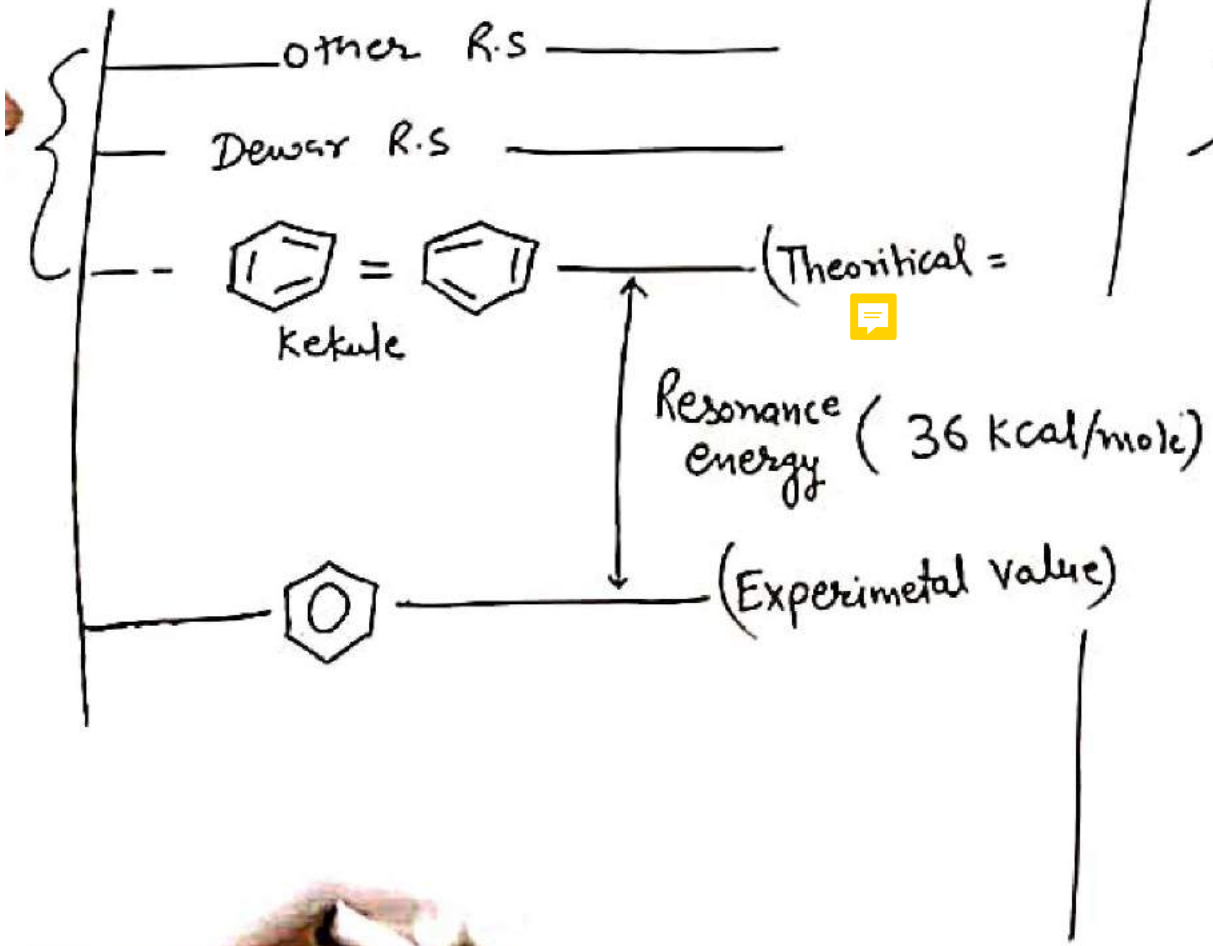
Energy difference between most stable resonating structure and resonance hybride.

$$E_{RE} = E_{\substack{most \\ stable(R.S)}} - E_{hybride}$$

Resonance energy ∝ Stability of comp.

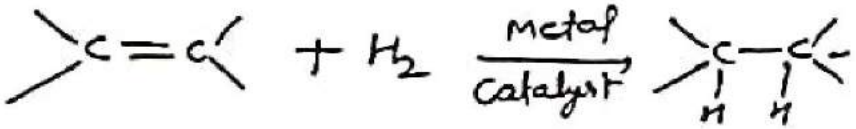
GENERAL ORGANIC CHEMISTRY - I

Benzene



Calculation of R.E. for benzene.

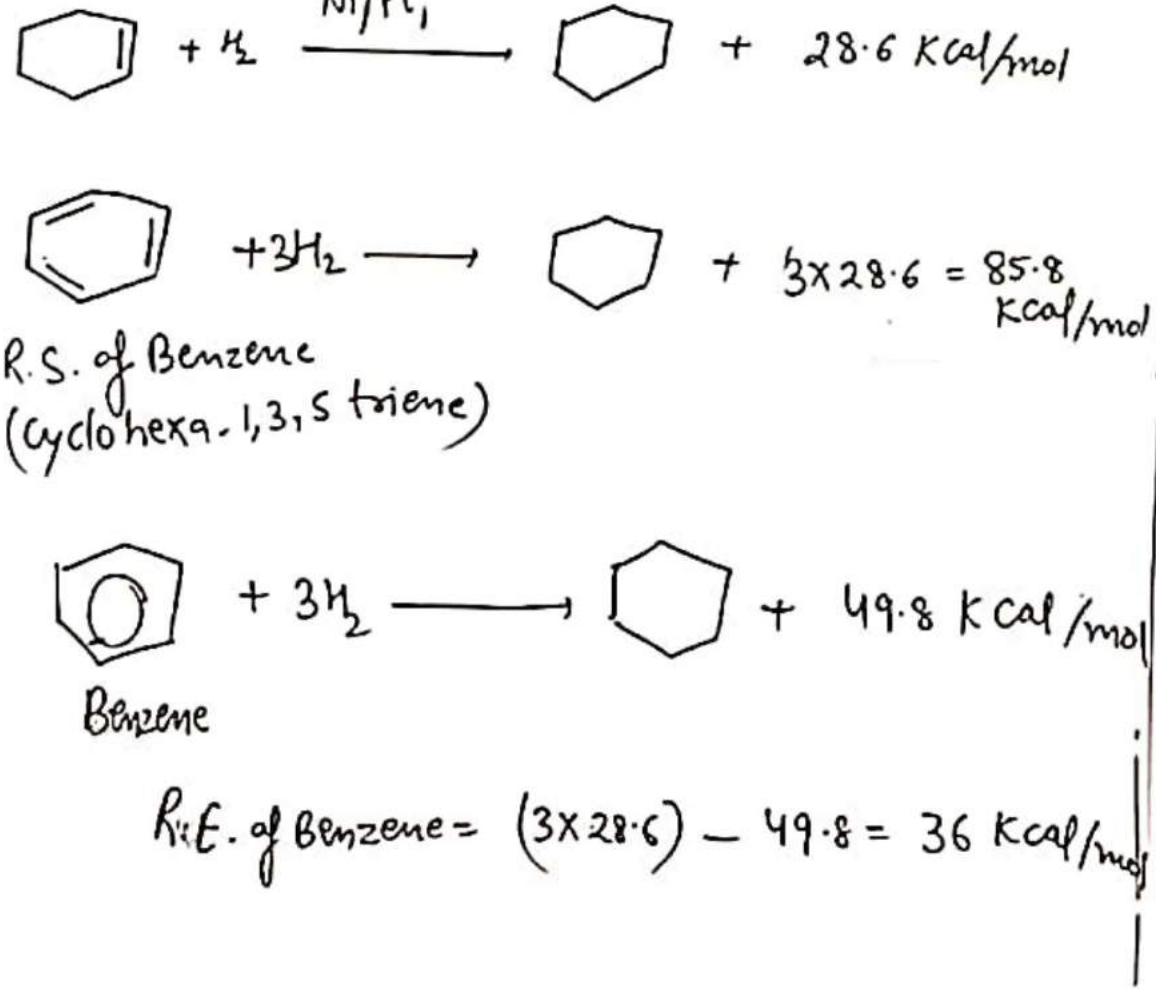
= on the basis of heat of hydrogenation.

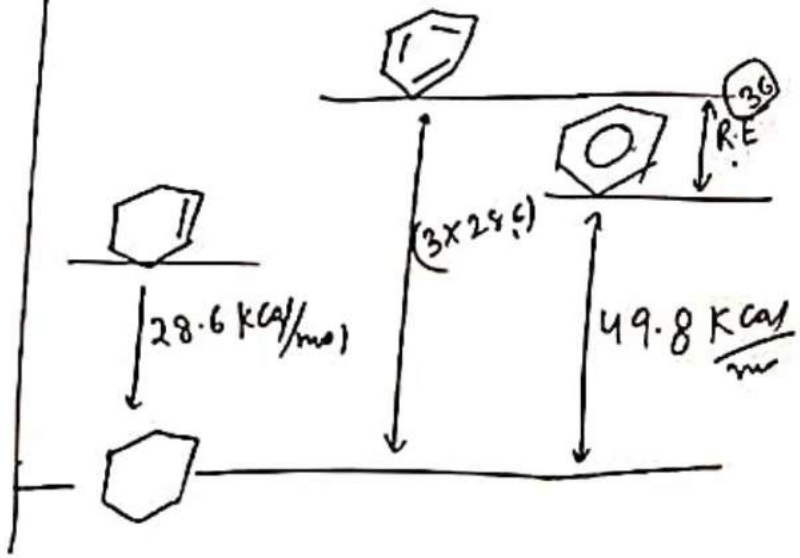


+

Heat of Hydrogenation

GENERAL ORGANIC CHEMISTRY - I





GENERAL ORGANIC CHEMISTRY - I

① R.E. Benzene = 36 kcal/mol

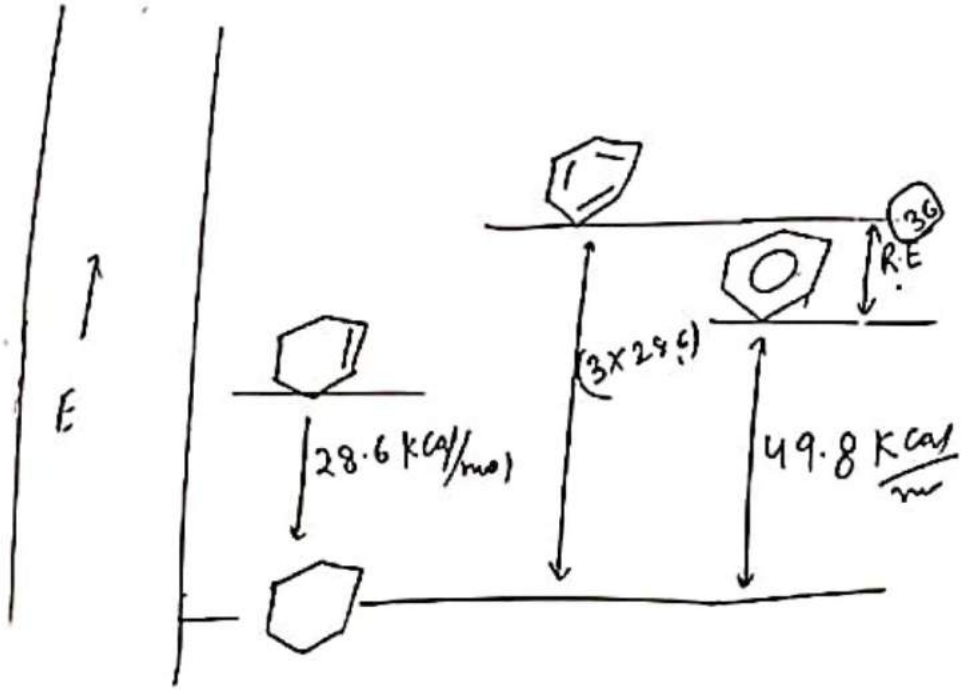
② ,  = 61 " "

③ "  = 84 " "

④  = 92 " "

⑤ $CH_2=CH_2$ = 0

⑥  ⇒ 4 kcal/mol



GENERAL ORGANIC CHEMISTRY - I

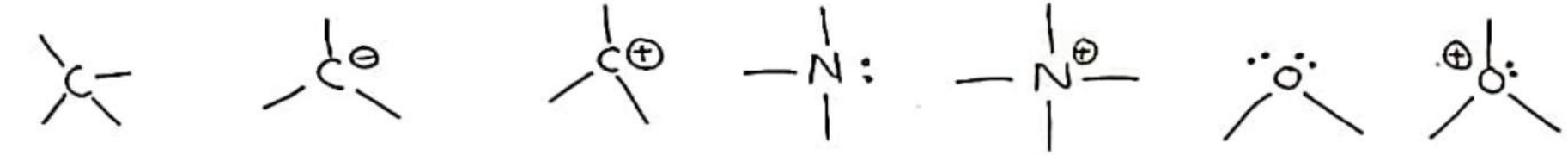
Rules For Resonating Structures:-

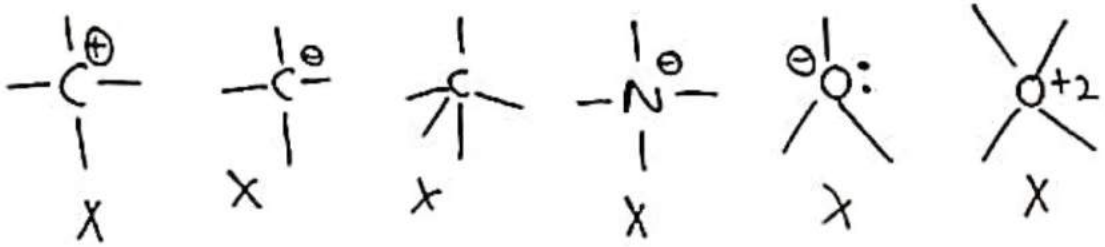
1. Resonance possible only in planar / nearly planar conjugated systems in which adjacent P/d orbitals are present in same plane with parallel arrangement.
2. The net charge remain same on each resonating structure of given compound.
3. Only π electrons of conjugative system can participate in resonance. (Not σ bond e⁻s)
4. All resonating structures must have the same arrangement of atomic nuclei.
5. In all resonating structures no. of paired electrons are same similarly no. of unpaired electrons are also same (unchange)
6. In all resonating structures, hybridization of perticular atom remain same (unchange)

GENERAL ORGANIC CHEMISTRY - I

Rules For Resonating Structures:-

⑦ In all resonating structure atom of 2nd period should follow octate rule.

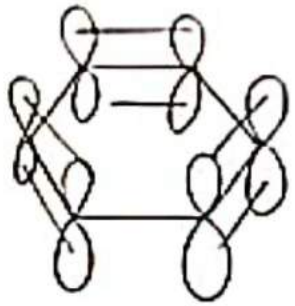




GENERAL ORGANIC CHEMISTRY - I

Rules For Resonating Structures:-

eg 1

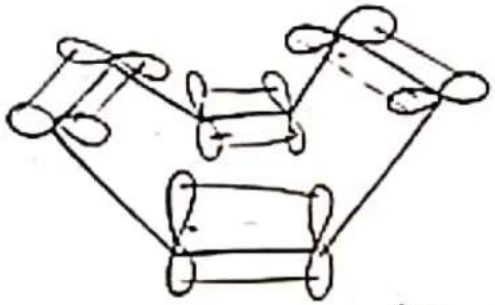


Resonance possible



Cyclooctatetraene

=

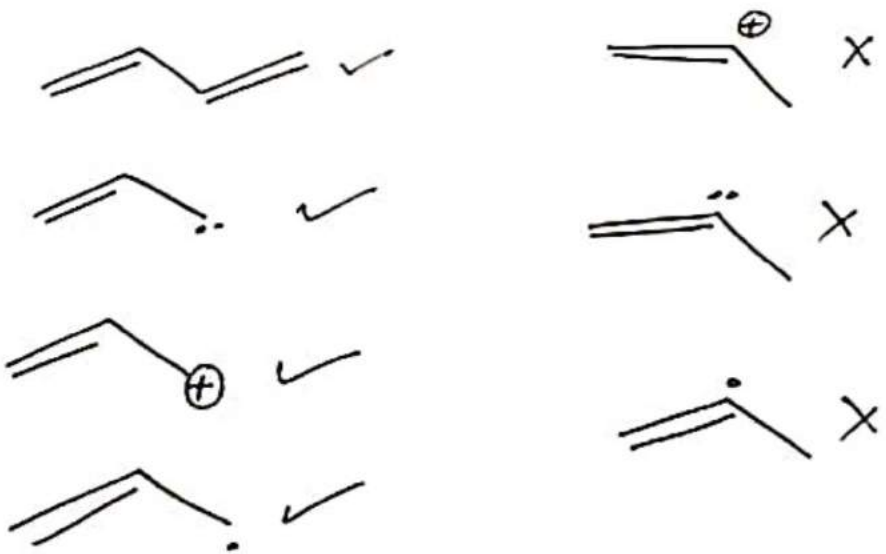


Tub Shape. (Non planar str.)

No Resonance

GENERAL ORGANIC CHEMISTRY

Types of Conjugation:-



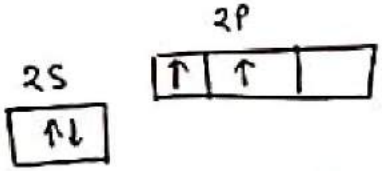
GENERAL ORGANIC CHEMISTRY

Types of conjugation:-

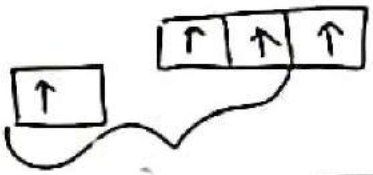
(A) Multiple bond - multiple bond conjugation:-

eg: $\underset{sp^2}{CH_2}=\underset{sp^2}{CH}-\underset{sp^2}{CH}=\underset{sp^2}{CH_2}$

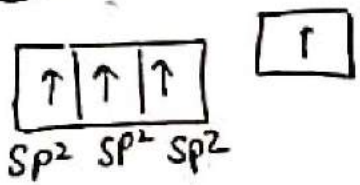
→ C. (G.S)

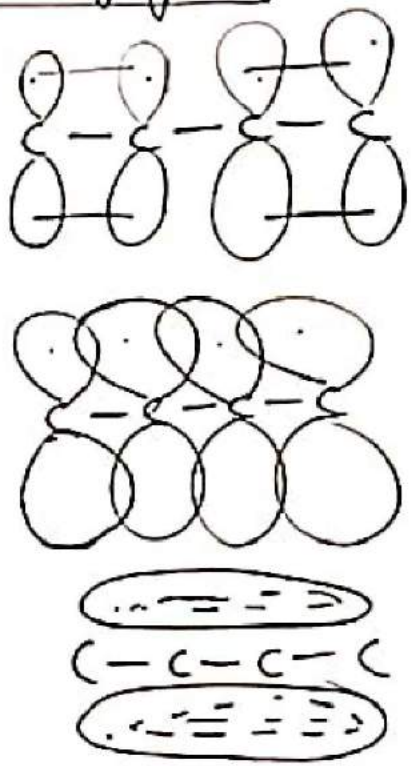


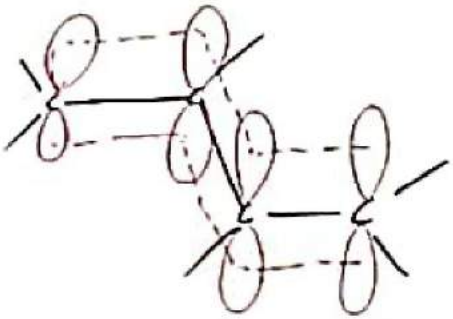
⇒ C in (E.S)



⇒ C in (H.S)





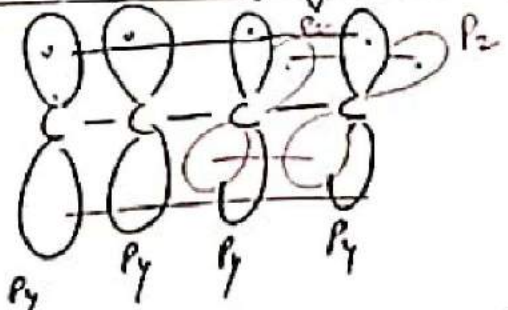


GENERAL ORGANIC CHEMISTRY

Types of Conjugation:-

(A) Multiple bond- multiple bond Conjugation:-

Eg 2 $\overset{sp^2}{CH_2}=\overset{sp^2}{CH}-\overset{sp}{C}\equiv\overset{sp}{CH}$

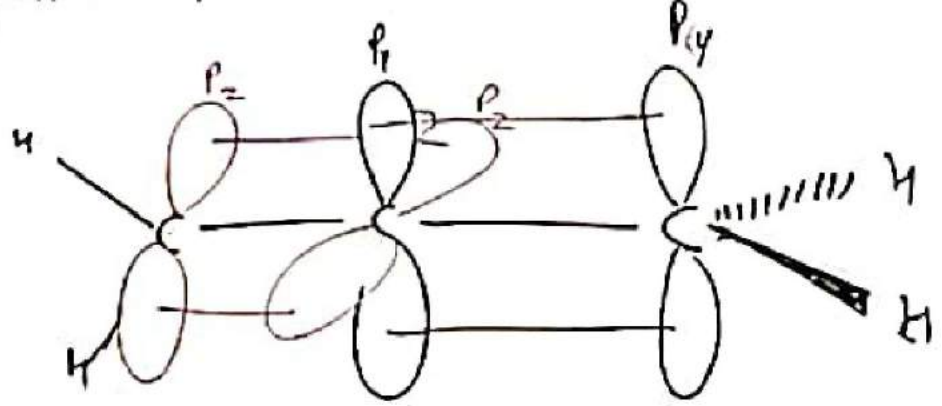


⇒ $6\pi e^-$

⇒ $4\pi e^-$ in conjugation

Eg 3 $CH_2=C=CH_2$

Sp^2 SP Sp^2



Eg 4 $CH_2=C=C=CH_2$

$4\pi e^-$ in conjugation

GENERAL ORGANIC CHEMISTRY

Types of Conjugation:-

Multiple bond - multiple bond Conjugation:-

5

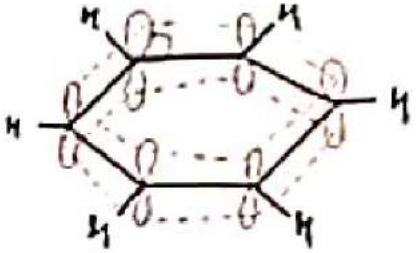


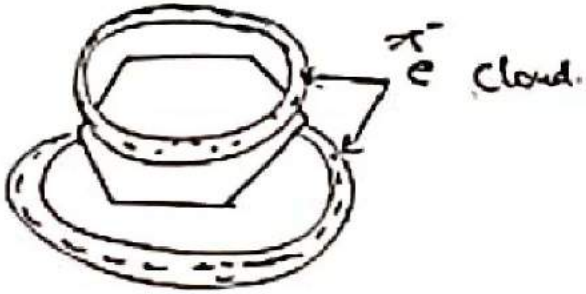
$6\pi e^-$ in conjugation

6

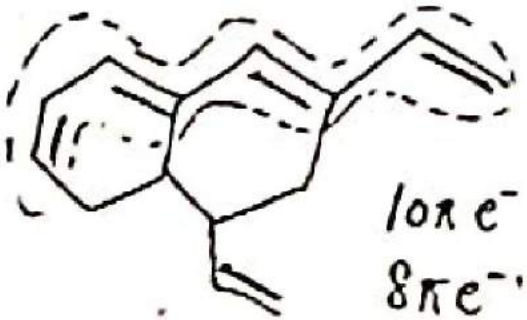


$6\pi e^-$





7



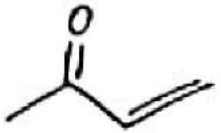
$10\pi e^-$ in compound

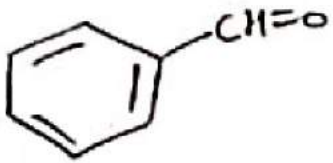
$8\pi e^-$ in conjugation

GENERAL ORGANIC CHEMISTRY

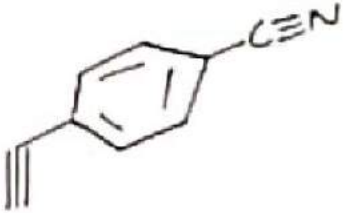
Types of conjugation:-

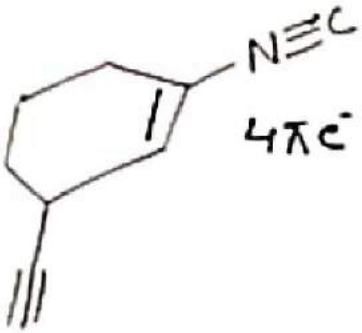
(A) Multiple bond - Multiple bond conjugation :-

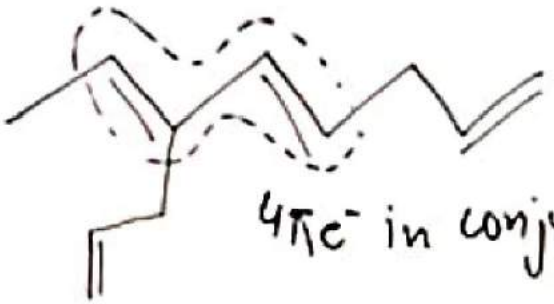
eg 8  4π e in conju.

eg 9  8π e⁻ in conjugation.

eg 10  12π e in conjugation

eg 11  10π e⁻ in conjugation.

eg 12  4π e⁻ in conj.

eg 13  4π e⁻ in conjugation

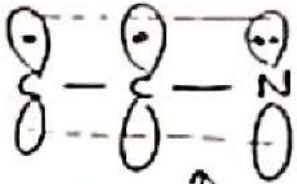
GENERAL ORGANIC CHEMISTRY

Types of conjugation:-

(B) Multiple bond - lone pair conjugation:-

eg1 $CH_2 = CH - \ddot{N}H_2$

SP^2 SP^2 SP^2



$4\pi e^-$ in conjugation

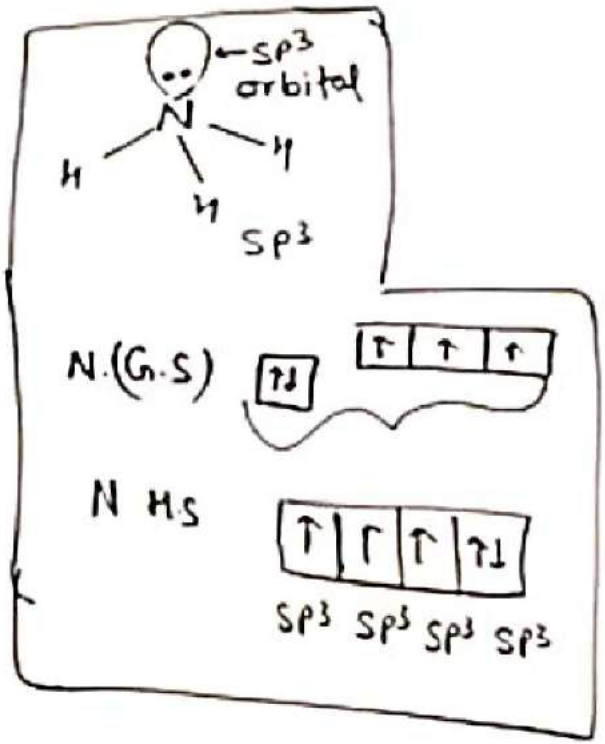
N in G.S. 2S ↑↓ 2P ↑ ↑ ↑

N in E.S ↑ ↑ ↑ ↑↓

N in H.S ↑ ↑ ↑ ↑↓

Sp^2 sp^2 Sp^2

Unhybridised P orbital



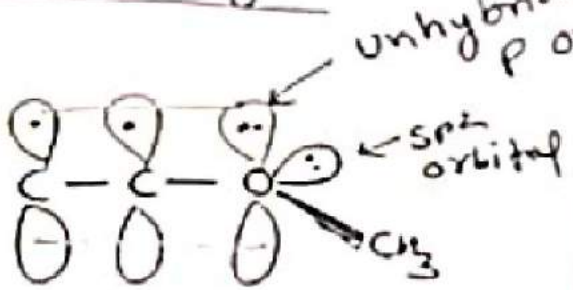
GENERAL ORGANIC CHEMISTRY

Types of conjugation:-

(B) Multiple bond - lone pair conjugation:-

eg 2 $CH_2=CH-\ddot{O}-CH_3$

sp^2



$4\pi e^-$ in conjugation.

eg 3 $CH_2=C=\ddot{O}$ ✓

eg 4 $CH_2=\ddot{O}:$ X

eg 5 $CH_2=CH-\ddot{\underset{\ddot{}}{N}}:$

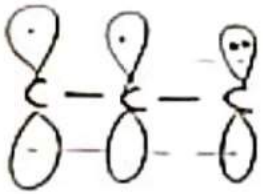
eg 6 CH_2

GENERAL ORGANIC CHEMISTRY

Types of conjugation:-

(B) Multiple bond - lone pair conjugation:-

eg 6 $CH_2=CH-\ddot{C}H_2$ ($\ominus$, SP^2)

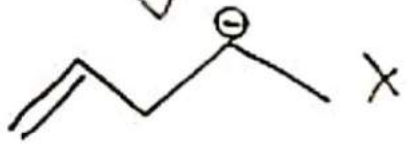


$4\pi e^-$ in conjugation.

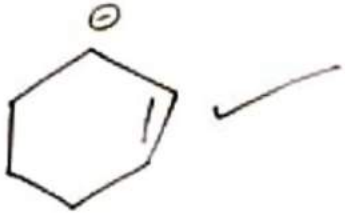
eg 7 $CH_2=\overset{\ominus}{C}H$ No conjugation.

eg 8 $CH_2=C=\overset{\ominus}{C}H$ ✓

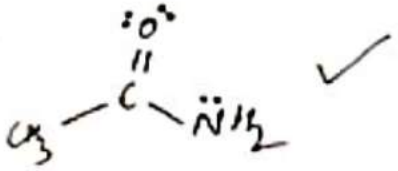
eg 9



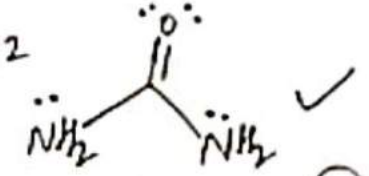
eg 10



eg 11



eg 12



eg 13



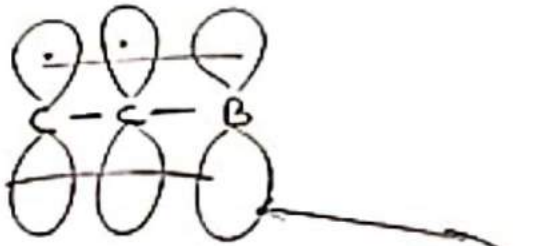
$8\pi e$ in conju.

GENERAL ORGANIC CHEMISTRY

Types of conjugation:-

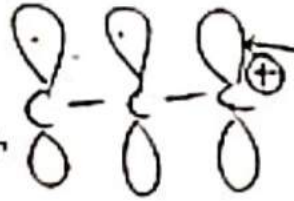
(c) Multiple bond - Vacant orbital conjugation:-

Eg 1 $CH_2=CH-BH_2$



$2\pi e^-$ in 3 orbitals

Eg 2 $CH_2=CH-\overset{\oplus}{C}H_2$



$2\pi e^-$ in conjugate

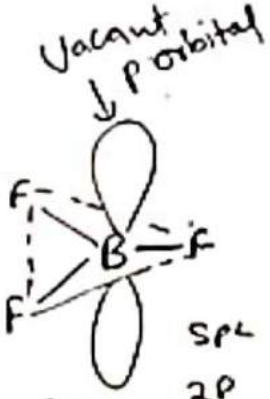
vacant

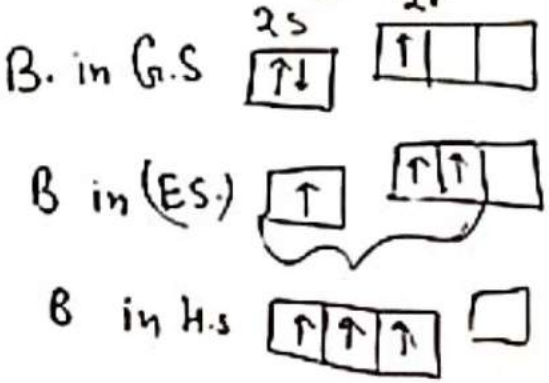
Eg 3 $CH_2=CH-\overset{\oplus}{N}H_3$

sp^3

NO vacant orbital

No Resonance

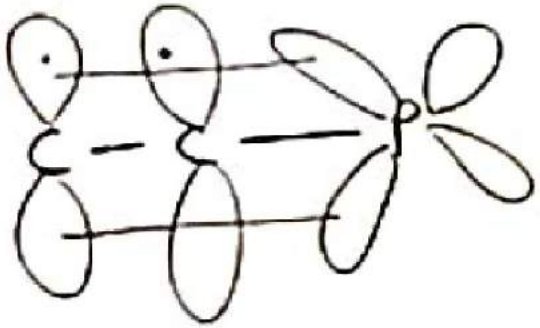




Types of conjugation:-

(c) Multiple bond - Vacant orbital conjugation:-

eg 4 $CH_2=CH-\overset{\oplus}{P}R_3$ ✓



eg 5 $CH_2=CH-\ddot{\underset{\cdot\cdot}{O}}:$ ✓

eg 6 $CH_2=\overset{\oplus}{C}H$ X

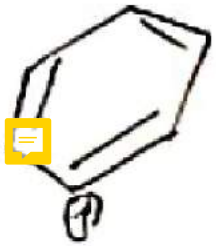
eg 7 $CH_2\overset{\curvearrowright}{=}C=\overset{\oplus}{C}H$ ✓

eg 8



X

eg 9



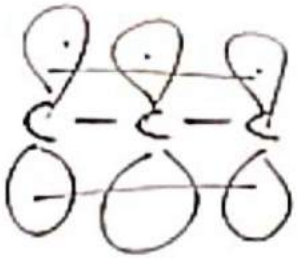
✓

GENERAL ORGANIC CHEMISTRY

Types of conjugation:-

(D) multiple bond - free radical conjugation:

eg1 $CH_2=CH-\dot{C}H_2$



$3\pi e^-$ in conjugation.

eg $\cdot$  X

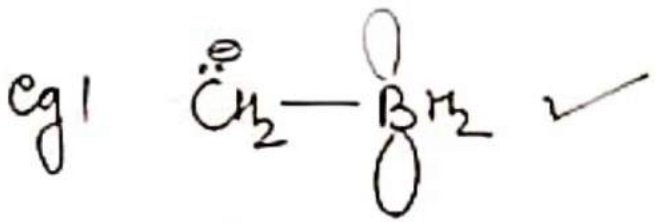
eg2 $CH_2=\dot{C}H$ X

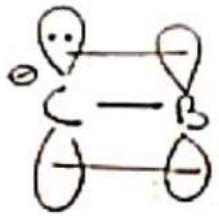
eg3 $CH_2=C=\dot{C}H$ ✓

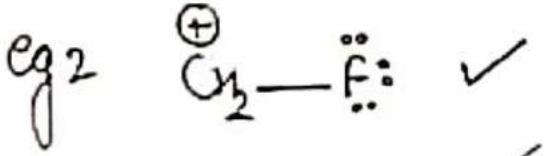
GENERAL ORGANIC CHEMISTRY

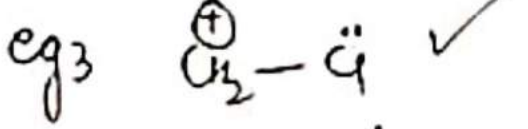
Types of conjugation:-

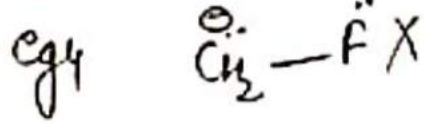
(E) lone pair - vacant orbital conjugation:-

eg 1 

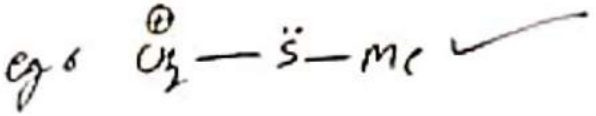


eg 2 

eg 3 

eg 4 

eg 5 

eg 6 

eg 7 

GENERAL ORGANIC CHEMISTRY

Rules For Identification Of Stability of Resonating Structures :-

1. Greater number of π bond containing resonating structures are more stable.
2. Resonating structures in which all atoms have complete Octet, are comparitively more stable.
3. Neutral resonating structures are generally more stable than charged resonating structures.
4. For charged resonating structures :- (a) Negative charge is more stable on larger size containing atom (if comparision between atoms of same group)
   (b) Negative charge is more stable on more electronegative atom and vice-versa. (When size of atoms are approx. same.

GENERAL ORGANIC CHEMISTRY

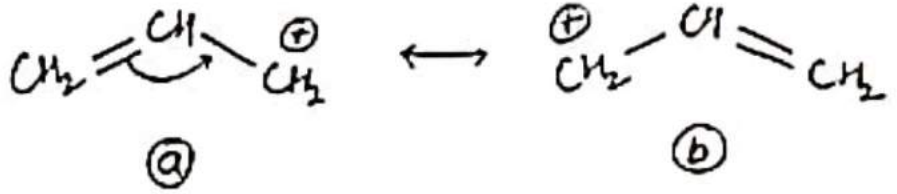
Rules For Identification Of Stability of Resonating Structures :-

5. Aromatic resonating structures are generally more stable than non aromatic reso. structures.

6. -ve charge is more stable on adjacent position of e^- withdrowing group, Similarly +ve charge is more stable on adjacent position of e^- donating group.

7 When conjugated no. of π e^-s are same then linear conju. containing structures are more stable than crossed conjugative Structures.

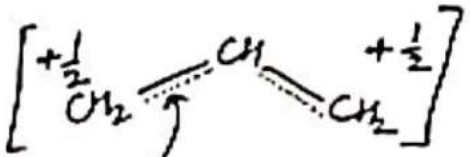
GENERAL ORGANIC CHEMISTRY -I

eg! 

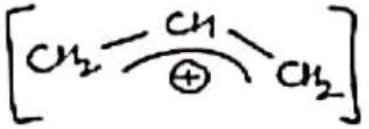
ⓐ ⟷ ⓑ

⇒ Equally Stable R.S.

⇒ Both C–C bond length are same



or



Bond order = $\frac{\text{no. of bonds}}{\text{no. of resonating Structures}} = \frac{3}{2} = 1.5$

$= 1 + \frac{1}{2} = 1.5$

For Single bond (B.O.) = 1

For double bond (B.O.) = 2

" Triple bond (B.O.) = 3

⇒ Bond order ∝ Bond Strength

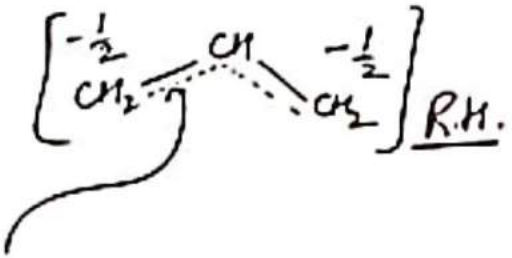
⇒ Bond order ∝ $\frac{1}{\text{Bond length}}$

GENERAL ORGANIC CHEMISTRY -I

eg 2.

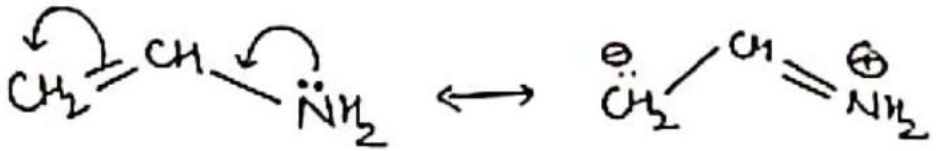


Equally stable R.S.

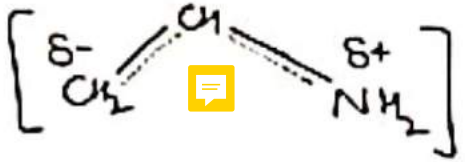


Bond order = 1.5

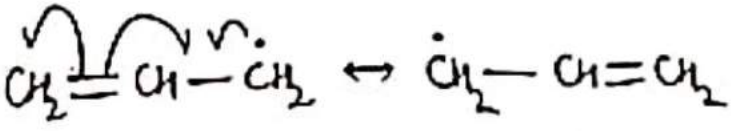
eg 3



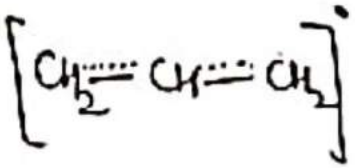
Neutral R.S
(more Stable)



eg 4



Equally Stable

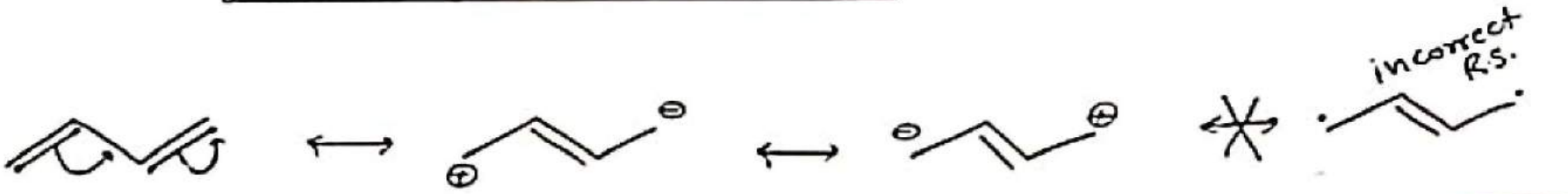


↷ for movement of 2 e⁻s

↶ " " " 1 e⁻

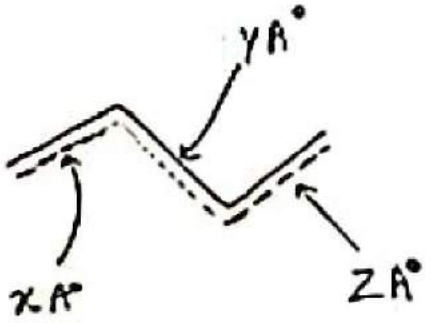
GENERAL ORGANIC CHEMISTRY -I

Eg 5



ⓐ ⓑ ⓒ

Stability ⇒ $\boxed{a > b = c}$

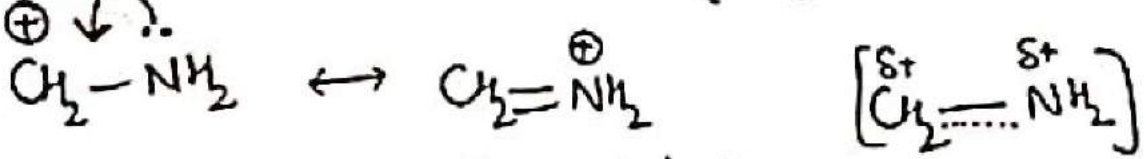


⇒ $y > x = z$

⇒ $y < 1.54\,A°$

⇒ $(x = z) > 1.34\,A°$

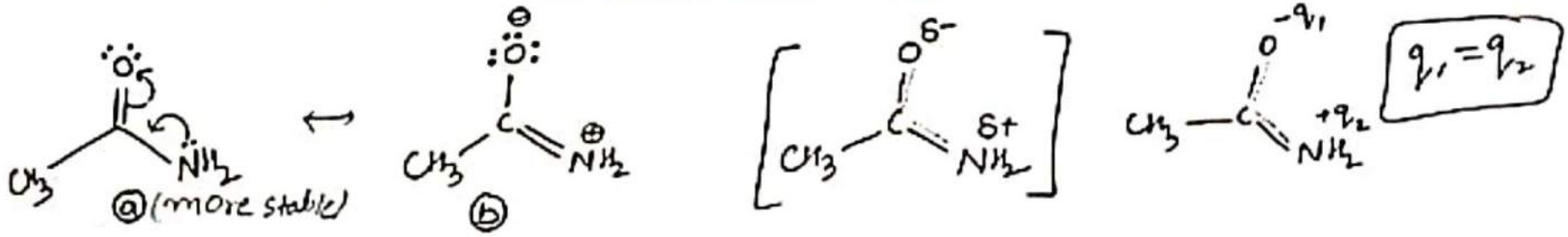
Eg 6



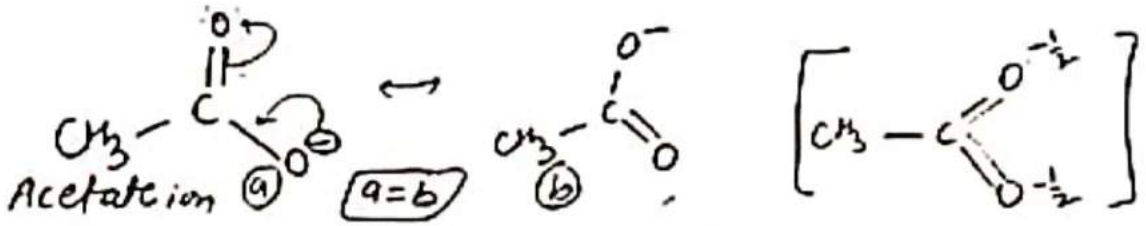
(more Stable)

GENERAL ORGANIC CHEMISTRY -I

eg 7

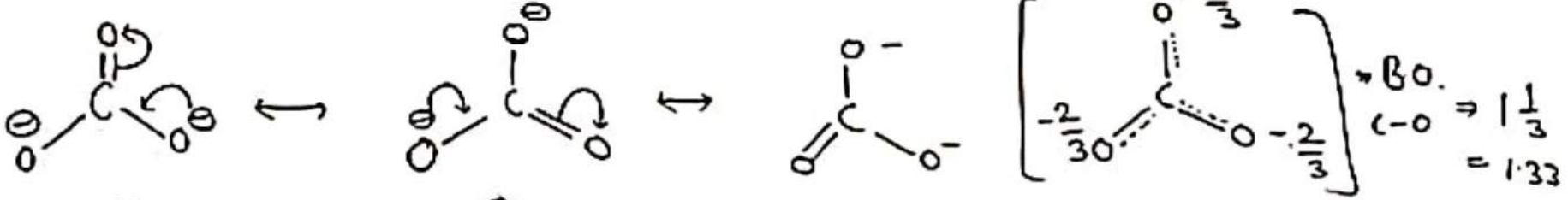


eg 8



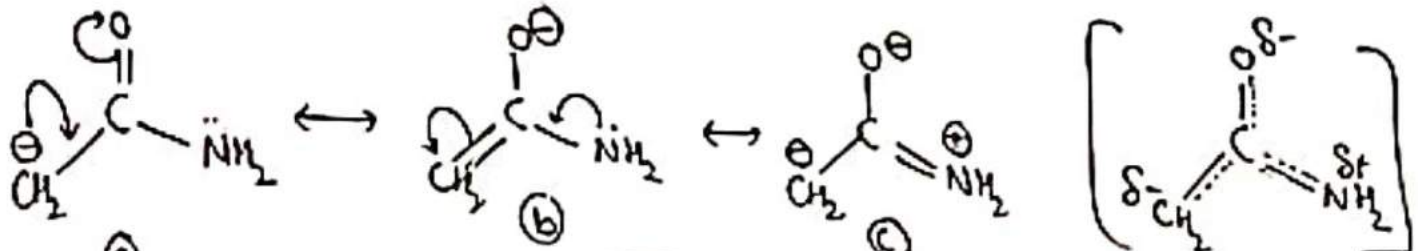
- C-O Bond order = 1.5
- Both C-O bond length = equal.

eg 9



- BO. C-O $\Rightarrow 1\frac{1}{3} = 1.33$

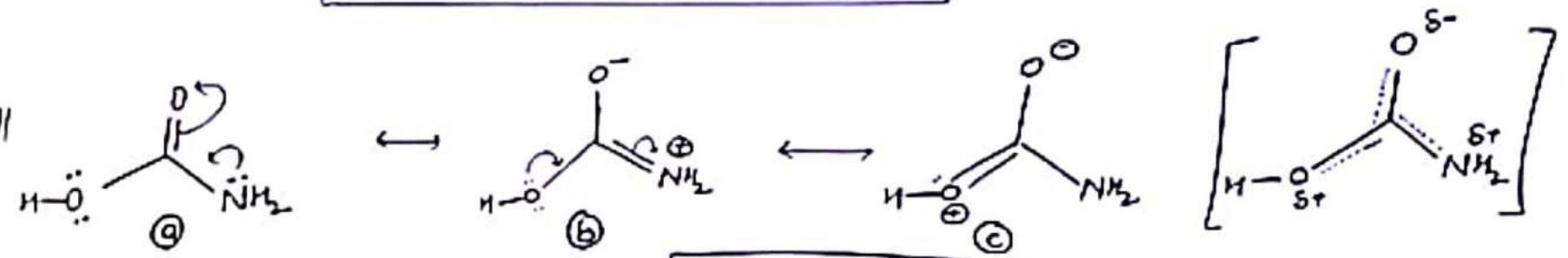
eg 10



b > a > c Stability

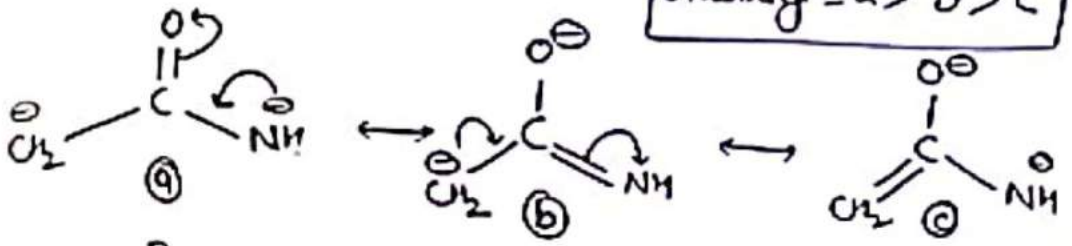
GENERAL ORGANIC CHEMISTRY -1

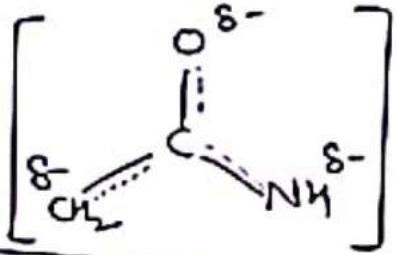
eg 11



Stability = a > b > c

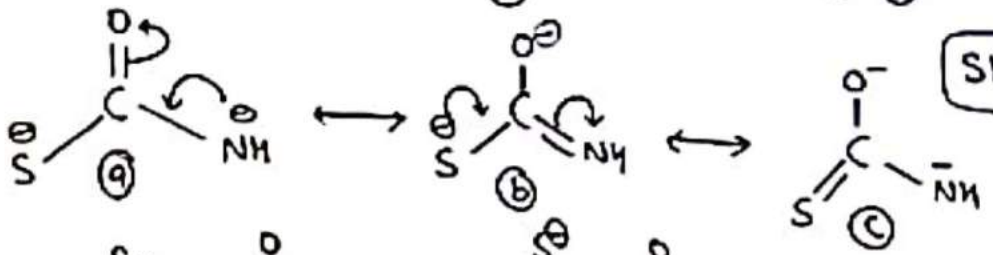
eg 12

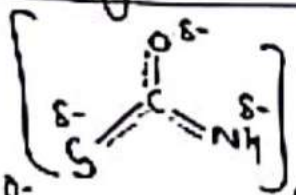




Stability = c > b > a

eg 13



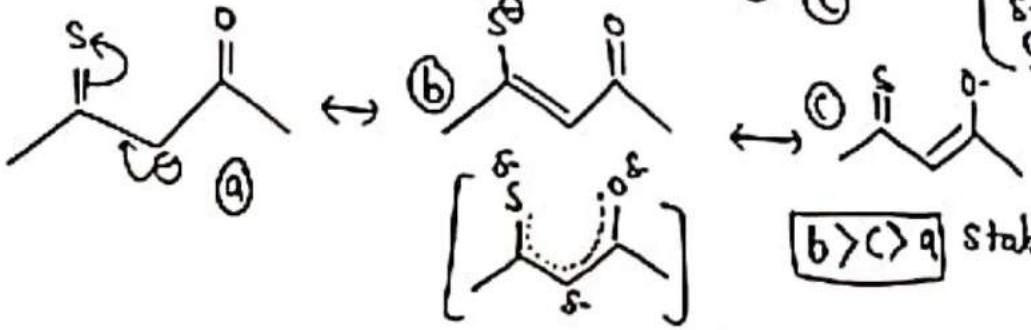


E.N O > N > C

Stability = b > a > c

$S^- > O^- > N^-$

eg 14



b > c > a stability.

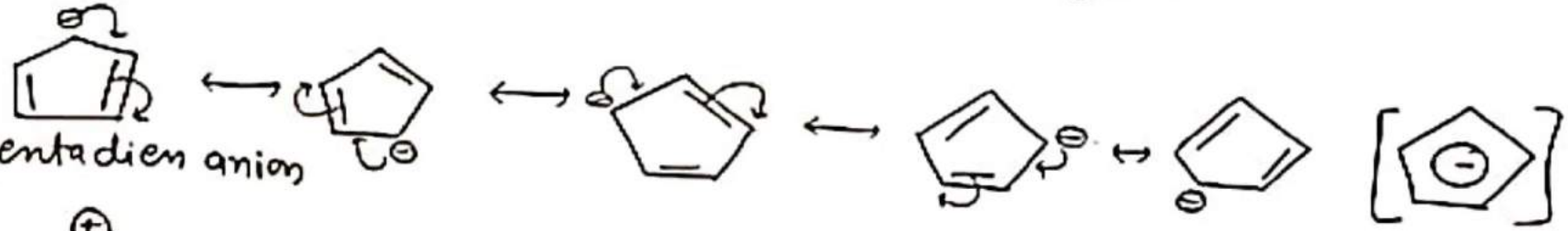
GENERAL ORGANIC CHEMISTRY -I

eg 15



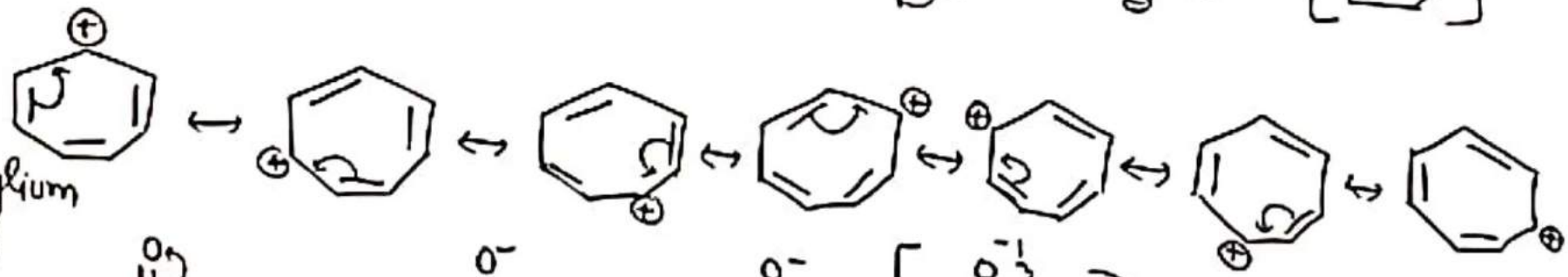
cyclopropenium ion

eg 16



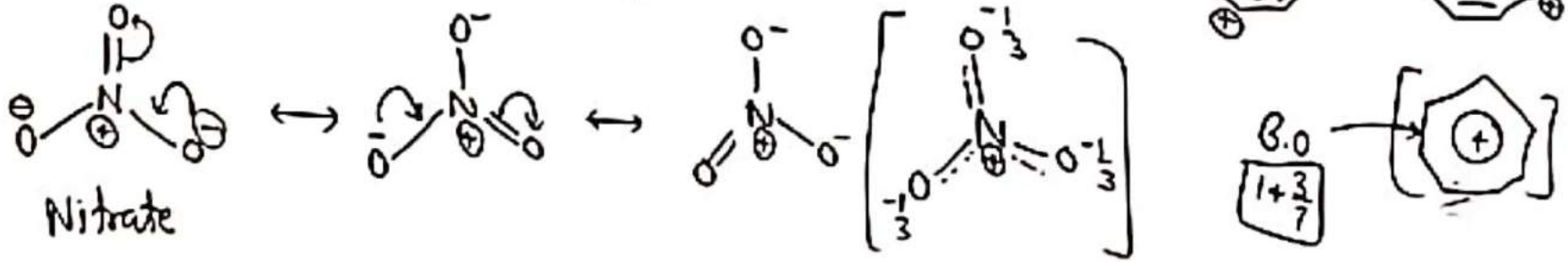
cyclopentadien anion

eg 17



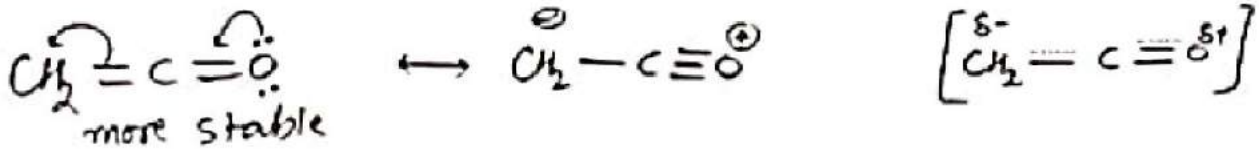
Tropylium ion

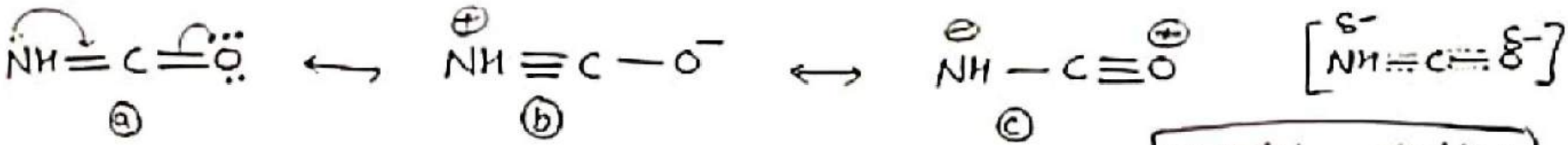
eg 18

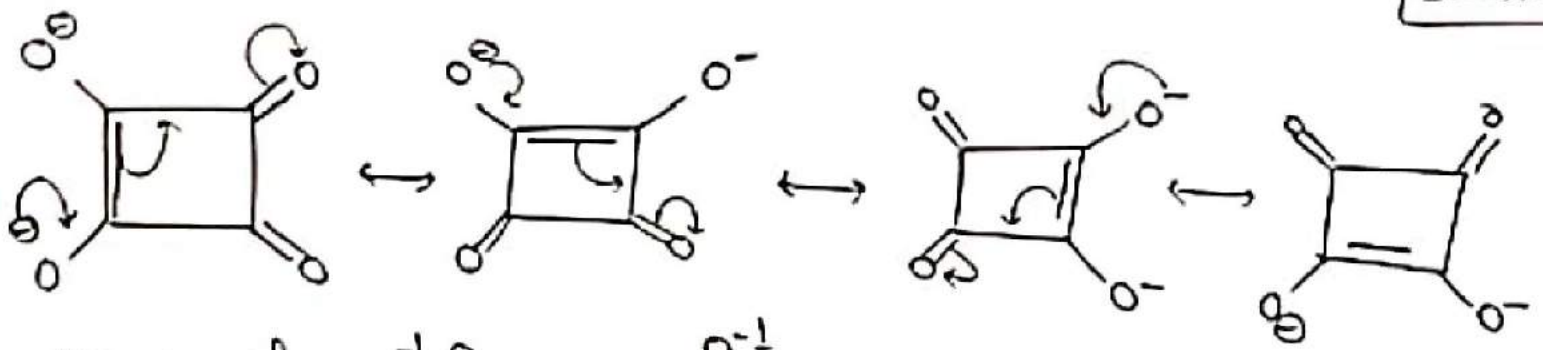


Nitrate

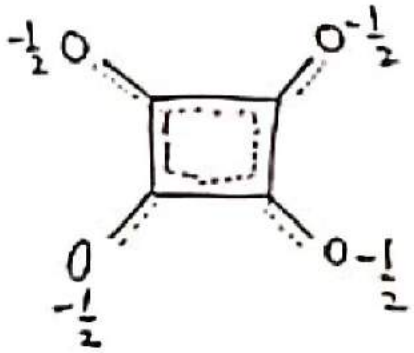
GENERAL ORGANIC CHEMISTRY -I

g 19 

g 20 

g 21 

Dianion of Squaric acid



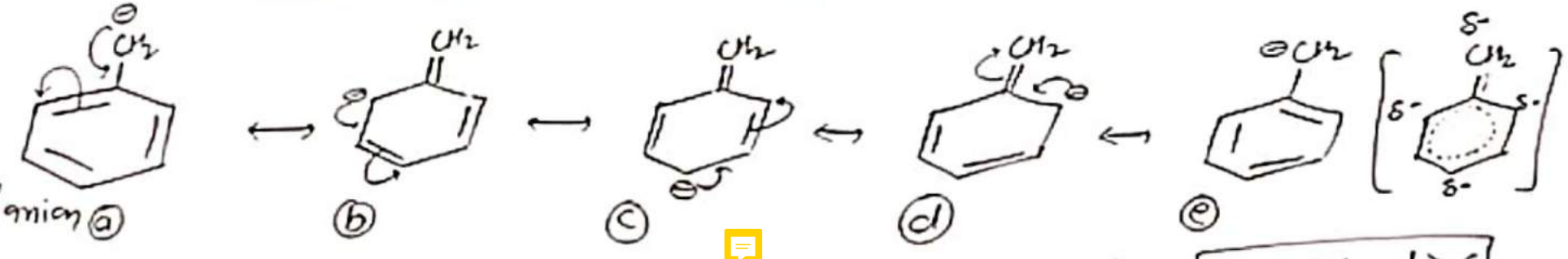
= Bond order for C–C = $1\frac{1}{4}$

= " " " C–O = $\frac{6}{4} = 1.5$

$\left(1\frac{2}{4}\right)$

GENERAL ORGANIC CHEMISTRY -I

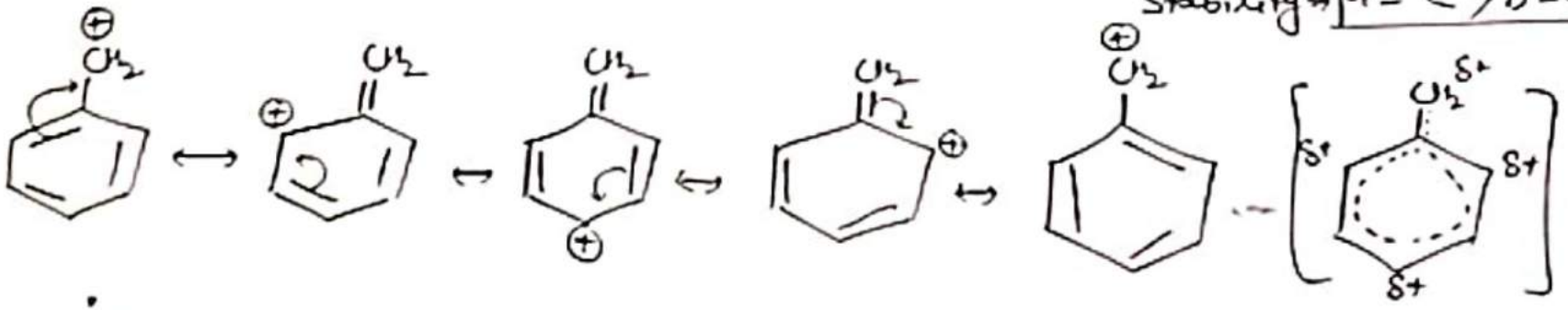
eg 22



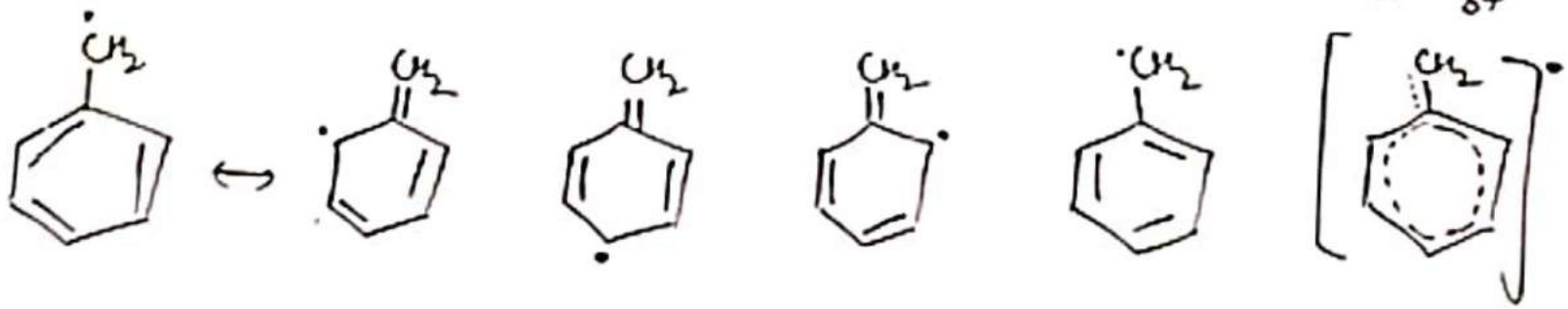
Benzylanion ⓐ ⓑ ⓒ ⓓ ⓔ

Stability → a = e > b = d > c

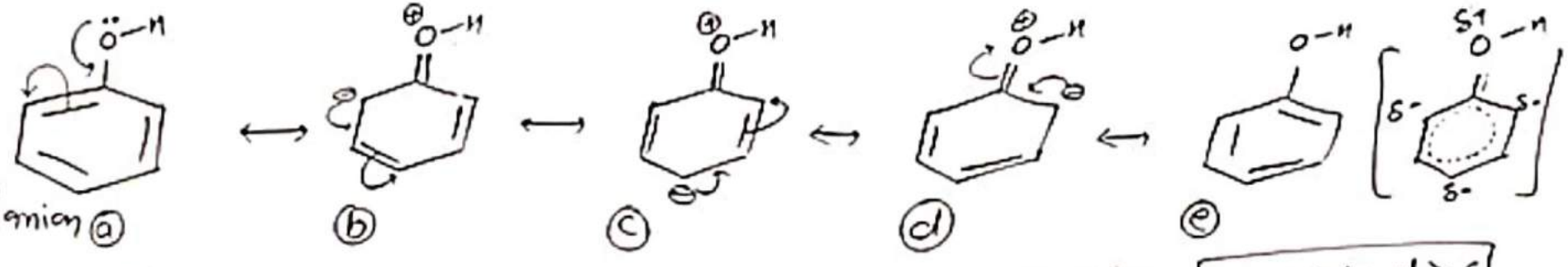
eg 23



eg 24



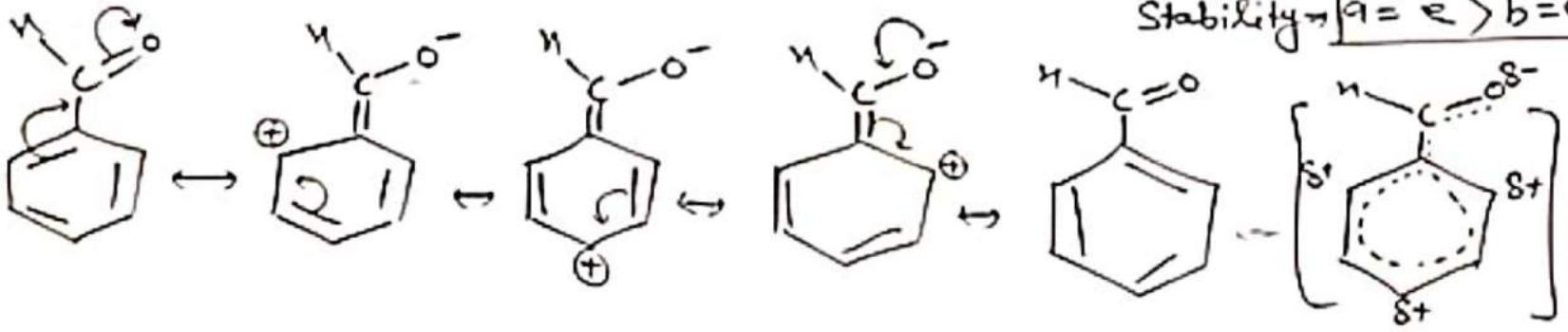
eg 25



Benzylanion ⓐ ⓑ ⓒ ⓓ ⓔ

Stability → a = e > b = d > c

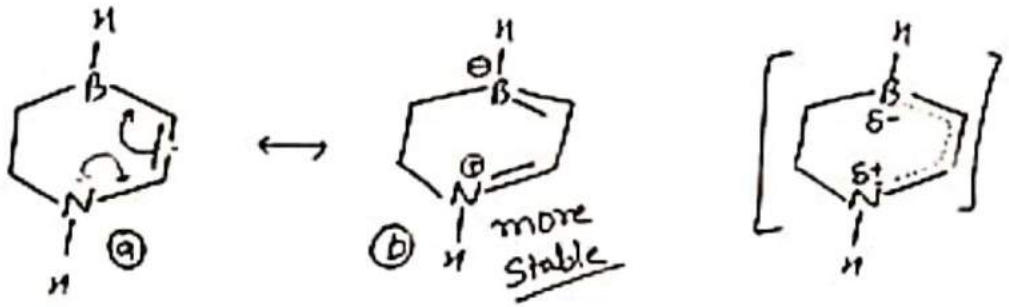
eg 26



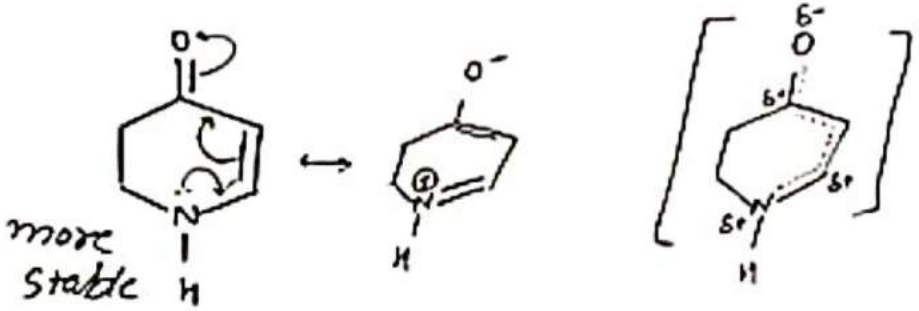
Resonance effect is equally effective of O, P. position for benzene ring (Benzylic resonance group)

GOC - I

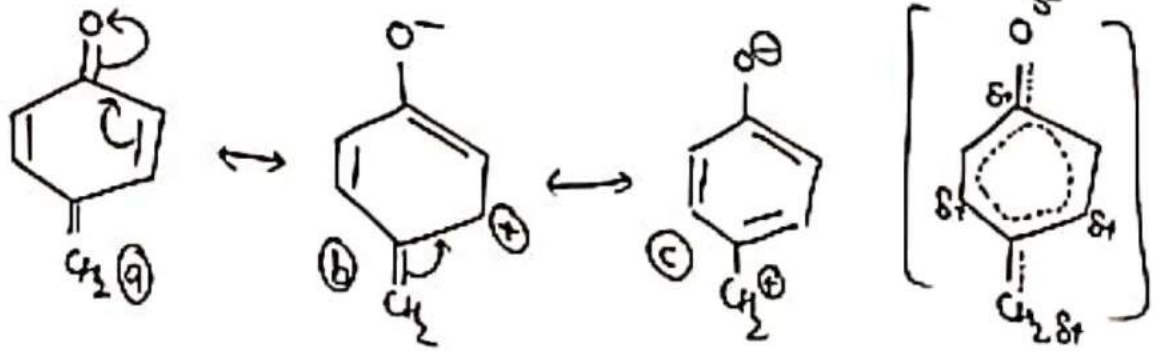
Eg 27



Eg 28



Eg 29

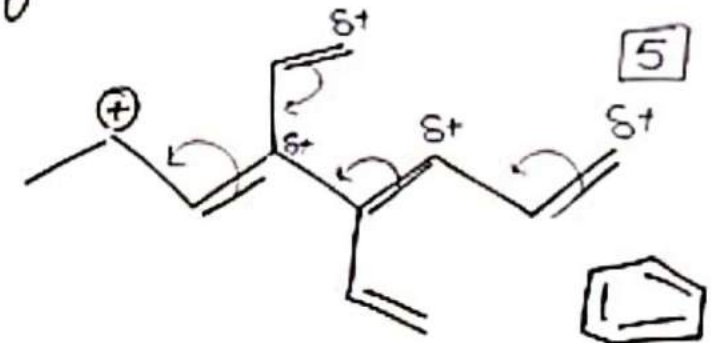


Stability

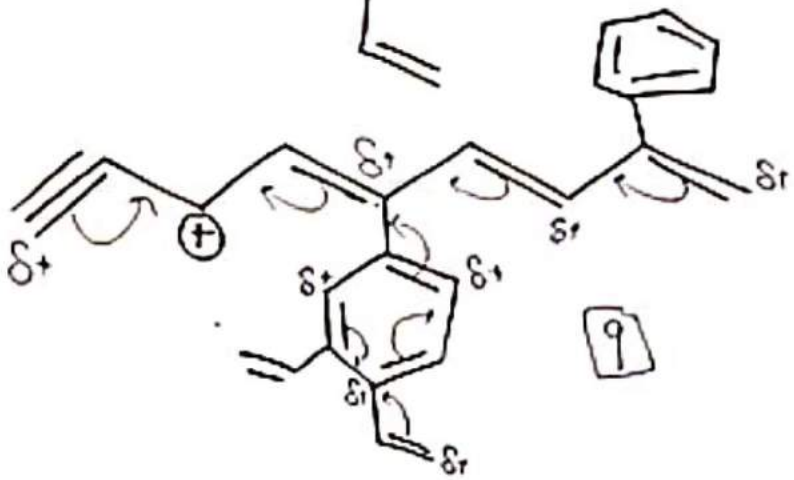
a > c > b

GOC - I

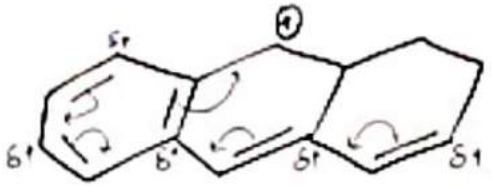
Q Identify no. of position in resonance hybrid which have δ+ for given structures.

① 

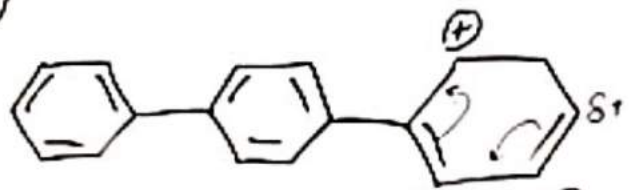
5

② 

9

③ 

6

④ 

3

GOC - I

Q Draw Neutral resonating Structures for given compounds.

① Benzene

② Napthalene

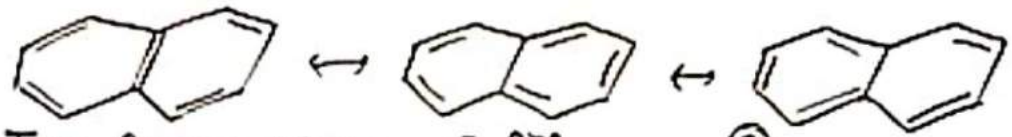
③ Anthracene

④ Phenanthrene

Solution 1



Sol 2.



Two Benzanoid ring (a) | one Benzanoid ring (b) | Three R.S. (c)

Stability a > b = c

GOC - I

Q Draw Neutral resonating Structures for given compounds.



Benzene

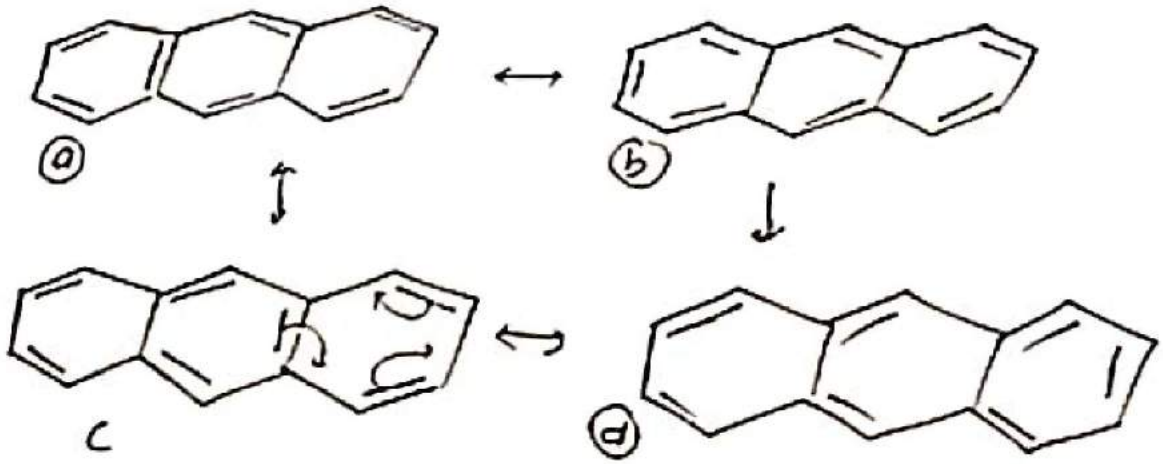
Napthalene

3) Anthracene

Phenanthrene



Sol. 3



Stability $(a = c) > (b = d)$

GOC - I

Q. Draw Neutral resonating Structures for given compounds.

① Benzene → 2

② Napthalene → 3

③ Anthracene → 4

④ Phenanthrene → 5

Sol. Phenanthrene

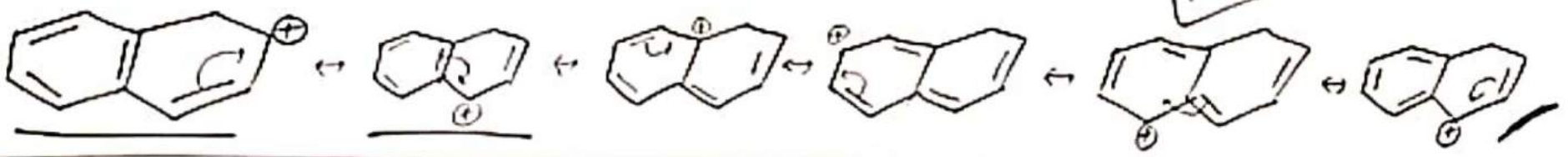


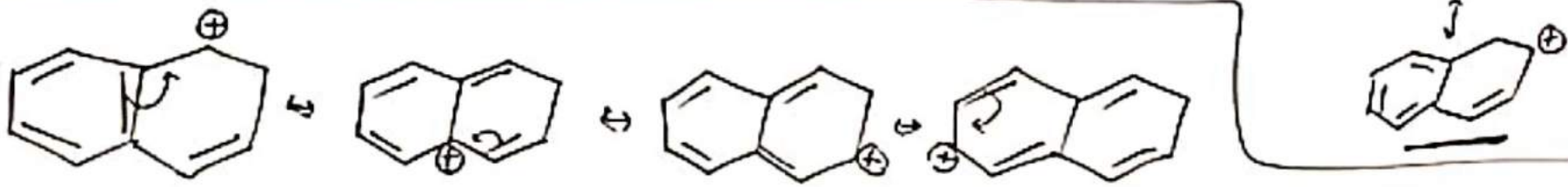
GOC - I

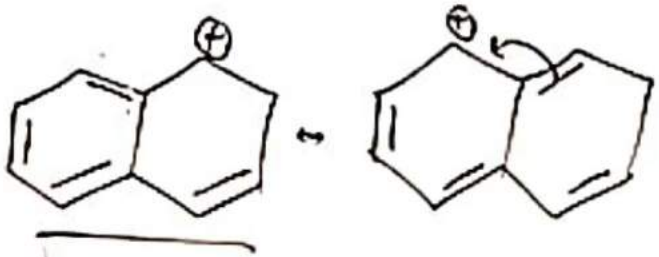
Q Draw possible Resonating structures.

4 R.S. have benzanoid ring.

7 Reso. Stru

① 

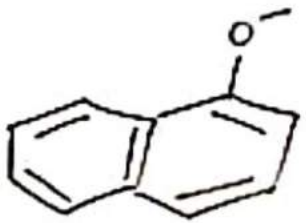
② 



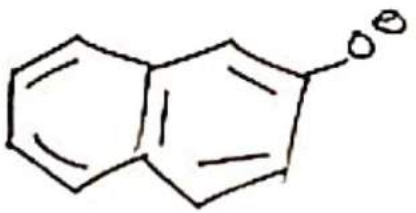
6 R.S = 2 R.S. have Benzanoid ring

GOC - I

Q Draw possible Resonating structures.

(3)  ?

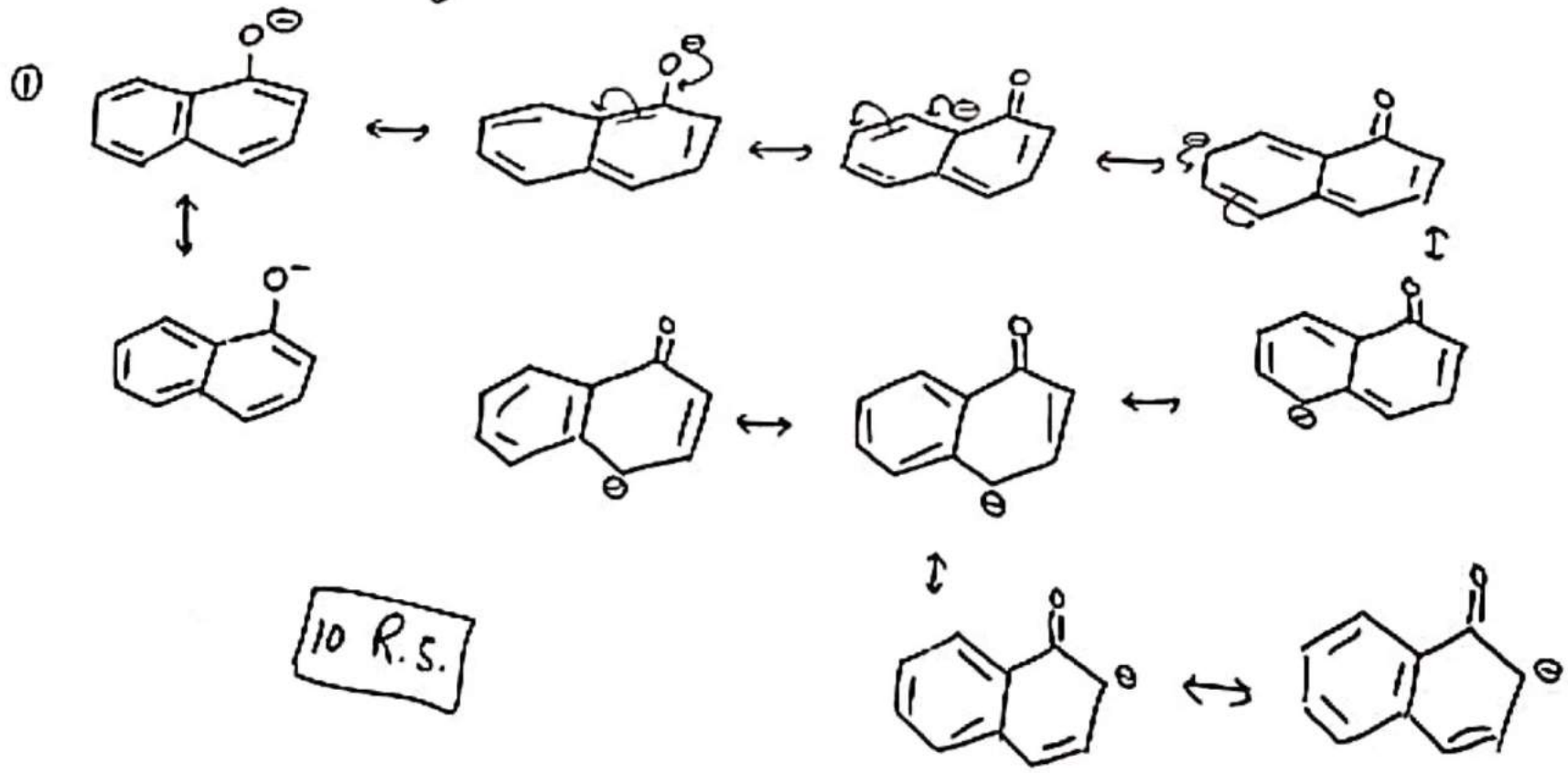
anion of α-Naphthol

(4)  ?

anion of β Naphthol

GENERAL ORGANIC CHEMISTRY - I

Q Draw resonating structures.

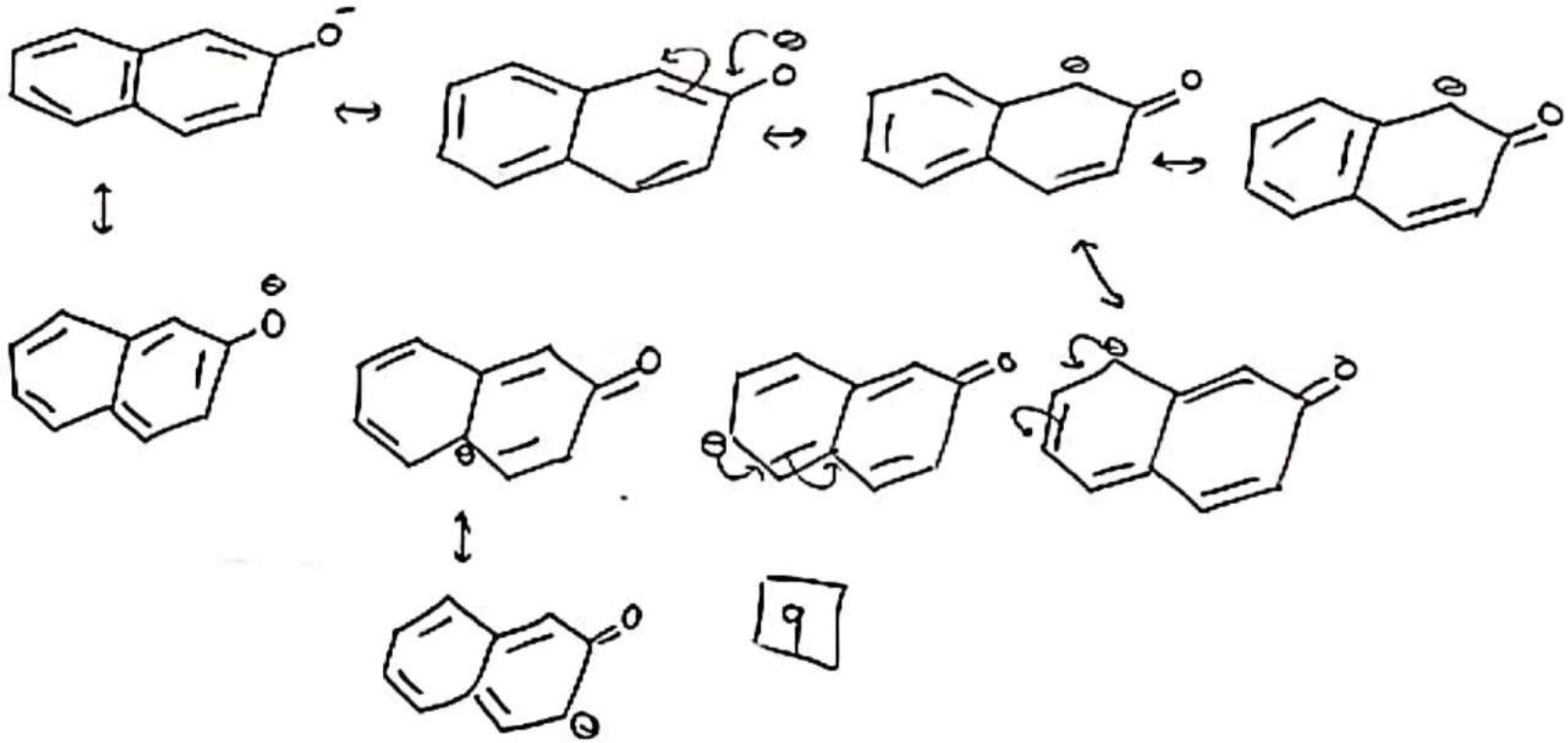


10 R.S.

GENERAL ORGANIC CHEMISTRY - I

Q Draw resonating structures. (IIT JEE)

(2)



9

GENERAL ORGANIC CHEMISTRY - I

Mesomeric effect (M) = Resonance effect (R)

Effects which are produced by resonance/ delocalisation of πe^- in conjugate system.

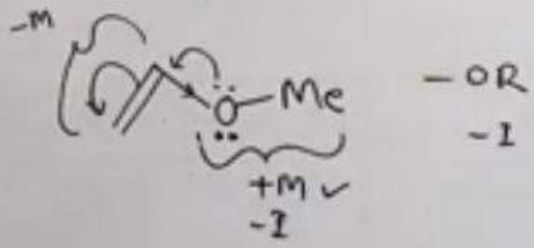
Types of mesomeric groups → +M, −M

GENERAL ORGANIC CHEMISTRY - I

Mesomeric effect (M) = Resonance effect (R)

(A) **+M / +R Groups** :- π e^- donar in conjugate system.

⇒ They have lone pair on first atom of conjugate system



-OR
-I

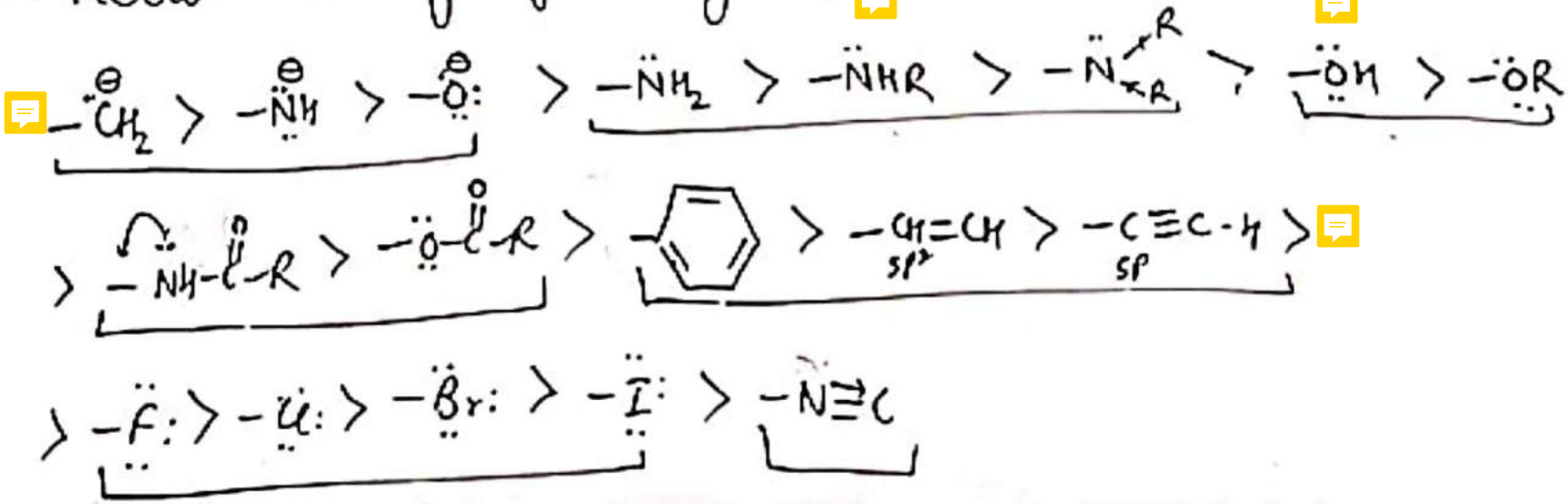
GENERAL ORGANIC CHEMISTRY - I

Mesomeric effect (M) = Resonance effect (R)

(A) **+M / +R Groups** :- π e⁻ donor in conjugate system.

⇒ They have lone pair on first atom of conjugate system

⇒ Relative strength of +M groups



GENERAL ORGANIC CHEMISTRY -I

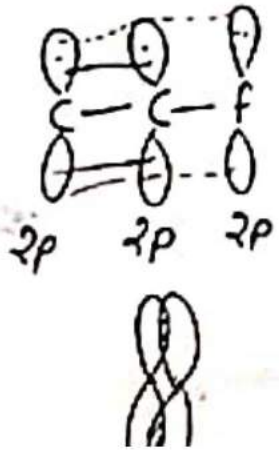
Mesomeric effect (M) = Resonance effect (R)

(A) +M / +R Groups.

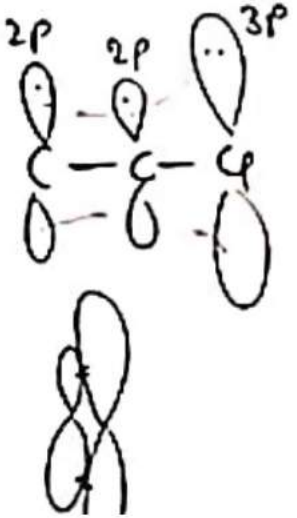
$-\ddot{O}H > -S-H$ +M

⇒ For X

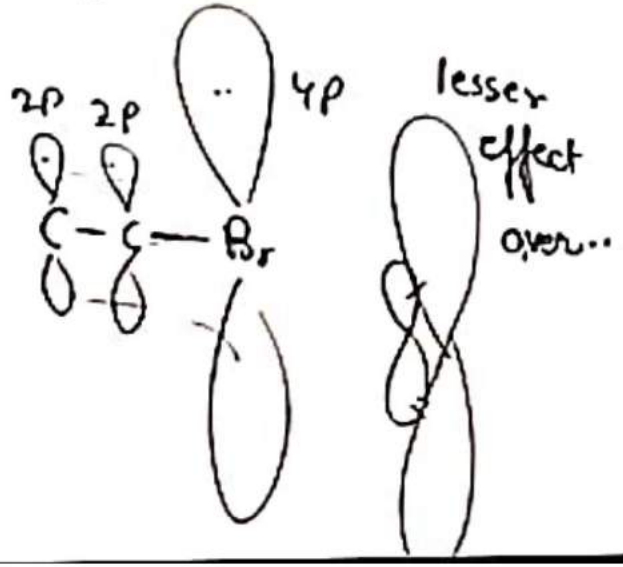
$CH_2=CH-\ddot{F}$



$CH_2=CH-\ddot{Cl}$



$CH_2=CH-\ddot{Br}$



GENERAL ORGANIC CHEMISTRY - I

⁂ Mesomeric effect (M) = Resonance effect (R)

B) -M / -R Group :- ⇒ πe^- withdrowing groups in conjugate system.

⇒ They have vacant orbital / multiple bond (with more electro-negative atom) on first atom of group.

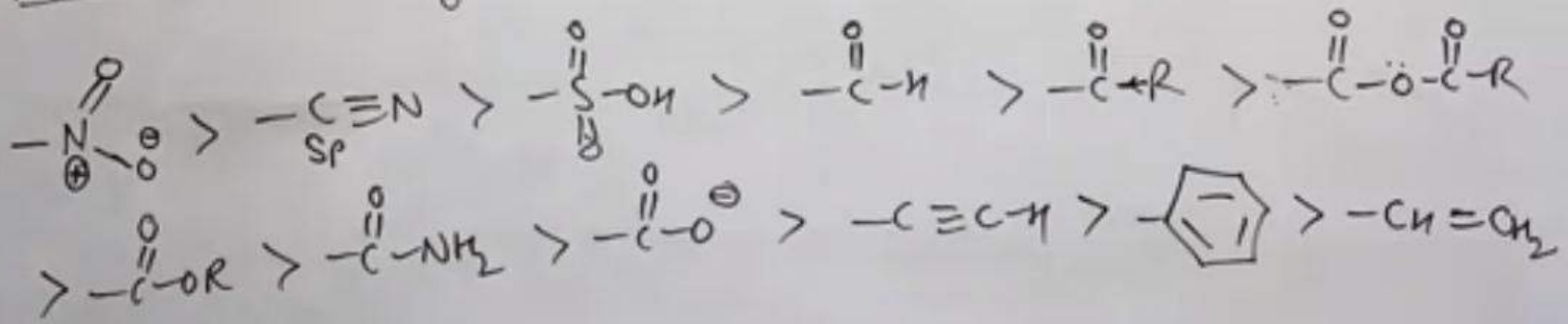


GENERAL ORGANIC CHEMISTRY - I

Mesomeric effect (M) = Resonance effect (R)

(B) -M / -R Group :- ⇒ πe^- withdrowing groups in conjugate system.

⇒ Relative Strength :-



GENERAL ORGANIC CHEMISTRY - I

Mesomeric effect (M) = Resonance effect (R)

(B) -M / -R Group :- ⇒ πe^- withdrowing groups in conjugate system.

⇒ Relative Strength :-



GENERAL ORGANIC CHEMISTRY - I

Mesomeric effect (M) = Resonance effect (R)

(B) -M / -R Group :- ⇒ πe^- withdrowing groups in conjugate System.

⇒ Relative Strength :-



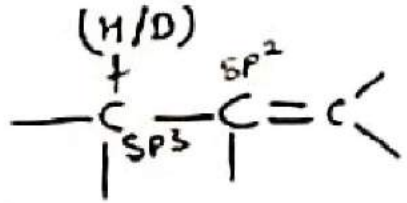
(III) Hyperconjugation / Baker-Nathan effect / NO Bond Resonance:-

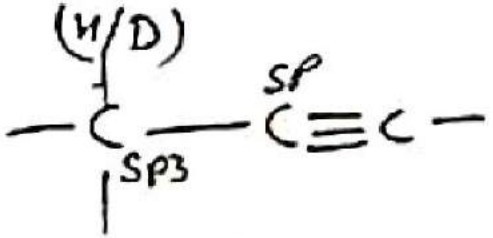
When min. one C–H / C–D σ-bond(s) of SP^3 carbon is/are present on adjacent position of carbon-carbon multiple bond / +ve charbon / carbon free radical, then these C–H / C–D σ bond orbitals is/are overlap with adjacent $P / \pi / \pi^*$ orbital, this special delocalisation is called hyper conjugation.

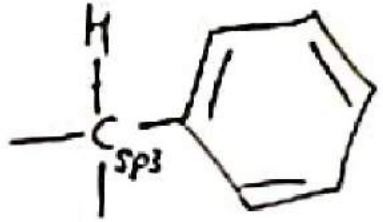
GENERAL ORGANIC CHEMISTRY-1

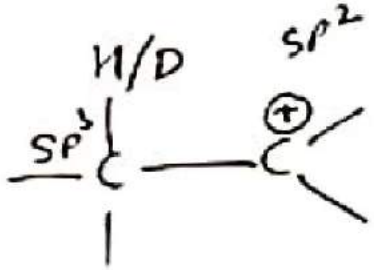
I) Hyperconjugation / Baker-Nathan effect / No Bond Resonance:-

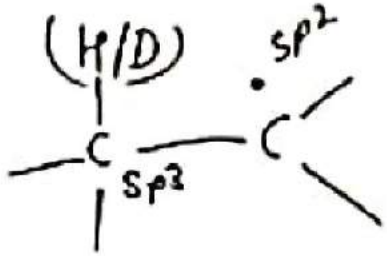
H.C. Possible in folliwing structures.

① 

② 

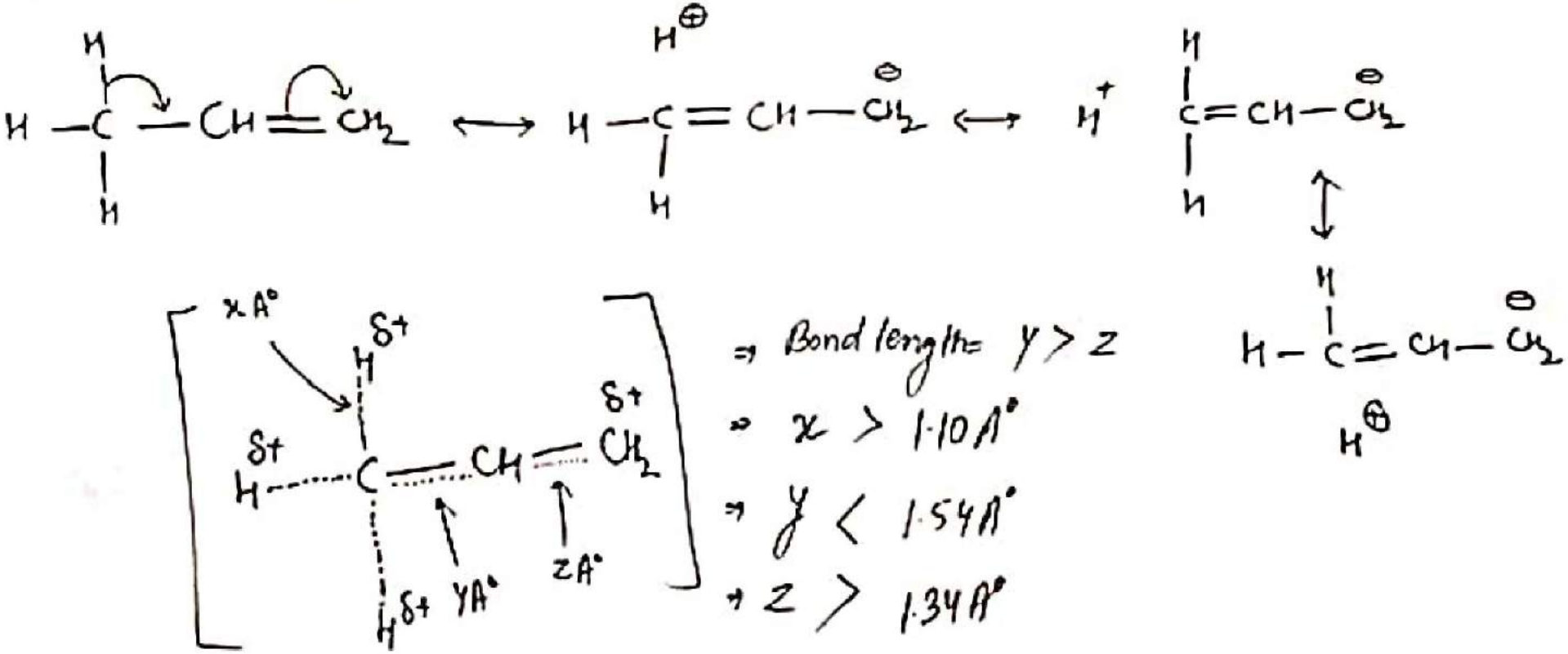
③ 

④ 

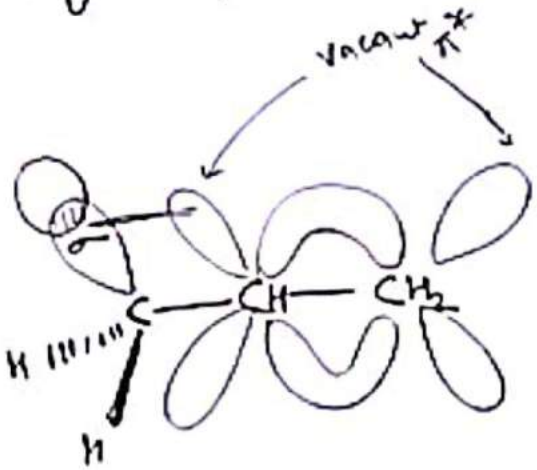
⑤ 

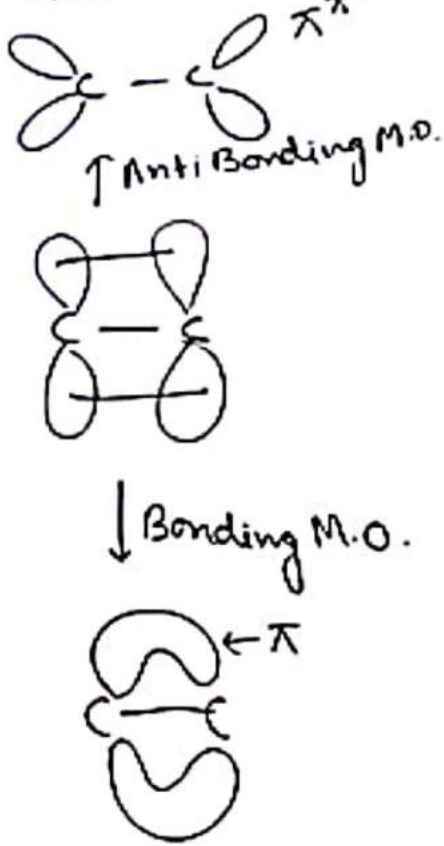
(III) Hyperconjugation / Baker-Nathan effect / NO Bond Resonance:-

eg: H.C. in alkene:-



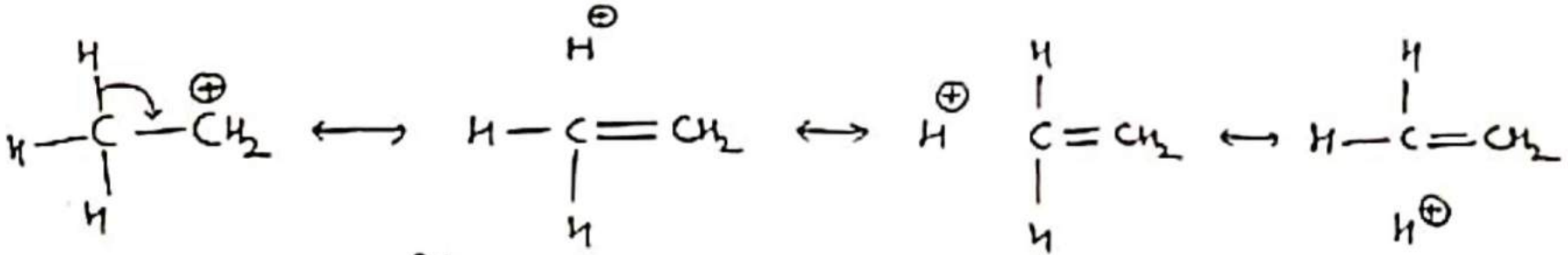
(III) Hyperconjugation / Baker-Nathan effect / No Bond Resonance:-

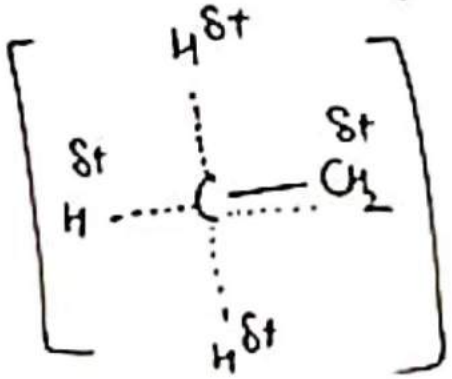




(III) Hyperconjugation / Baker-Nathan effect / No Bond Resonance

(b) H.C. in carbocation (carbonium ions)

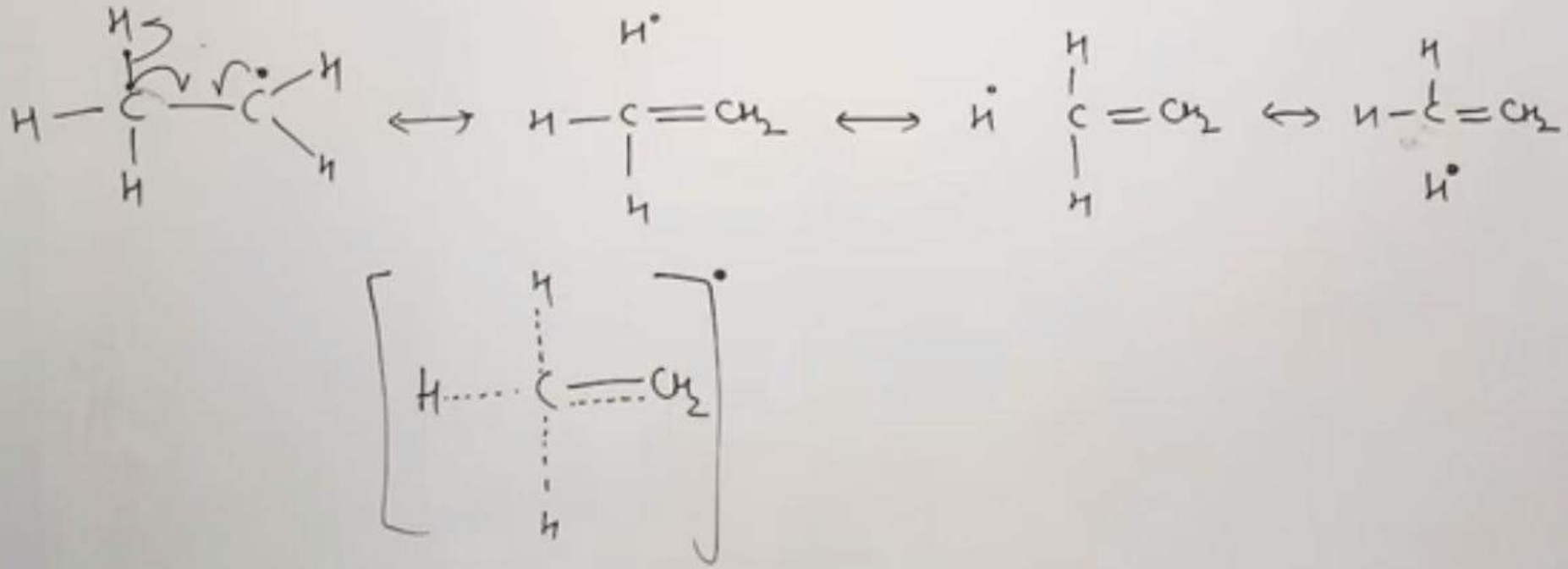




GENERAL ORGANIC CHEMISTRY-I

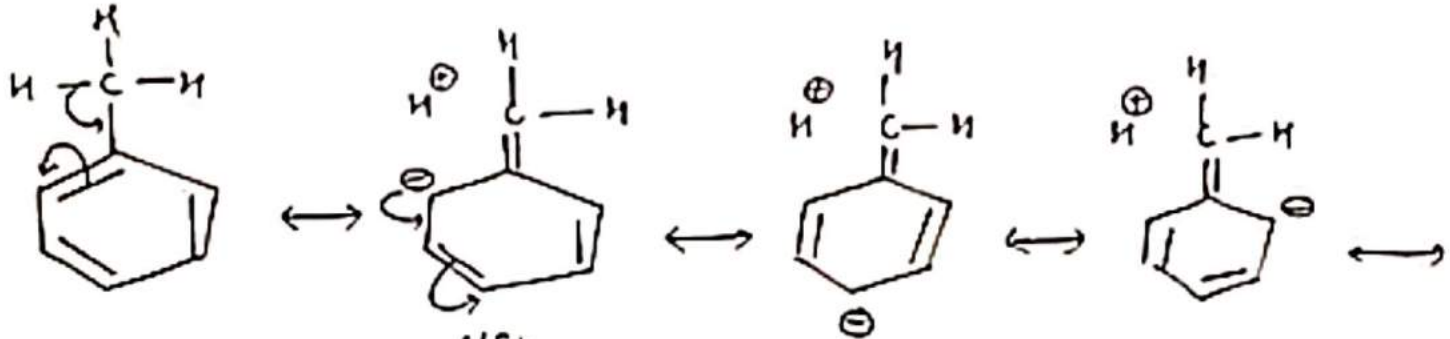
(III) Hyperconjugation / Baker-Nathan effect / No Bond Resonance

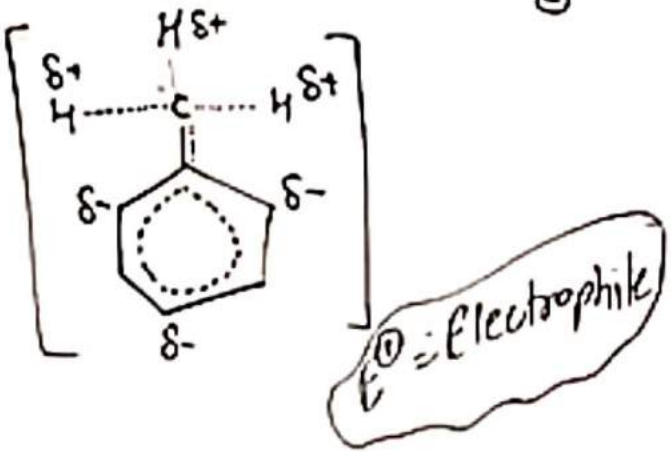
(C) H.C. in free radicals :-



(III) Hyperconjugation / Baker-Nathan effect / No Bond Resonance

(d) H.C. in Toluene ($Ph-CH_3$)





⊕ = Electrophile

Subscribe

(III) Hyperconjugation / Baker-Nathan effect / No Bond Resonance

Note

① Stability of alkene & Alkyne $\propto$ no. of Hyperconjugative Structure

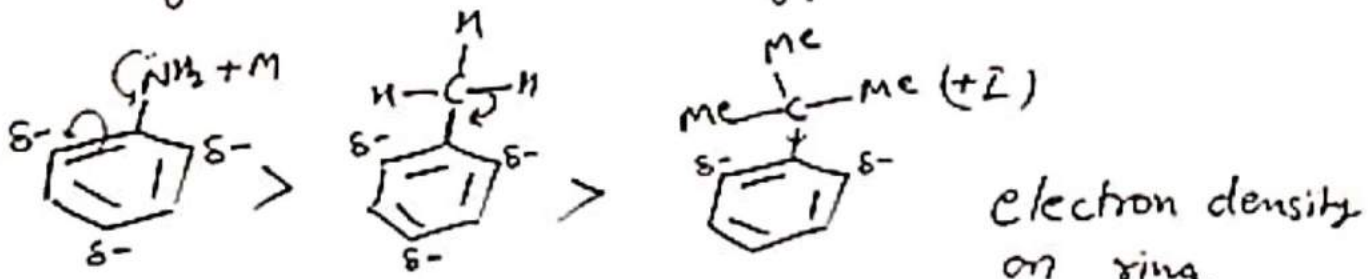
② " " Carbocations $\propto$ " " " "

③ " " free radicals $\propto$ " " " "

(4) Hyperconjugation is electron donating effect

- Effectivity ⇒ Resonace effect > Hyperconjugation > Inductive effect
- e⁻ donating nature ⇒ $\boxed{+M > +H > +I}$

(III) Hyperconjugation / Baker-Nathan effect / NO Bond Resonance

eg 

electron density on ring

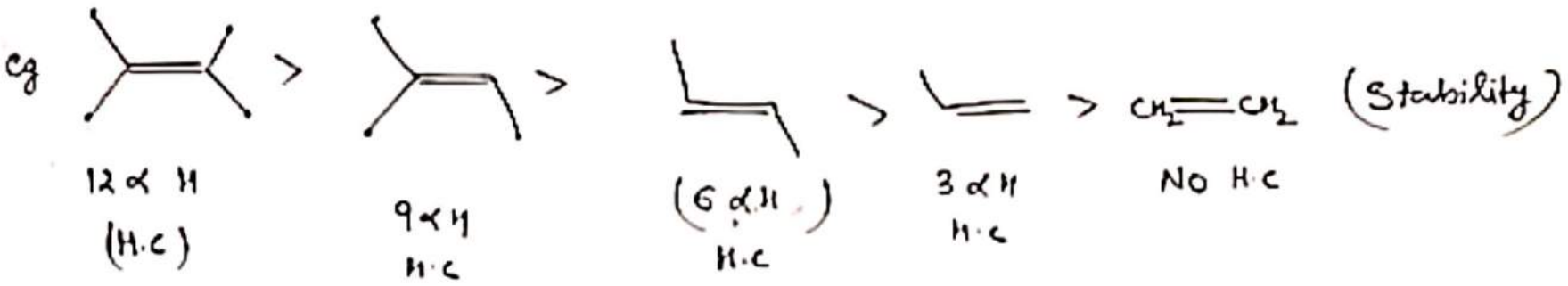
(5) on aromatic ring H.c. effect is applicable on ortho and para position (equally effect)

(6) C–H > C–D (Hyper conjugation effect)

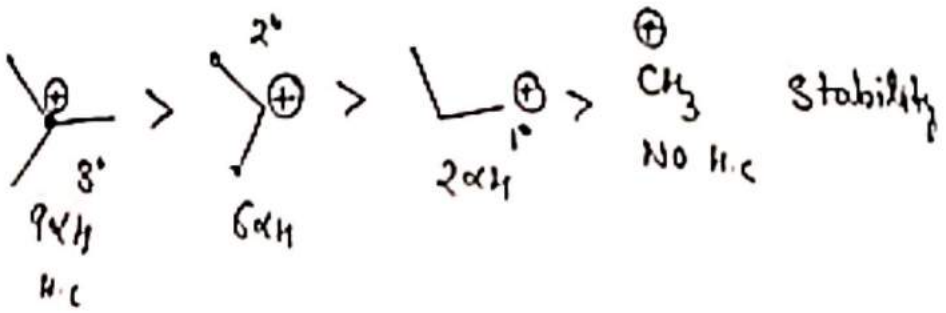
(because C–H bond is weaker than C–D bond)

Applications of Hyperconjugation (In Short)

① Stability of alkene :-



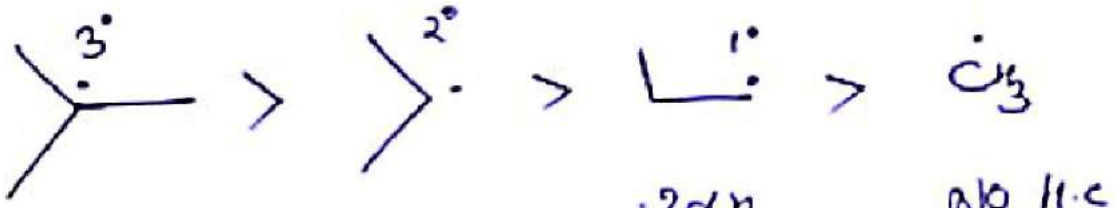
② Stability of carbocation :-



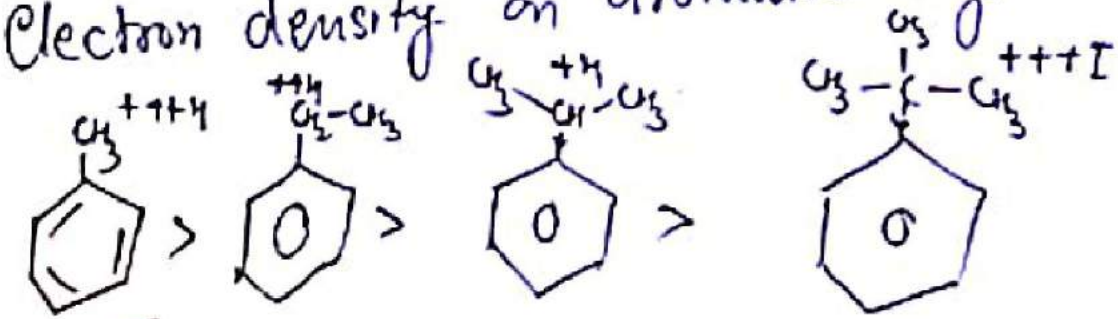
Subscribe

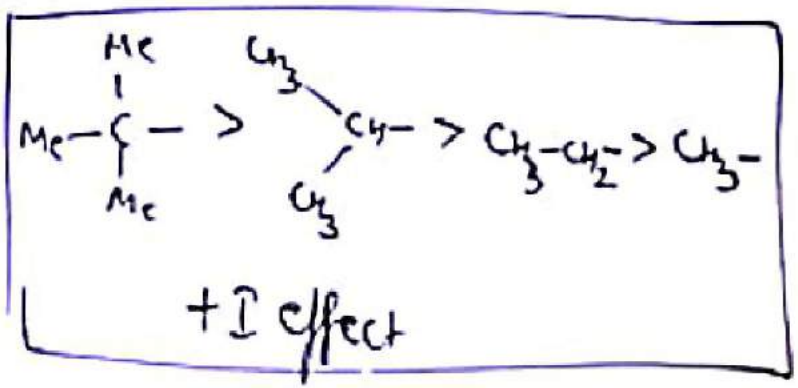
Applications of Hyperconjugation (in Short)

③ Stability of free radicals



④ Electron density on aromatic ring







Applications of Hyperconjugation (in Short)

⑤ Bond length of C=C in alkenes ∝ no. of H.C

No H.C

$CH_2=CH_2$

↓

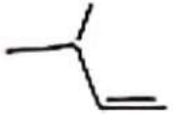
C=C Bond length → 1.34 A°



1.39 A°



1.37 A°



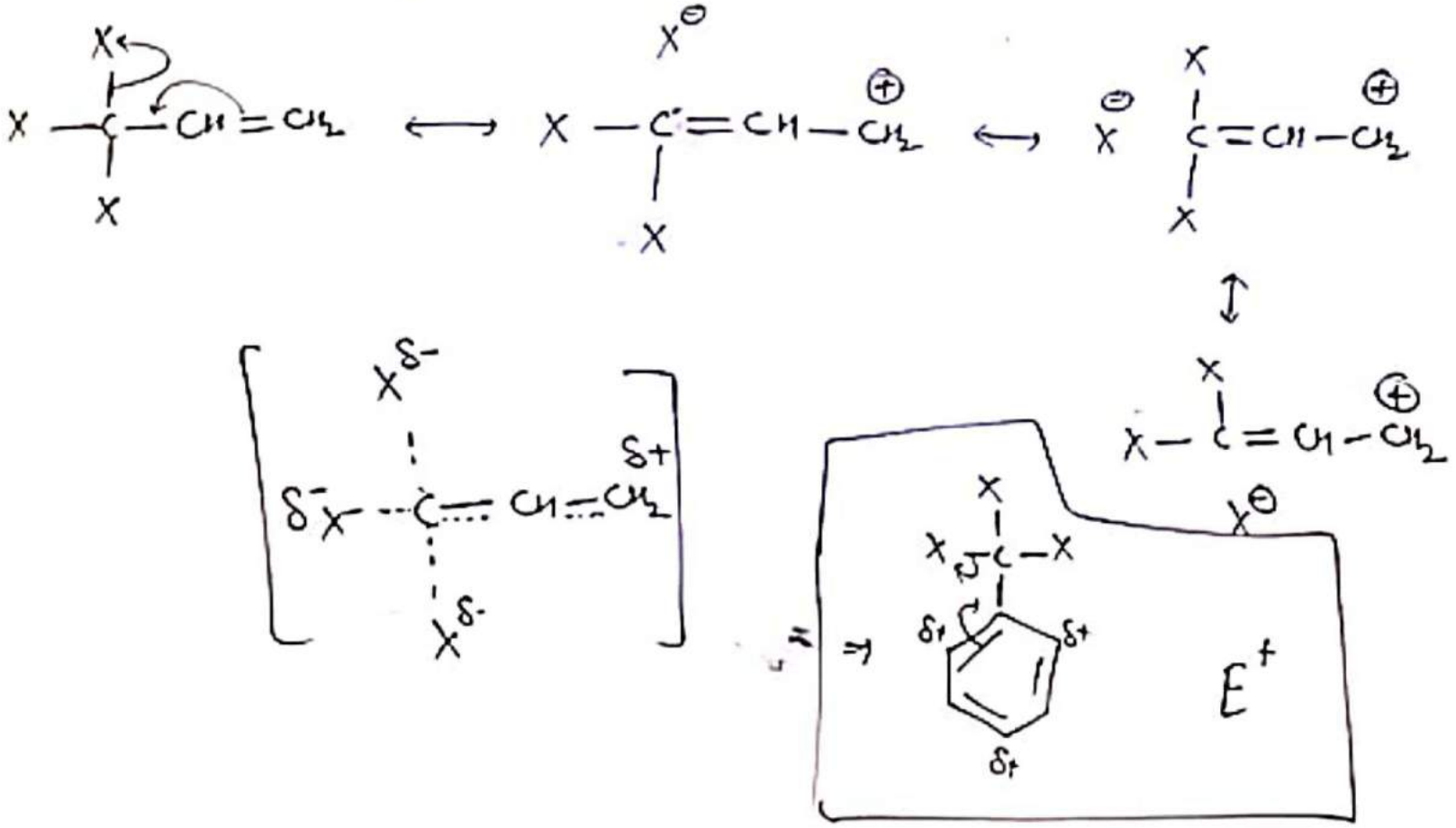
1.35 A°



↑

1.34 A°

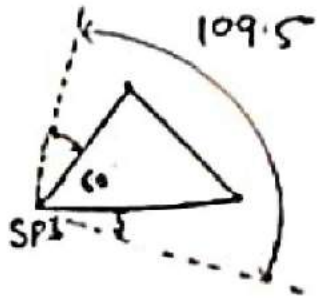
-ve H.C. / Reverce hyper conjugation:-

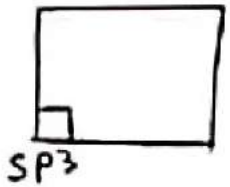

X
X
X

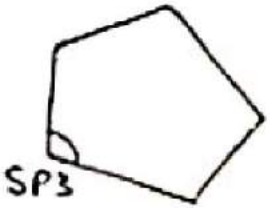
Applications of Electronic Effects

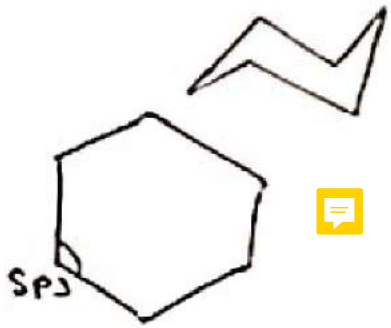
(I) Baeyer's angle strain / Ring Strain:-

It is represented by difference between ideal bond angle and actual bond angle in ring.





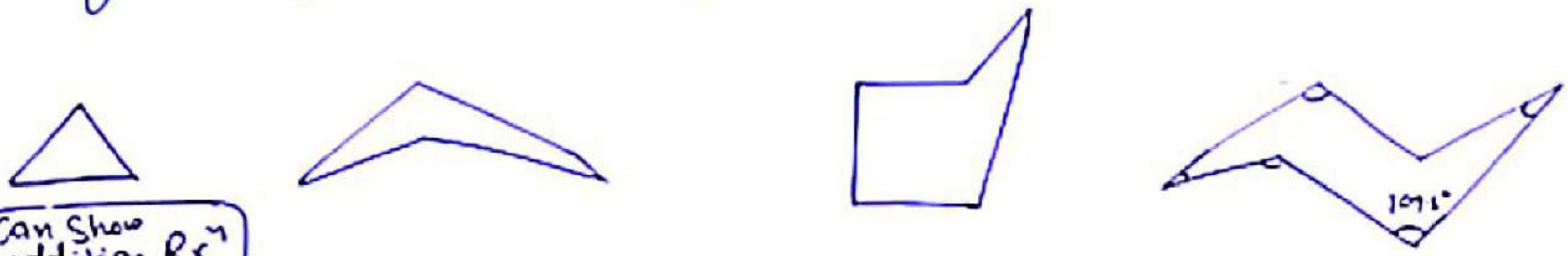




ideal angle	109·5°	109·5°	109·5°	109·5°
actual angle (if planar)	60°	90	108	120
	49·5°	19·5	1·5	10·5°

Applications of Electronic Effects

(I) Baeyer's angle strain / Ring Strain :-

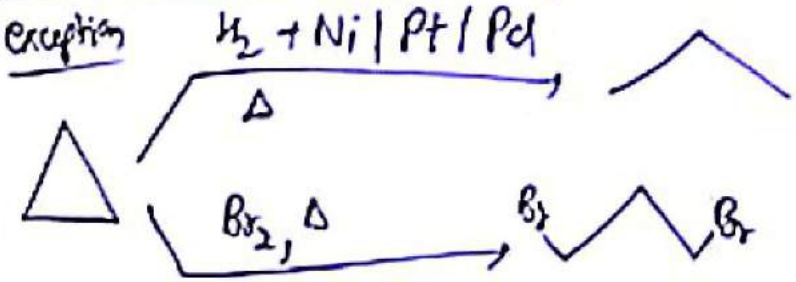


Can show addition Rx^n

⇒ Angle Strain :- Cyclo propane > Cyclobutane > Cyclo pentane > Cyclohexane

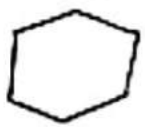
⇒ Stability = " < " < " < "

Exception



Applications of Electronic Effects

(I) Baeyer's angle strain / Ring Strain :-

Heat of combustion per mol = H_6  > H_5  > H_4  > H_3 

" " " per CH_2 = $\frac{H_6}{6} < \frac{H_5}{5} < \frac{H_4}{4} < \frac{H_3}{3}$

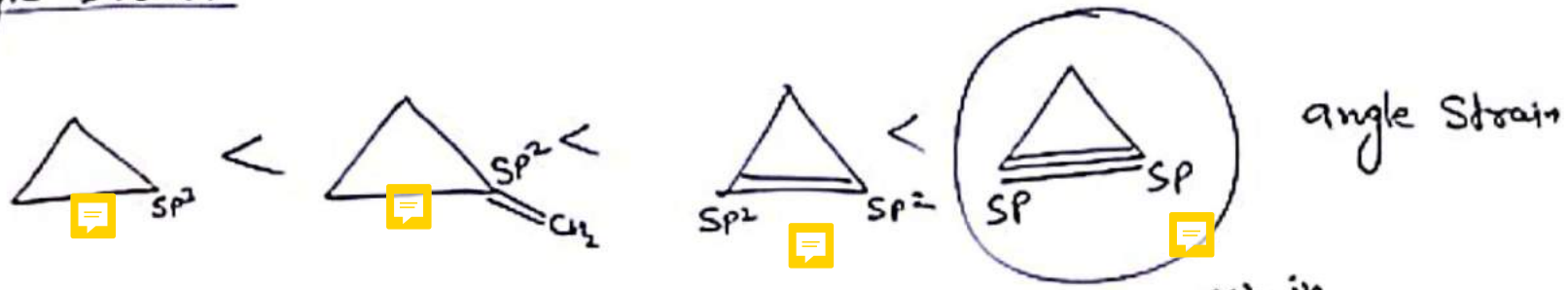
⇒ Heat of comb. $\propto$ no. of C atoms

" " " $\propto \frac{1}{\text{Stability}}$ For iso.

Applications of Electronic Effects

(I) Baeyer's angle strain / Ring Strain :-

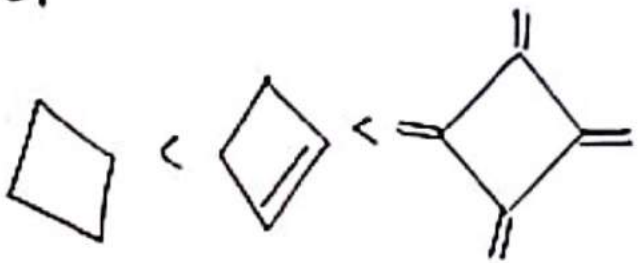
Q. Angle strain

① 

angle Strain

Not exist in nature

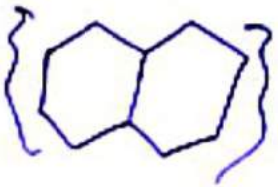
Note Sp atom is not possible in Small rings (up to 7 memberd)

② 

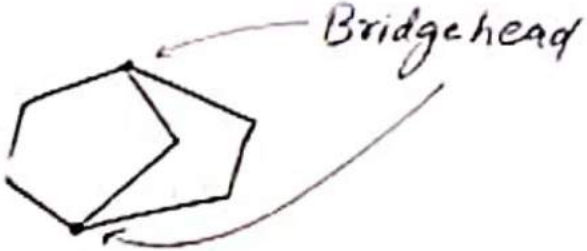
Applications of Electronic Effects

(II) BRADT's Rule :-

Acc. to this rule conditions of planarity are not possible on bridge head atom of small (up to 7 membered main ring) bicyclic compound (It is due to ring strain)

⇒ This rule is not applicable for fused ring containing bicyclic compounds. 

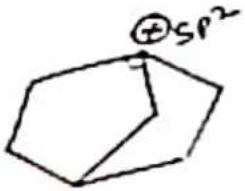
(II) BRADT's Rule :-



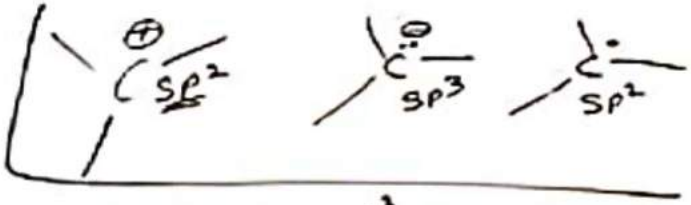
(Norbornane)

Bicyclo (2, 2, 1) Heptane

eg 1



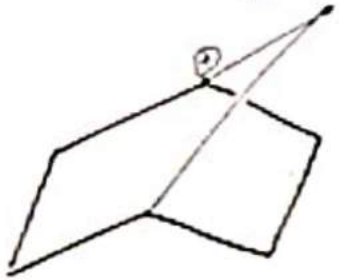
unstable



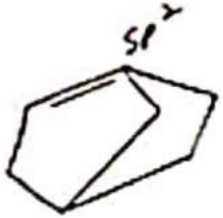
eg 2



exist in nature



eg 3



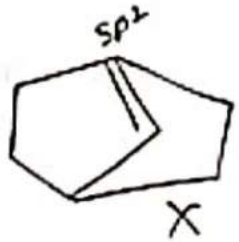
unstable

X

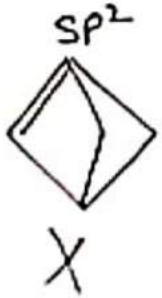
Applications of Electronic Effects

(II) BRADT's Rule :-

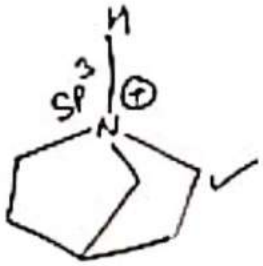
eg 4



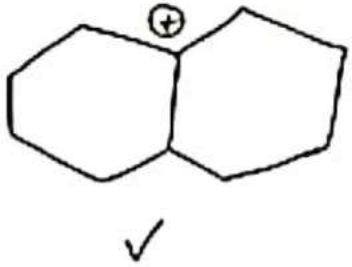
eg 5



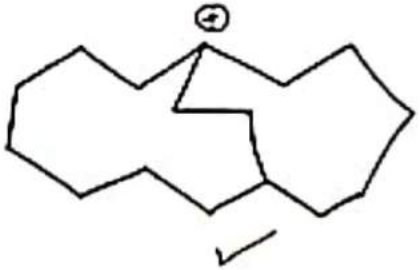
eg 6



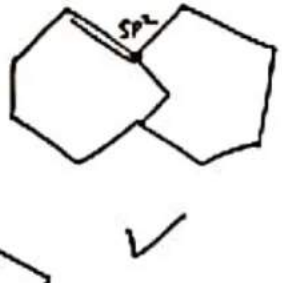
eg 7



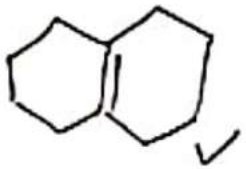
eg 8



eg 9

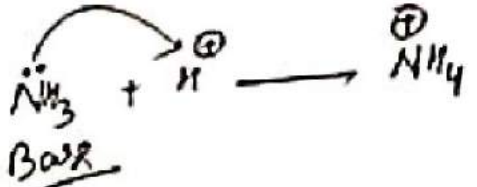


eg 10



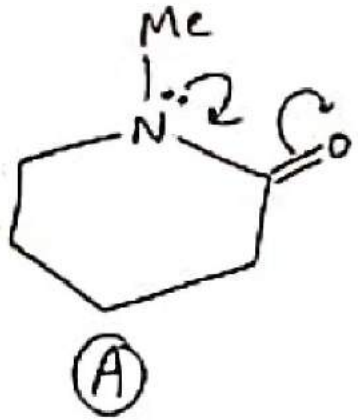
Applications of Electronic Effects

(II) BRADT's Rule :-

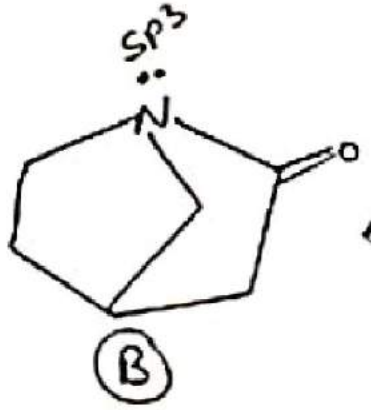


Q

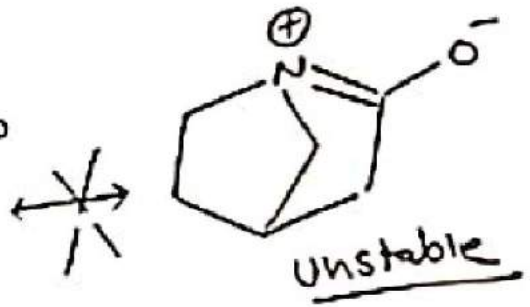
(1)



(A)



(B)



Unstable

⇒ Stability → A > B

⇒ delocalised l.P. → A

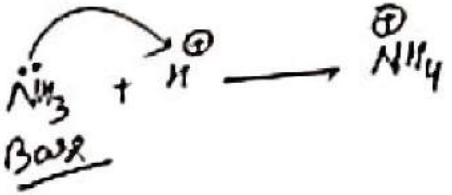
⇒ C=O Bond length → A > B

⇒ C–N Bon length → B > A

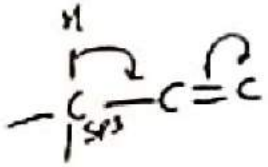
• e^- density on nitrogen → B > A

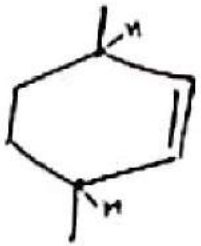
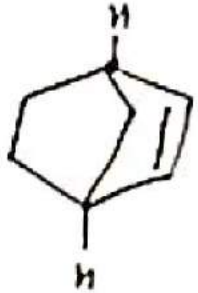
• Basic strength (B > A)

Applications of Electronic Effects



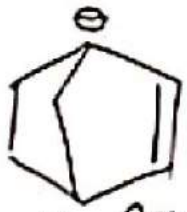
(II) BREDT's Rule :-



②  >  Stability

2 H.C. Hydrogens.

No H.C.

③  >  Stability

No Resonance

Applications of Electronic Effects

(III) SIR (Steric Inhibition in Resonance) effect

{⇒ Steric hindrance :- It is repulsion between bulky groups.}

⇒ SIR effect:- In some compounds planarity of conjugated system is disturbed by Steric hindrane, due to this resonance is also disturbed. This effect is called SIR effect:-

⇒ SIR effect is applicable for bulky groups like, $-NR_2$, $-COOH$, $-NO_2$, $-Br$, I etc

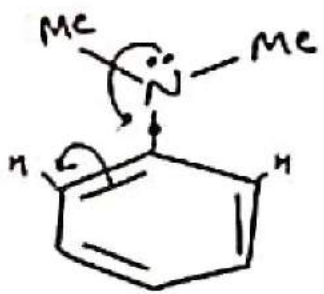
Applications of Electronic Effects

(III) SIR (Steric Inhibition in Resonance) effect

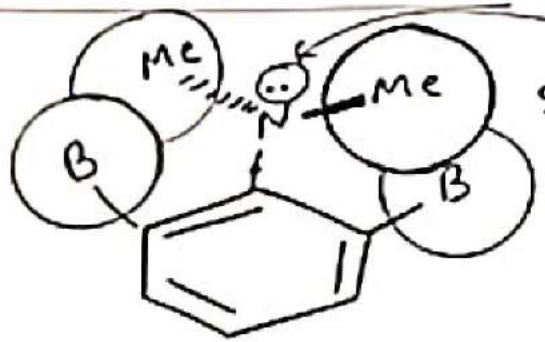
localised $N^{\oplus}$

SIR

eg



(A)



(B)

$E^{\oplus}$ ↗ electrophile

B = Bulky group

$B \neq H \neq D \neq F \neq OH \neq -NH_2$

⇒ Resonance energy ⇒ A > B

⇒ C–N Bond length ⇒ B > A

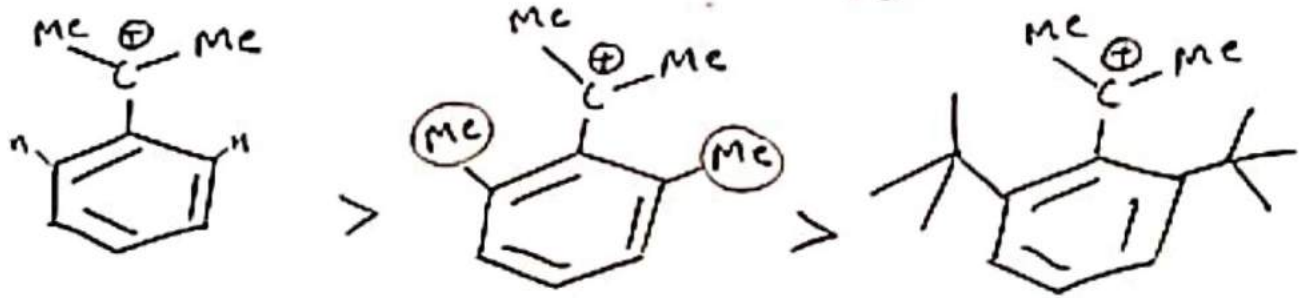
⇒ C^- density on aromatic ring ⇒ A > B

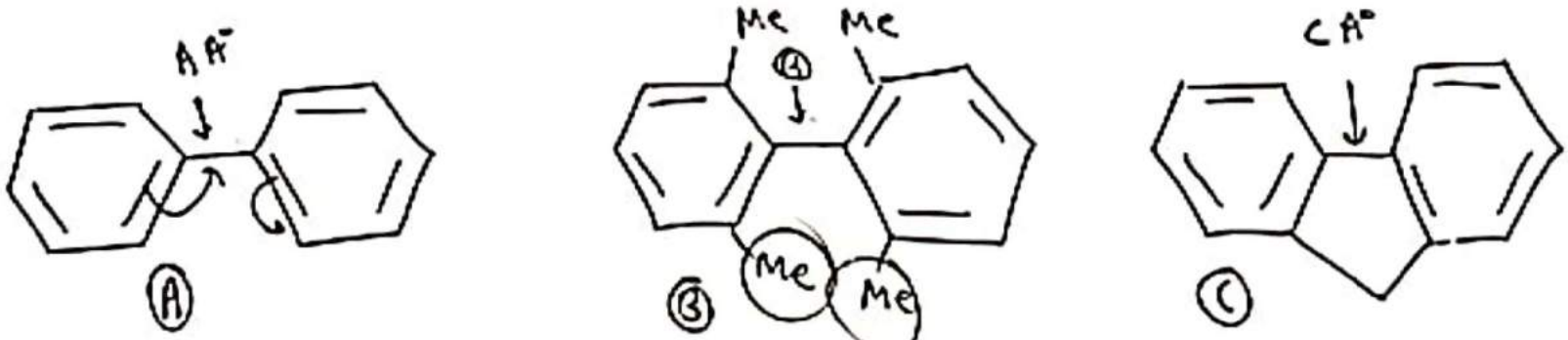
⇒ " " Nitrogen ⇒ B > A

⇒ Basic Strength ⇒ B > A

Applications of Electronic Effects

(III) SIR (Steric Inhibition in Resonance) effect

② 

③ 

⇒ Resonance energy → C > A > B

⇒ Bond length - B > A > C

APPLICATIONS OF ELECTRONIC EFFECTS

(IV) **Dipole moment** (μ) = It is degree of polarity of bond of different atom.

eg. $\overset{+q}{H} \longrightarrow \overset{-q}{Cl}$

$\leftarrow d \rightarrow$

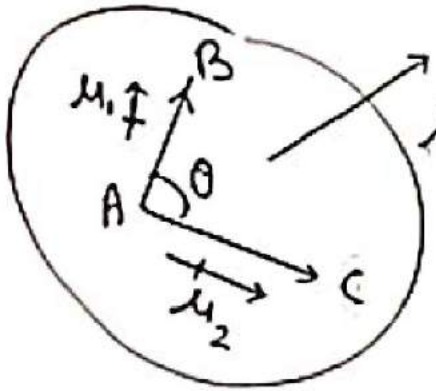
$$\mu = q \cdot d$$

Unit ⇒ Col. M (S.I)

⇒ esu. cm (CGS) = D

Debye

eg



$$\mu_{Net} = \sqrt{\mu_1^2 + \mu_2^2 + 2\mu_1\mu_2\cos\theta}$$

eg $(A + B)_{mix}$

μ_A μ_B

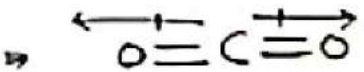
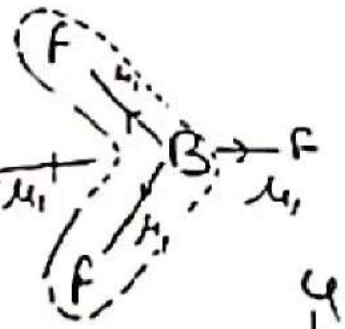
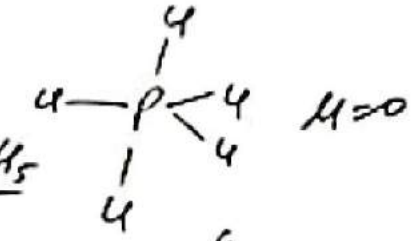
X_A X_B

$$\mu_{mix} = \mu_A X_A + \mu_B X_B$$

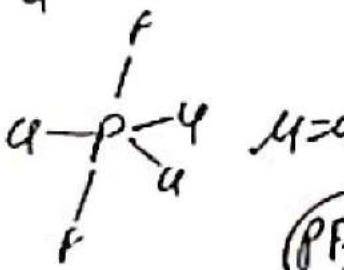
APPLICATIONS OF ELECTRONIC EFFECTS

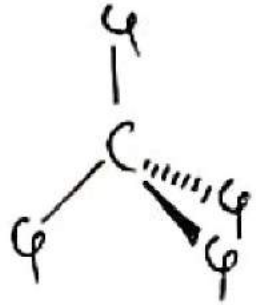
(IV) Dipole moment (μ) :-

Eg 4 Example of Non polar ($\mu = 0$) comp. (symmetrical comp)

-  $\mu = 0$
-  $\mu_{Net} = 0$
- $\Rightarrow sp^3$  $\mu = 0$

PH_5

PF_2H_3  $\mu = 0$

- $\Rightarrow sp^3$  $\mu = 0$

(PF_3H_2) $\mu \neq 0$

APPLICATIONS OF ELECTRONIC EFFECTS

(IV) Dipole moment (μ) :-

eg 5 $\underset{}{CH_3}\rightarrow\underset{-I}{NO_2} > \underset{}{CH_3}\rightarrow CN > \underset{}{CH_3}\rightarrow CHO > \underset{}{CH_3}\rightarrow Cl > \underset{}{CH_3}\rightarrow OH > \underset{}{CH_3}-\underset{}{CH_3}$ ($\mu = 0$)

eg 6 $\overset{SP^3}{CH_3}-\overset{SP^3}{CH_2}-\overset{SP^3}{CH_3} < \overset{SP^3}{CH_3}\rightarrow\overset{SP^2}{CH}=CH_2 < \overset{SP^3}{CH_3}\rightarrow\overset{SP}{C}\equiv CH$

($\mu = 0$)

E.N. = SP > SP^2 > SP^3

eg 7 $\underset{SP^3}{CH_3-CH_2}\rightarrow Cl > \underset{SP^2}{CH_2=CH}\rightarrow Cl > H-\underset{SP}{C}\equiv C-Cl$

$\sigma \rightarrow$, $\pi \leftarrow$

(vinyl chloride)

APPLICATIONS OF ELECTRONIC EFFECTS

(IV) Dipole moment (μ) =

$$\mu = q \cdot d$$

eg 8 $H-\overset{\delta-}{F} > H-Cl > H-Br > H-I$ (μ)

Imp: eg 9 CH_3-F CH_3-Cl CH_3-Br CH_3-I

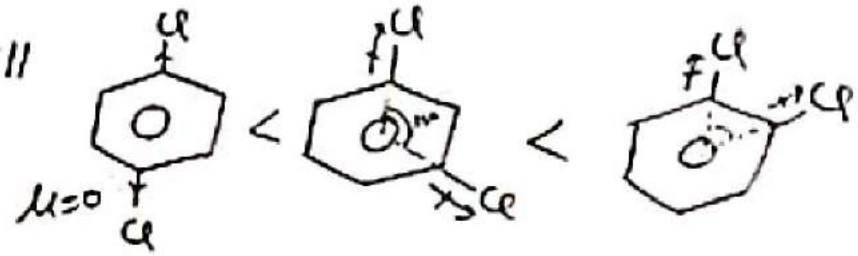
$CH_3-Cl > CH_3-F > CH_3-Br > CH_3-I$

eg 10 $C_2H_5-F > C_2H_5-Cl > C_2H_5-Br > C_2H_5-I$

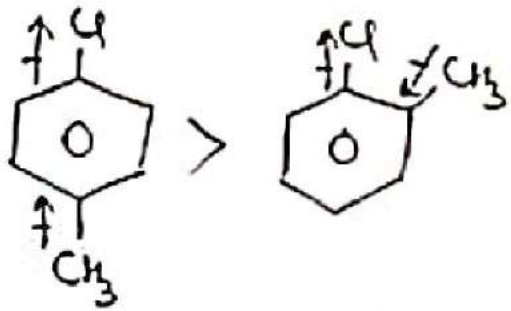
APPLICATIONS OF ELECTRONIC EFFECTS

(IV) Dipole moment (μ) =

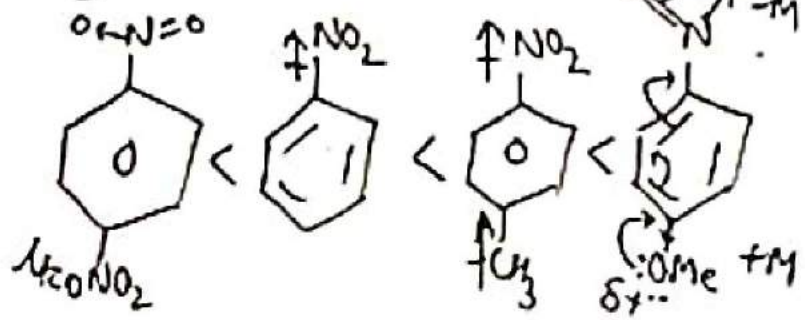
eg 11



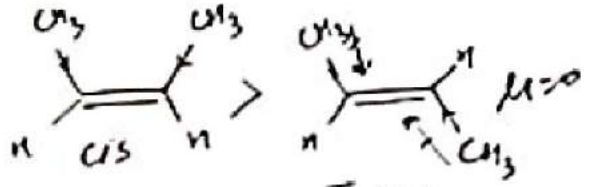
eg 12



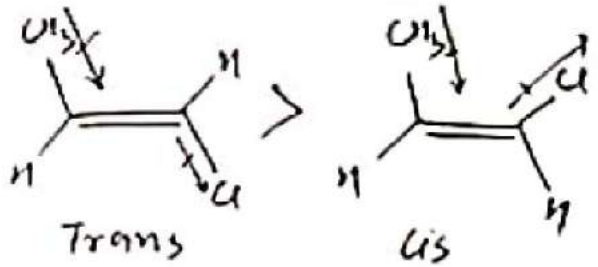
eg 13



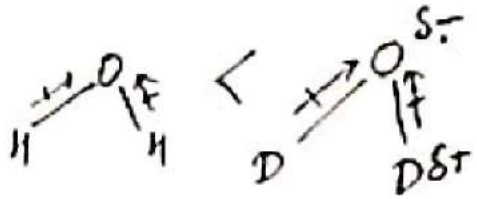
eg 14



eg 15

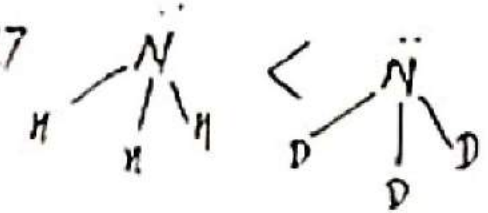


eg 16



D > H) +I

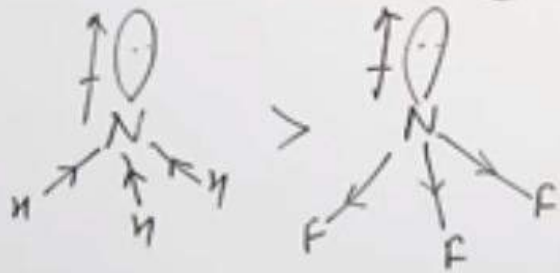
eg 17



APPLICATIONS OF ELECTRONIC EFFECTS

(IV) Dipole moment (μ) :

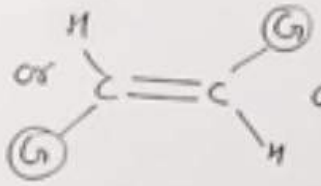
eg 18



Note



Para



Trans.

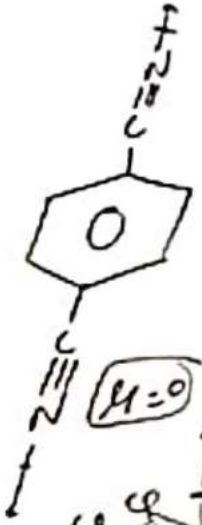


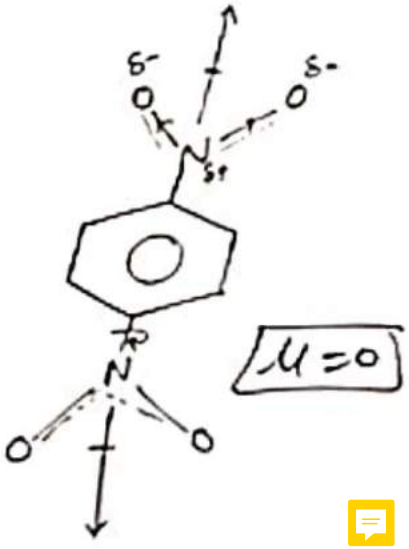
Anti

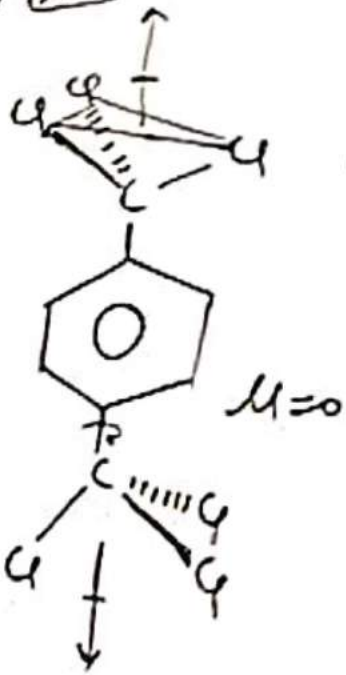
(I) if G = $-X$, $-CH_3$, $-CX_3$, $-NO_2$, $-COO^{\ominus}$, $-CN$ etc } $\mu = 0$

(II)

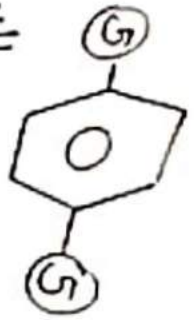
APPLICATIONS OF ELECTRONIC EFFECTS

eg  $\mu = 0$

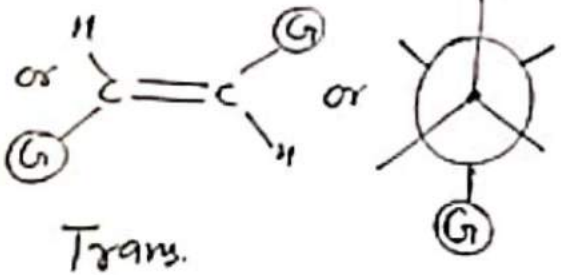
eg  $\mu = 0$

eg  $\mu = 0$

Note



Para



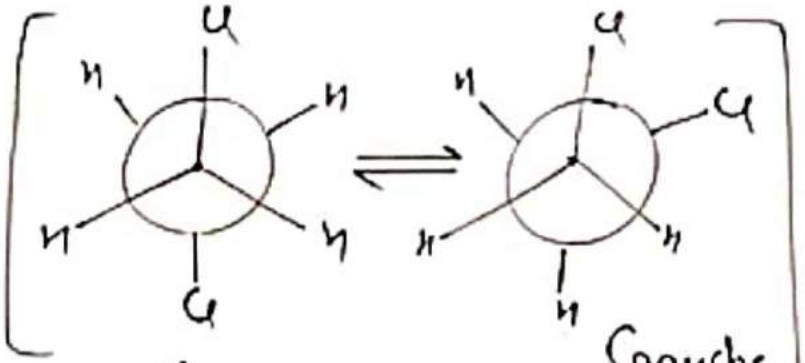
Trans. Anti

(I) if G ⇒ $-X$, $-CH_3$, $-CX_3$, $-NO_2$, $-COO^{\ominus}$, $-CN$ etc } $\mu = 0$

(II) if G ⇒ $-OH$, $-OR$, $-SH$, SR, $-COOH$, $-COOR$, $-COCl$, $-CH_2Cl$, $-CHCl_2$ etc } $\mu \neq 0$

APPLICATIONS OF ELECTRONIC EFFECTS

eg $Cl-CH_2-CH_2-Cl$



Anti ⇌ Gauche]$_{mix}$

Anti

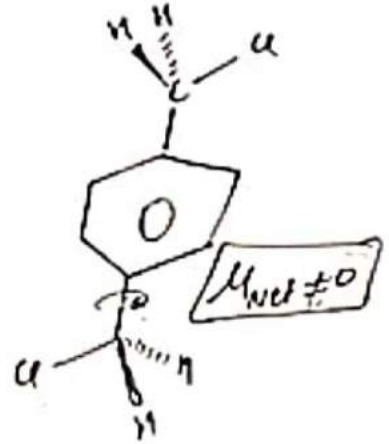
$\mu_{Anti} = 0$

Gauche

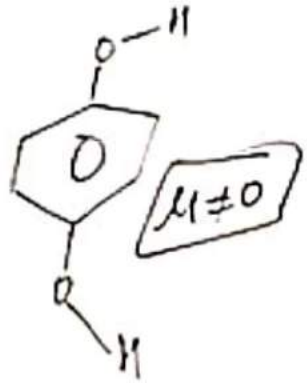
$\mu_G \neq 0$

$\mu_{mix} \neq 0$

eg

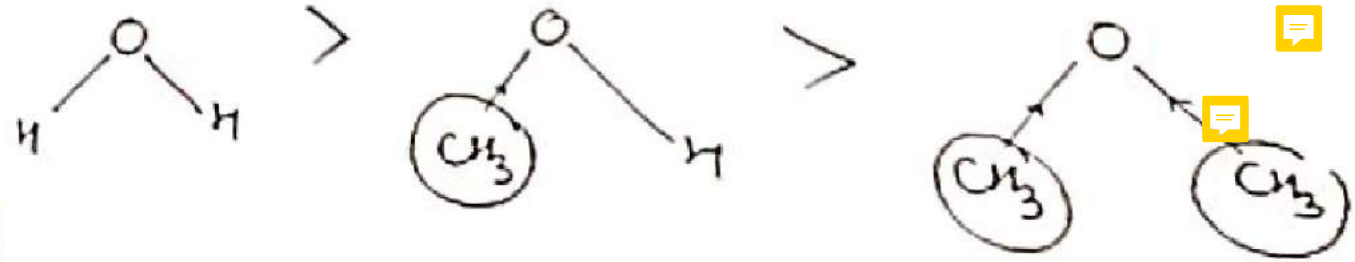


eg

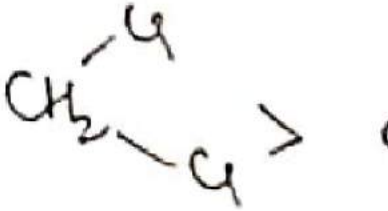
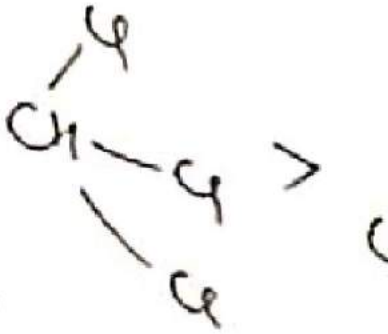
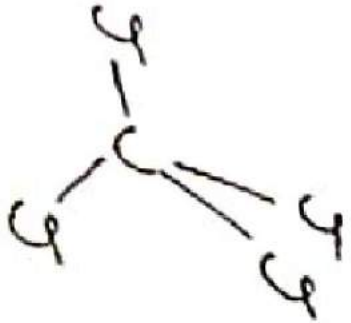


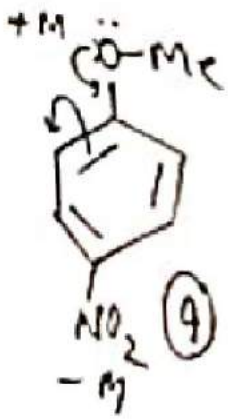
GENERAL ORGANIC CHEMISTRY – I

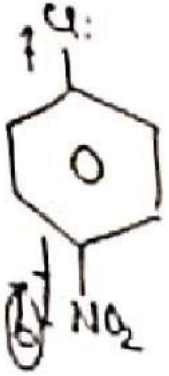
(IV) Dipole moment :-

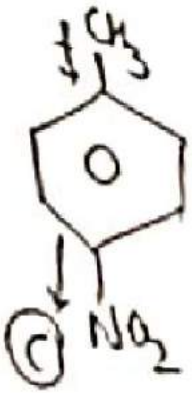
eg 

$-OMe$
$+M, -I$

eg CH_3-Cl >  >  > 

eg 





a > c > b

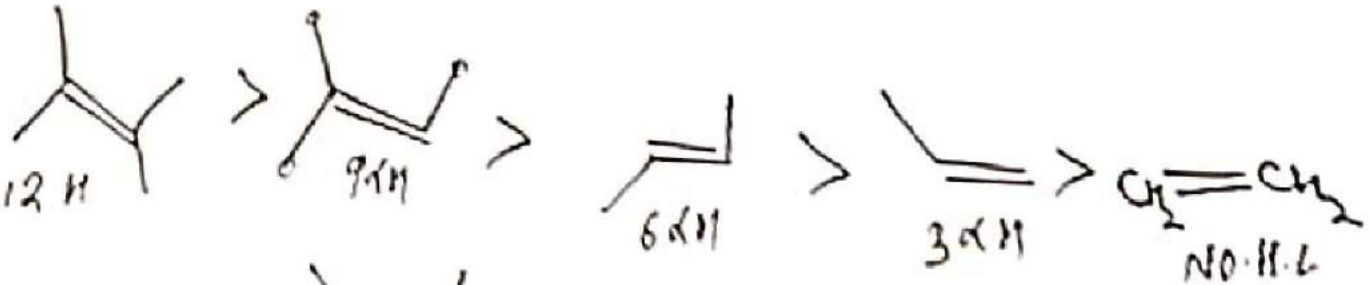
GENERAL ORGANIC CHEMISTRY – I

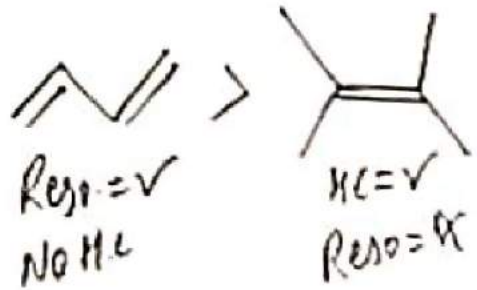
(V) Stability of alkenes:-

⇒ Stability of alkene ∝ delocalisation of π e⁻

⇒ " " " : ∝ Hyper conjugation

eg 1  (Stability)

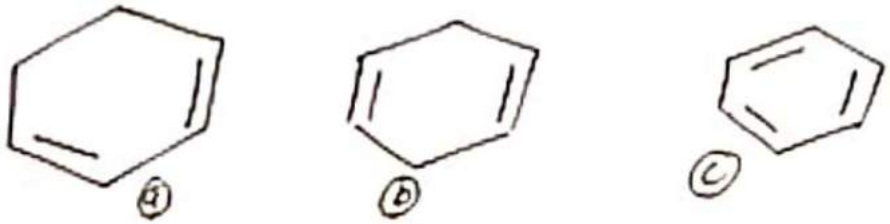
eg 2 

eg 3 

GENERAL ORGANIC CHEMISTRY – I

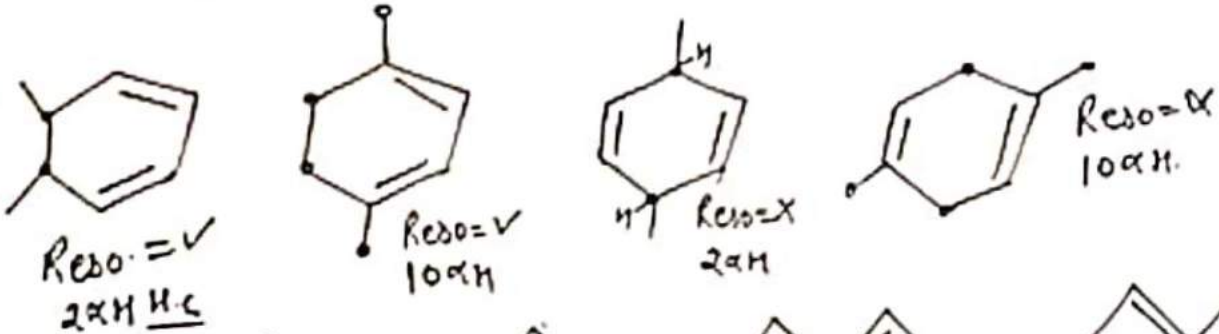
(V) Stability of alkenes:-

eg 4



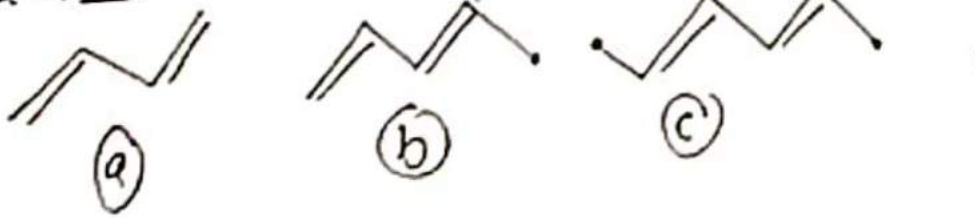
Stability → c > a > b

eg 5



Stability = b > a > d > c

eg 6



Stability = c > b > a > d

GENERAL ORGANIC CHEMISTRY – I

(V) Stability of alkenes :-

⇒ **Note** ⇒ When no. of α-Hydrogens are same then brached alkenes are more stable than unbranched alkenes.

eg7  (Stability)

6 α H.C 6 α H.C

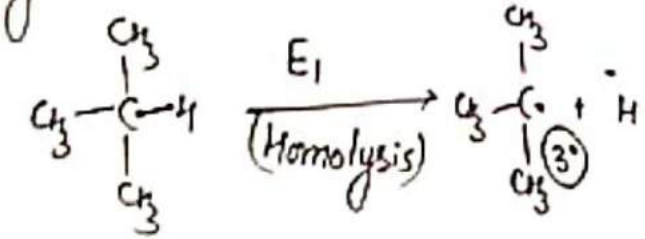
GENERAL ORGANIC CHEMISTRY – I

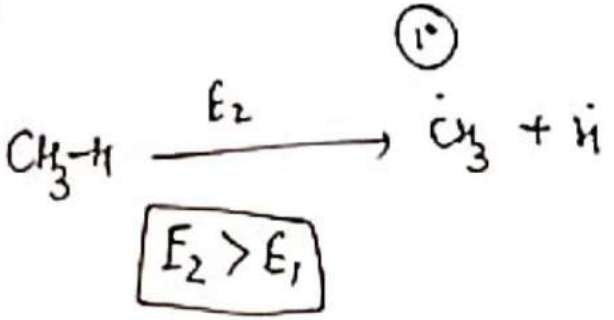
(V) Stability of alkenes:-

⇒ Brached alkane isomers are more stable than unbrached alkane.

Stability of alkane α branching

Brached isomers have greater no. of 1° C–H bonds which are stronger than 2° C–H and 3° C–H bonds.

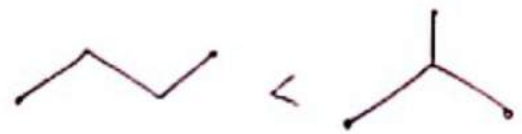




GENERAL ORGANIC CHEMISTRY – I

(I) Stability of alkenes:-

Bond Strength → 1° C–H > 2° C–H > 3° C–H

eg  Stability

six 1° C–H nine 1° C–H

eg  Stability

Six 1° C–H 9 1° C–H 12 1° C–H

Heat of formation $\propto$ no. of atom in one molecule

" " " " $\propto$ Stability (for isomers)

Heat of combustion $\propto$ no. of carbon atoms

" " " " $\propto \frac{1}{\text{Stability}}$ (for isomers)

Heat of Hydrogenation $\propto$ no of π bonds

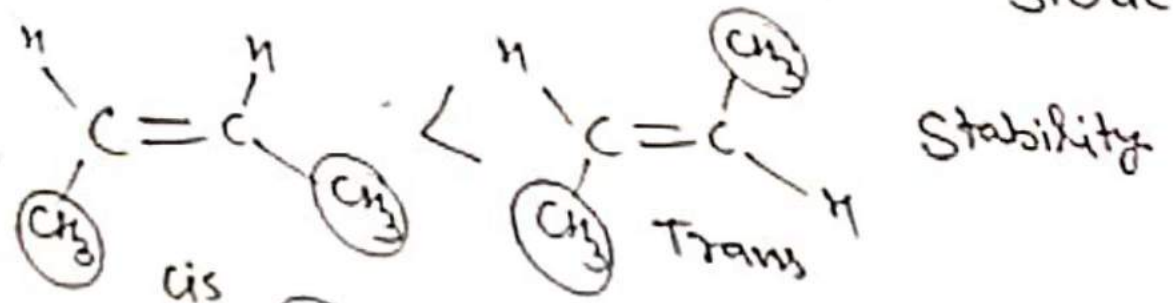
" " " " $\propto \frac{1}{\text{Stability}}$ (for isomers)

eg $CH_4 < C_2H_6 < C_3H_8$ Heat of formation and heat of combustion.

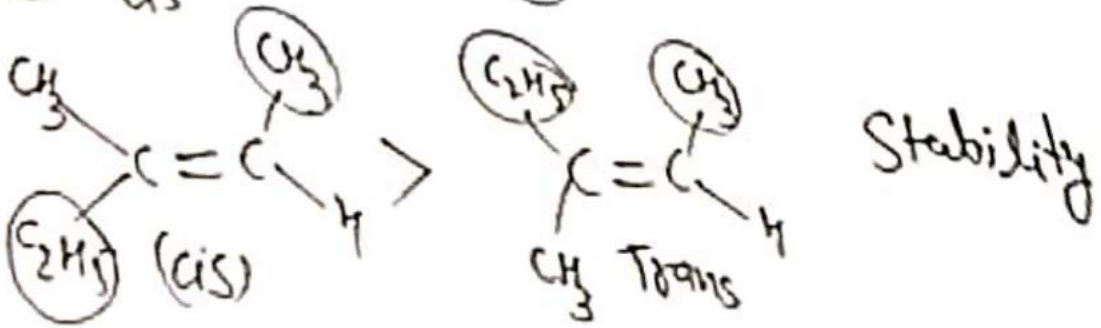
(IV) Stability of alkene:-

⇒ For geometrical isomers: stability $\propto \frac{1}{\text{Steric hindrance}}$

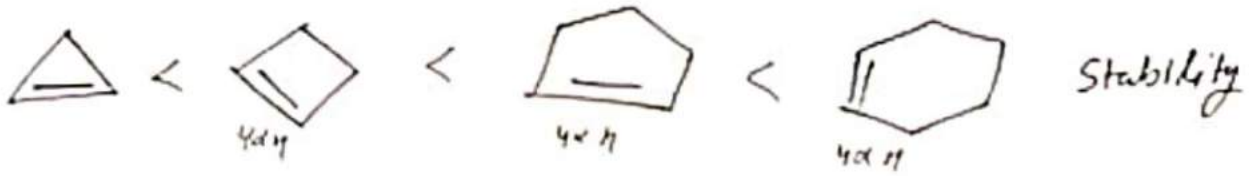
eg8



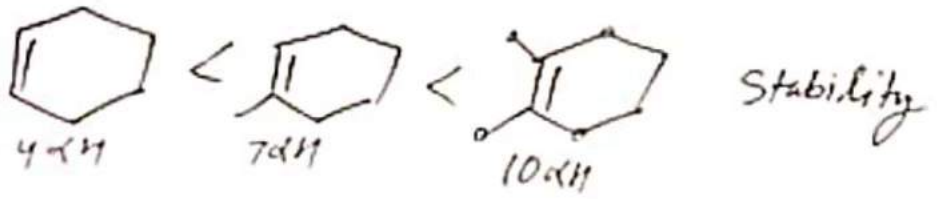
eg 9



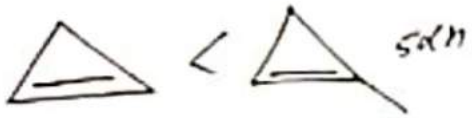
GENERAL ORGANIC CHEMISTRY – I

eg 10 

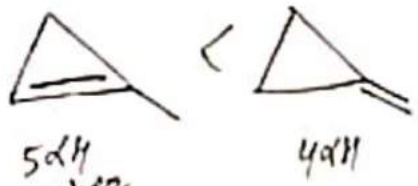
Stability

eg 11 

Stability

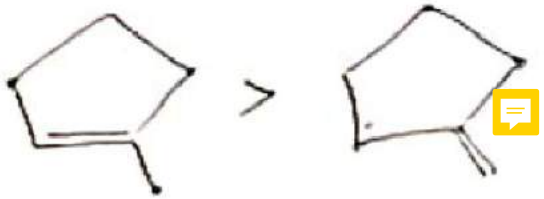
eg 12 

∴ :

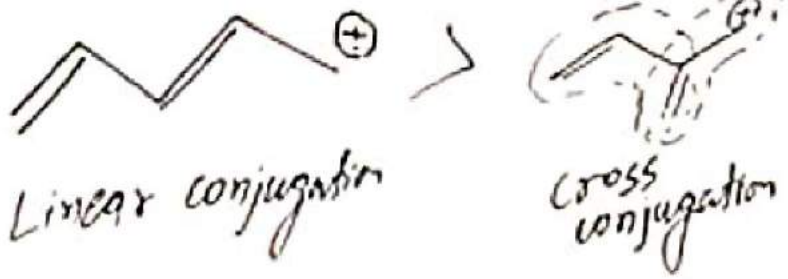
A eg 13 

∴

GENERAL ORGANIC CHEMISTRY – I

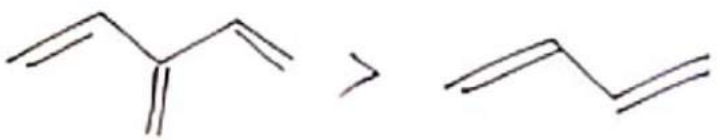
eg 

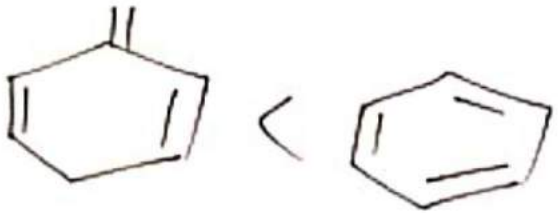
⇒ When no. of conjugated π e⁻ are same then Linear conjugated isomers are more stable than crossed conjugated structures.

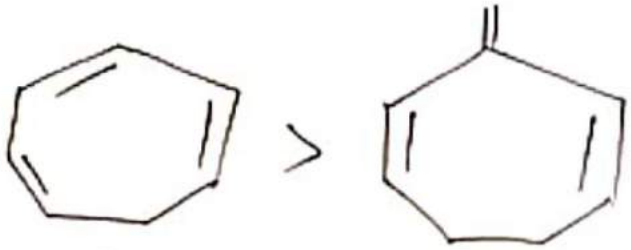
eg 

eg 

General Organic Chemistry – I

eg 

eg 

eg 

Linear Cross Conj.

GENERAL ORGANIC CHEMISTRY – I

(VI) Bond length

$$\text{Bond length} \propto \frac{1}{\text{multiplicity of bond.}}$$

eg 1 $CH_2=CH_2$ < < < < (C=C bond length)

eg 2

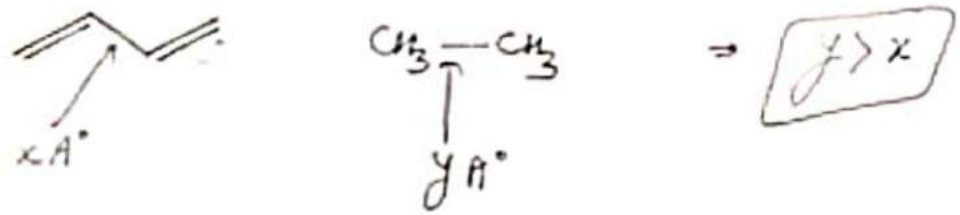
(Bond length C–C) $\propto z < y < x$

" " C–H) $\propto \alpha > \beta > \gamma$

GENERAL ORGANIC CHEMISTRY – I

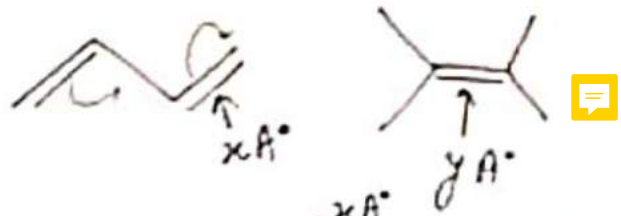
(VI) Bond length

Eg 3



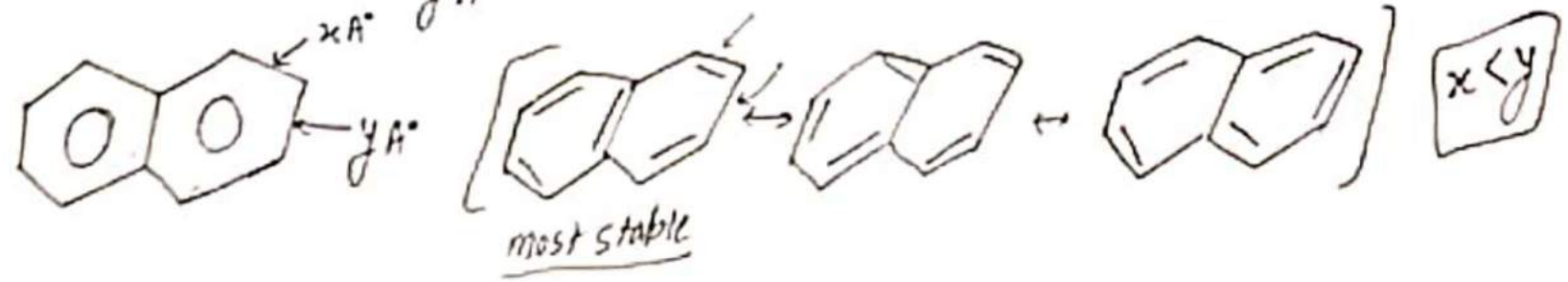
⇒ y > x

Eg 4



(C=C) B.L ⇒ x > y

Eg 5



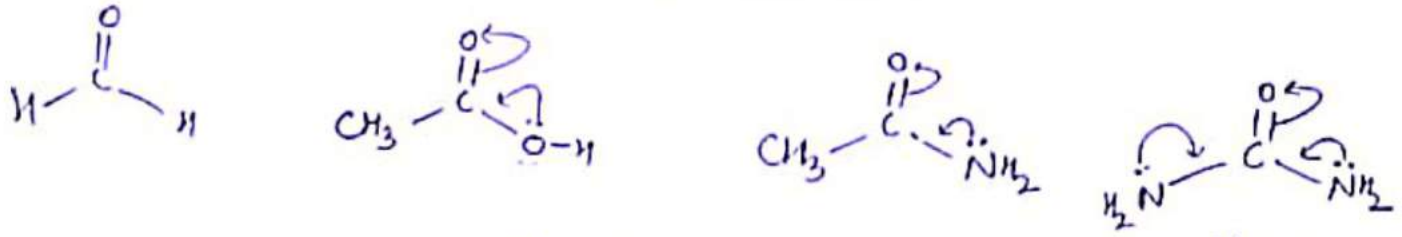
x < y

GENERAL ORGANIC CHEMISTRY – I

(VI) Bond length

$-OH < -NH_2$

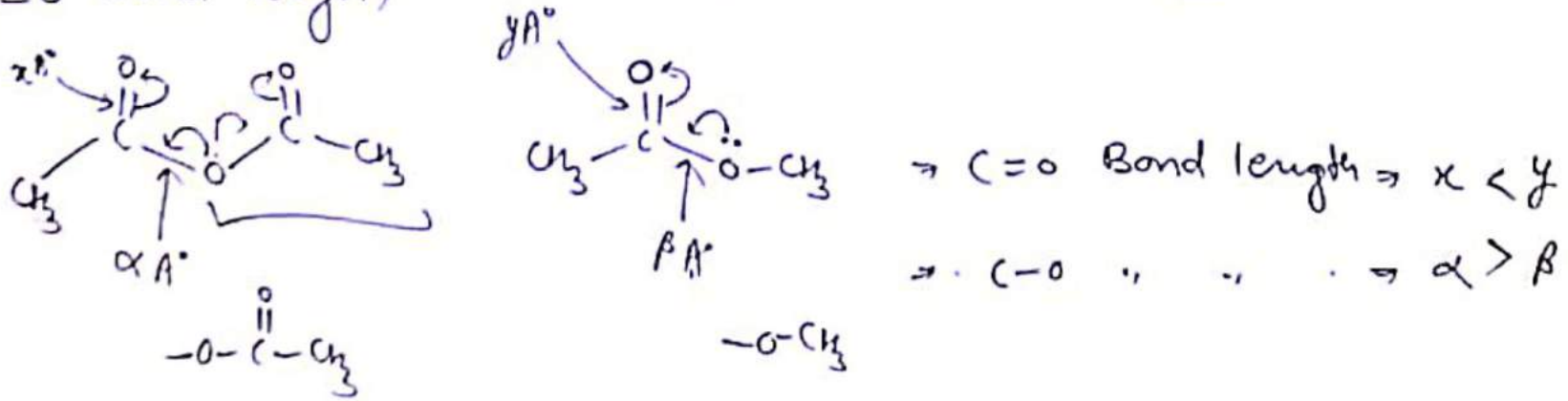
eg 6



Ureq

(C=O Bond length) → d > c > b > a

eg 7



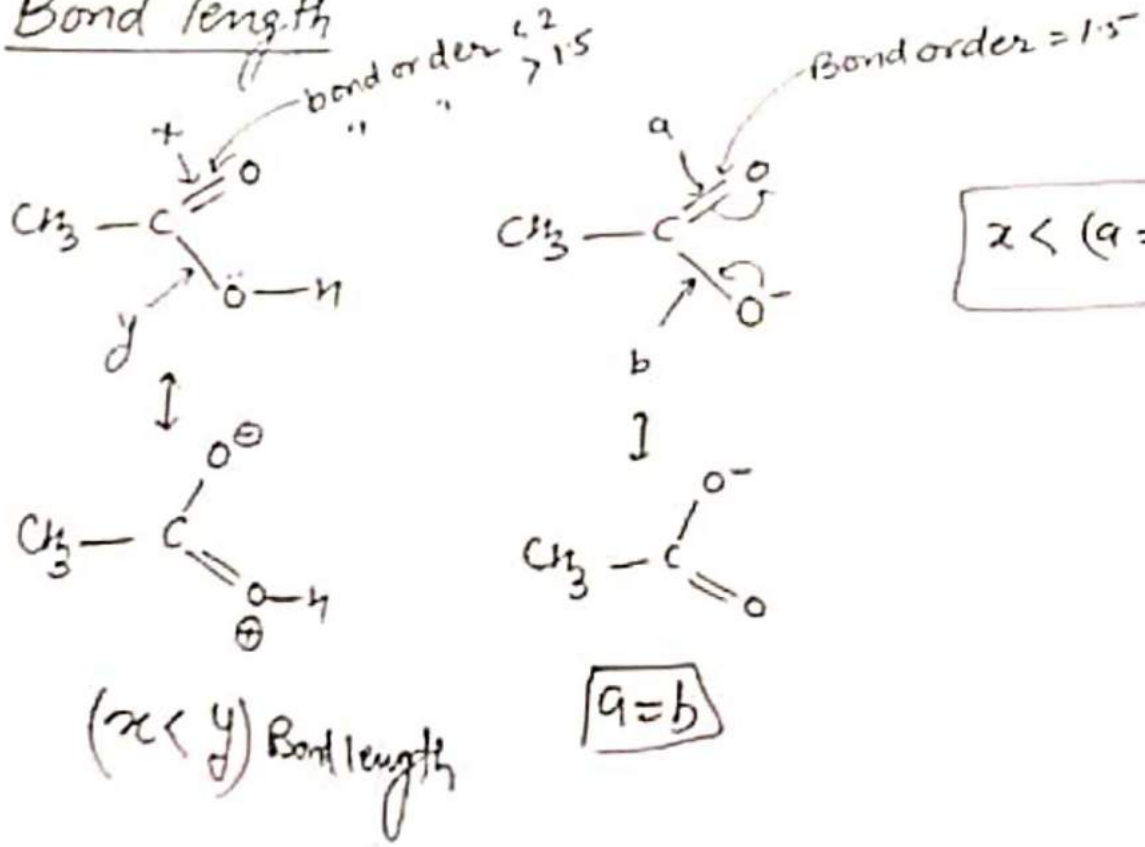
→ C=O Bond length → x < y

→ C–O " " " → α > β

GENERAL ORGANIC CHEMISTRY – I

(VI) Bond length

eg



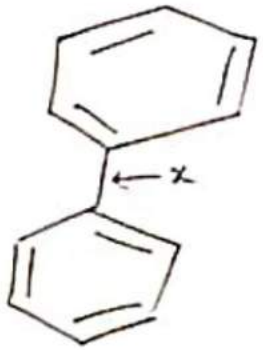
$x < (a = b) < y$ Bond length

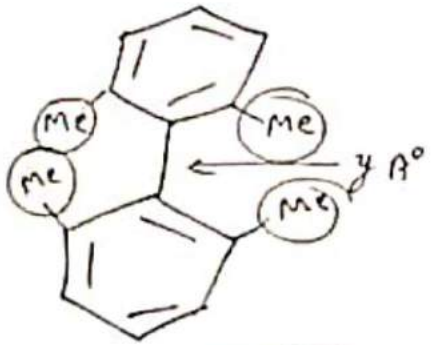
(x < y) Bond length

a = b

GENERAL ORGANIC CHEMISTRY – I

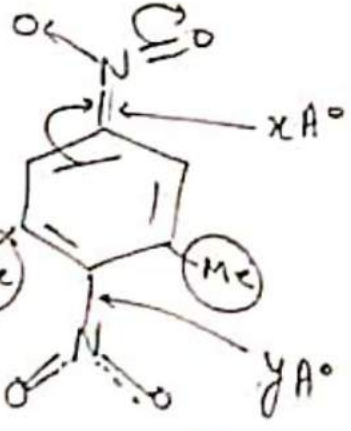
eg





x < y

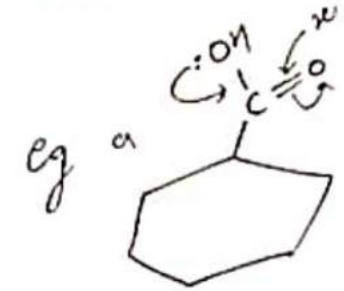
eg



SIR

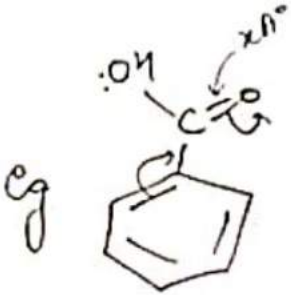
x < y

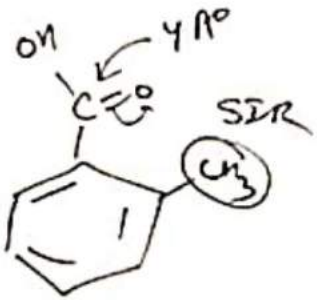
eg





eg





o-Tolulic acid

x > y

GENERAL ORGANIC CHEMISTRY – I

VII) AROMATICITY

⇒ Conditions of aromaticity:-

(1) Compound should be cyclic

(2) Ring should be planar, unsaturated and completly conjugated.

(3) Each atom of ring should sp^2 hybridized (sp)

(4) Ring must have $(4n+2)\pi e^-$ (Huckle's Rule)

$n = 0, 1, 2, 3 \cdots$

$(4n+2) \Rightarrow 2\pi e, 6\pi e, 10, 14, 18, 22 \cdots$



Subscribe

GENERAL ORGANIC CHEMISTRY – I

(VII) AROMATICITY

Conditions for antiaromatic comp

(1)

(2)

(3)

(4) $4n\pi c$ in ring for antiaromaticity

$n = 1, 2, 3, 4 - - - - - -$

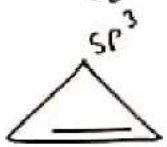
$4\pi c, 8\pi c, 12 - - -$

Stability order

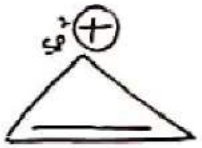
Aromatic > Non aromatic > Anti aromatic

GENERAL ORGANIC CHEMISTRY – I

Identify aromatic, non aromatic and Antiaromatic.

(1) 

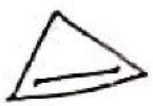
N.A.

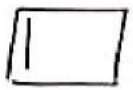
(2) 

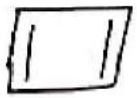
2πe⁻ Aro.

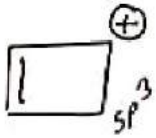
(3) 

4πe⁻ A.A.

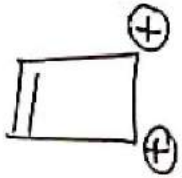
(4) 
3πe⁻ N.A.

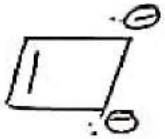
(5) 
2πe⁻ N.A.

(6) 
4πe⁻ A.A.

(7) 

N.A

(8) 
2πe⁻ Aro.

(9) 
6πe⁻

(10) 

N.A

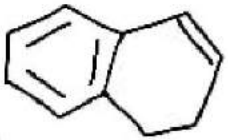
(11) 

6πe⁻ Aro.

(12) 
4πe⁻ A.A.

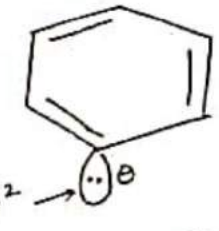
(13) 
Aro.

(14) 
6πe Aro.

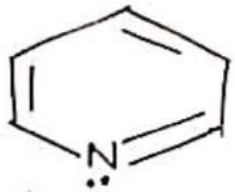
(15) 
Aro.

GENERAL ORGANIC CHEMISTRY – I

Q Identify aromatic, non aromatic and Antiaromatic.

16. 

Aromatic
6π e⁻ in ring

17. 

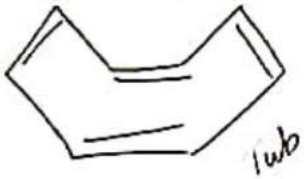
6π e⁻
aro.
Pyridine

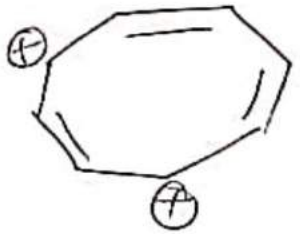
18 

N.A.

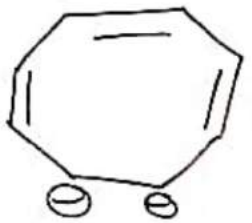
19 

6π e⁻
Aro.
Tropylium

20 
8π e⁻
Imp.
• Non planar
• Non Aro.

Tub shape

21 

6π e⁻
Aro.

(22) 

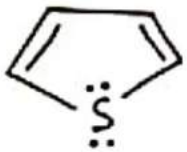
10π e⁻
Aro.

23 

6π e⁻
Aro.
Pyrrol

GENERAL ORGANIC CHEMISTRY - I

Q. Identify aromatic, Nonaromatic and antiaromatic compounds.

(24)  $6\pi e^-$ aromatic

Thiophene

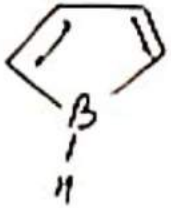
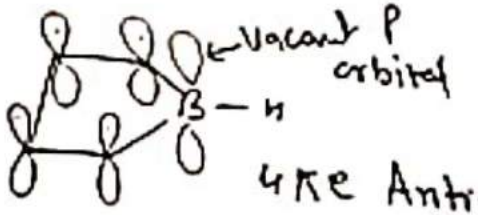
(25)  $6\pi e^-$ aromatic

Furan

(26)  $6\pi e^-$ aromatic

(27)  $8\pi e$ Anti aro.

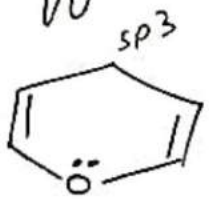
(28)  $4\pi e^-$ Antiaromatic

(29)   $4\pi e$ Antiaro.

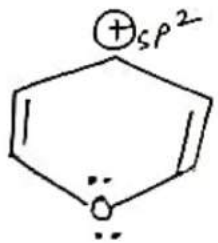
(30)  $6\pi e^-$ Aromatic

GENERAL ORGANIC CHEMISTRY - I

Q. Identify aromatic, Nonaromatic and antiaromatic compounds.

(31) 

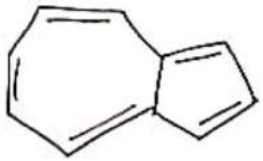
6 π e⁻
N.A.

(32) 

6 π e⁻
Aro.

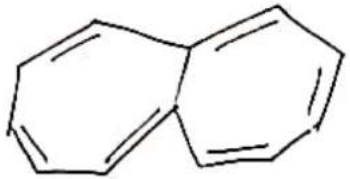
(33) 

8 π e⁻
antiaromatic

(34) 

10 π e⁻
aro.

Azulene

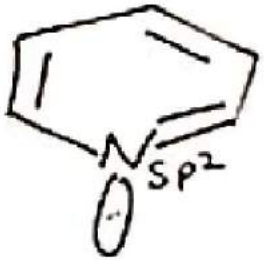
(35) 

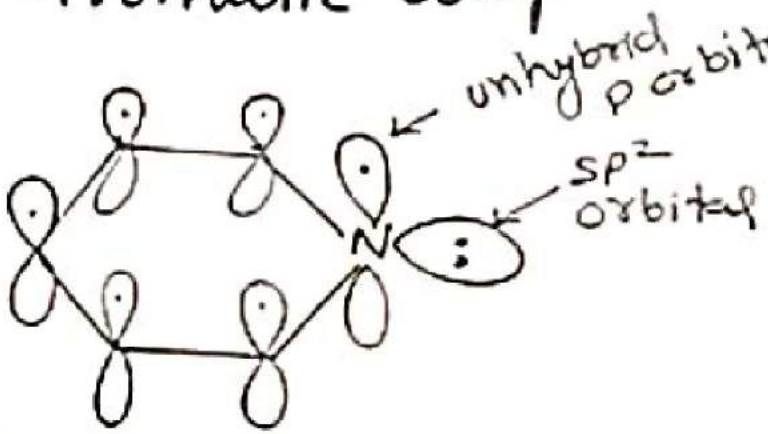
12 π e
Antiaromatic

GENERAL ORGANIC CHEMISTRY - I

Structures of important aromatic comp.

Pyridine

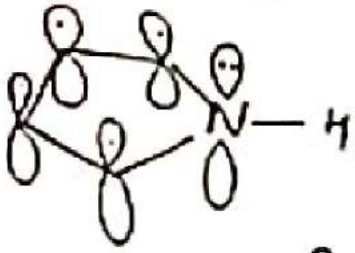




Pyrrol



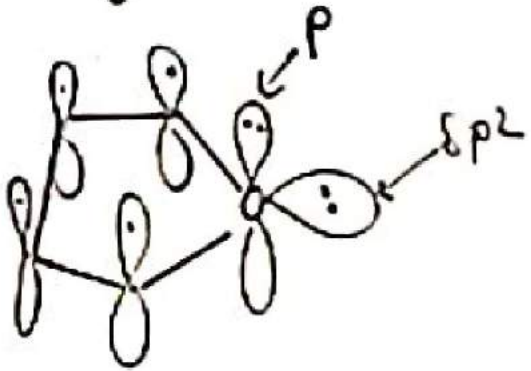
=



Furan



=



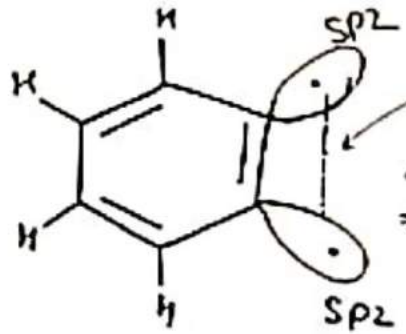
GENERAL ORGANIC CHEMISTRY - I

3D Structures of important aromatic comp.

(iv) **Benzyne**



(Reaction intermediate)



$(sp^2 - sp^2)$

- weaker than π bond
- ⇒ in same plane of ring
- ⇒ localised

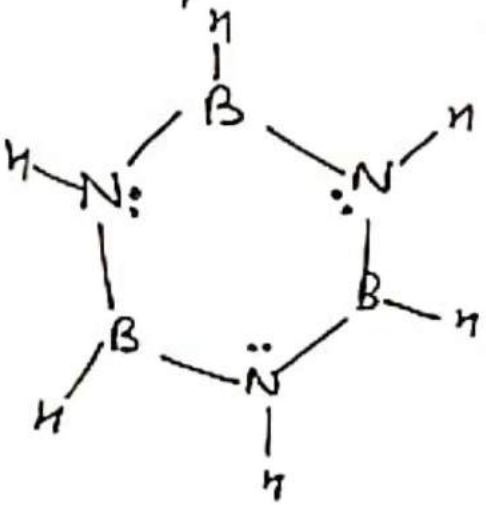
6 πe

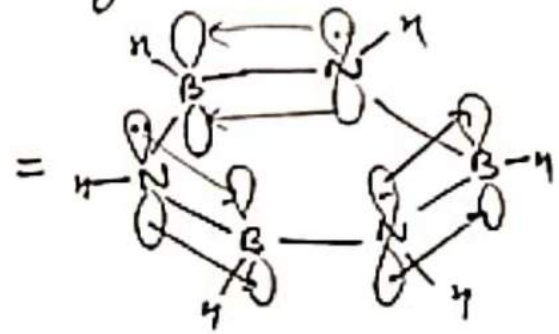
Aro.

GENERAL ORGANIC CHEMISTRY - I

3D Structures of important aromatic comp.

(i) Borazol / Borazine / inorganic benzene. $N_3B_3H_6$





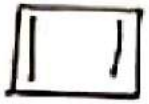
= 6πe aromatic

⇒ More reactive than benzene

⇒ lesses stable than benzene (Polar bond)

GENERAL ORGANIC CHEMISTRY - I

Annulene:- complete conjugated monocyclic polyene

Eg 1  [4] Annulene → Antiaromatic

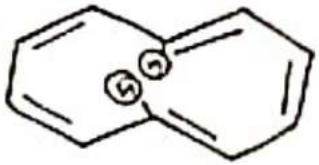
Eg 2  [6] Annulene → Aromatic

Eg 3  [8] Annulene → Non planar
→ Non aromatic

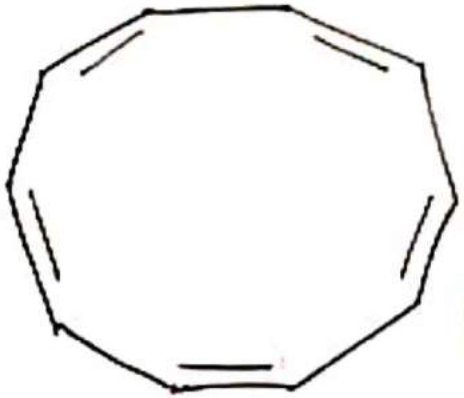


GENERAL ORGANIC CHEMISTRY - I

Annulene:- complete conjugated monocyclic polyene

eg 

[10] Annulene
cis, cis, Trans, cis, Trans → Non planar, Non aromatic

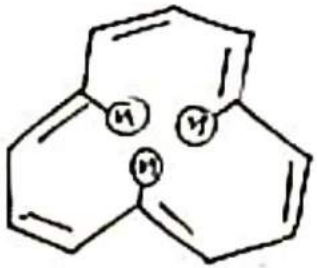


all cis
[10] annulene
(high ring strain)

GENERAL ORGANIC CHEMISTRY - I

Annulene:- Complete conjugated monocyclic polyene

(5)

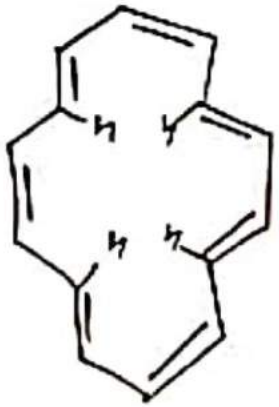


→ [12] Annulene

→ Non planar

→ Non aromatic

(6)

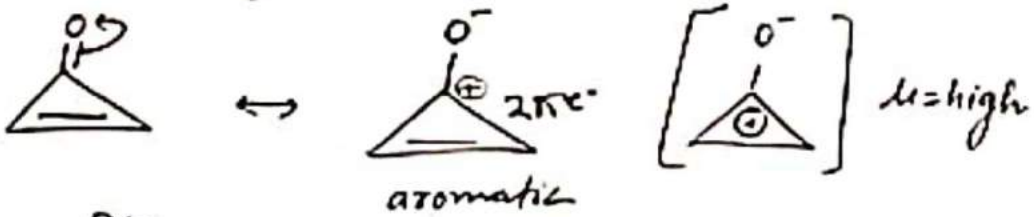


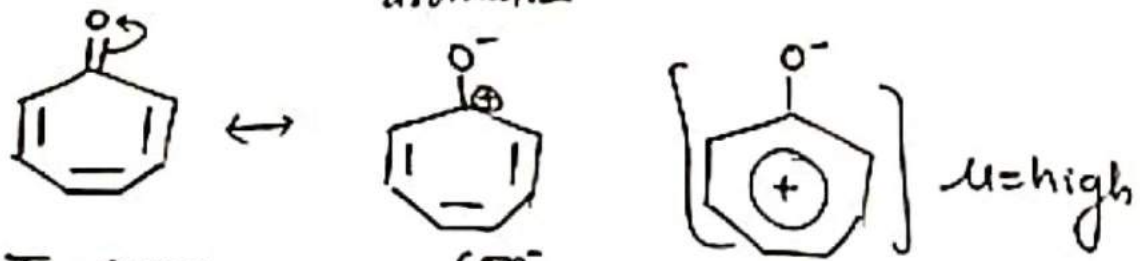
[14] Annulene

→ Partial aromatic

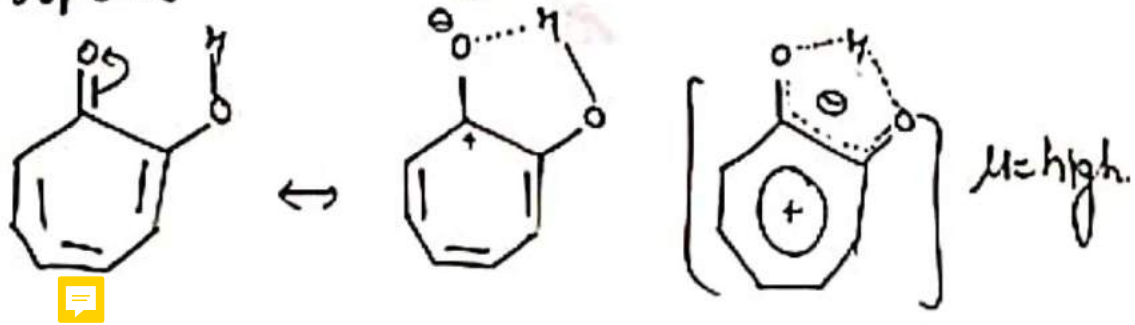
GENERAL ORGANIC CHEMISTRY - I

Examples of Polar aromatic comp

① 

② 

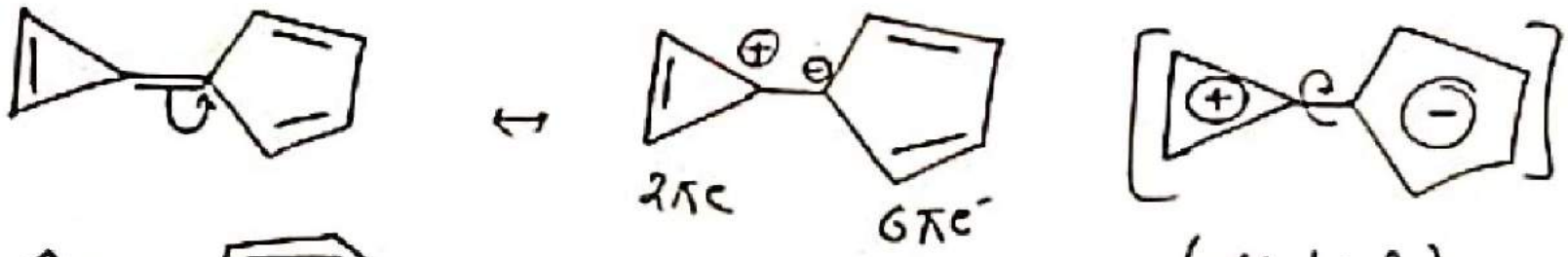
Tropone

(3) 

GENERAL ORGANIC CHEMISTRY - I

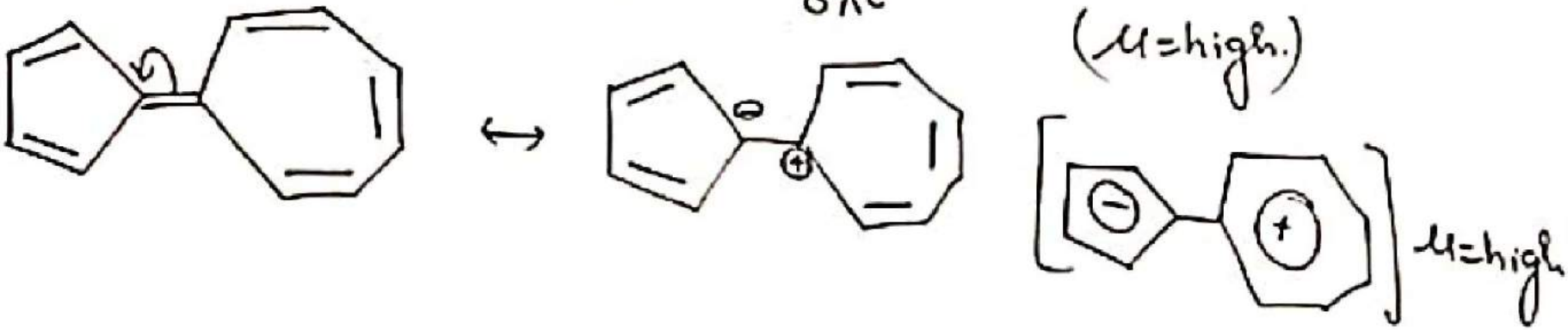
Examples of Polar aromatic comp

Eg 4



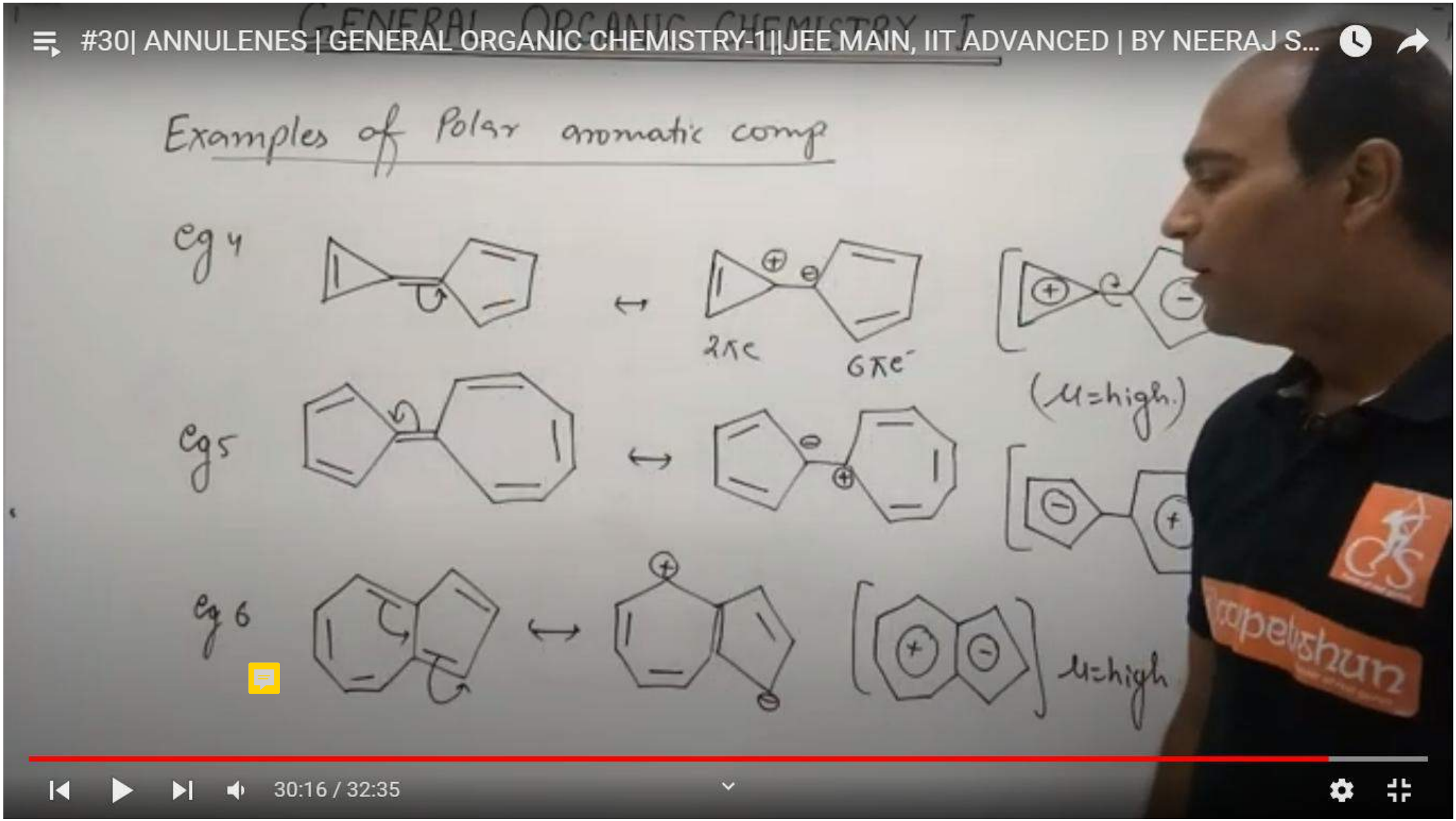
(μ = high.)

Eg 5



μ = high

#30| ANNULENES | GENERAL ORGANIC CHEMISTRY-1||JEE MAIN, IIT ADVANCED | BY NEERAJ S...

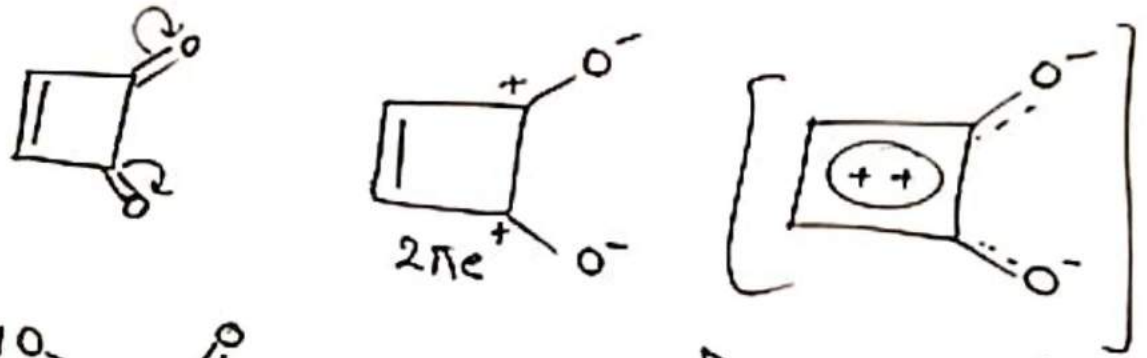


30:16 / 32:35

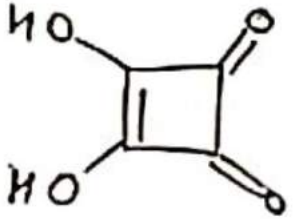
GENERAL ORGANIC CHEMISTRY - I

Examples of Polar aromatic comp

eg.

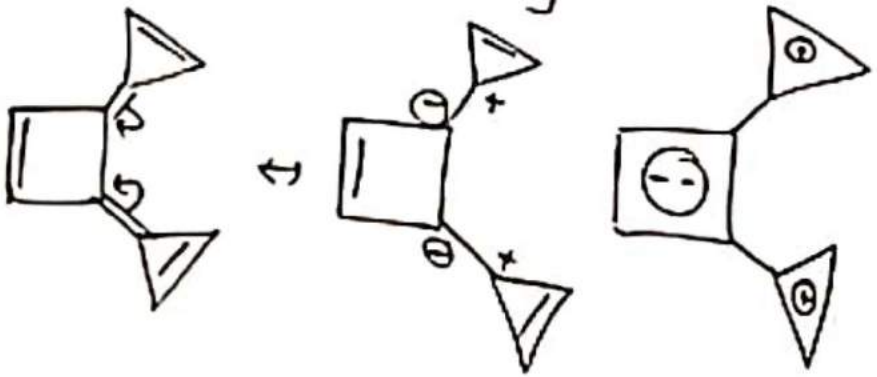


eg



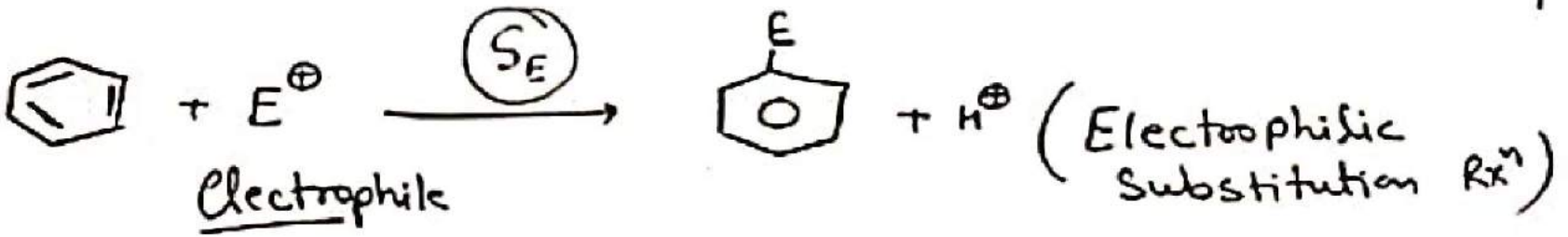
Squaric acid

eg



GENERAL ORGANIC CHEMISTRY - I

VIII) Electron density on aromatic rings / Reactivity of aromatic compounds.



Electrophile

(Electrophilic Substitution Rx^n)

- Reactivity of aromatic comp. $\propto$ e^- density on aro. ring of reactant

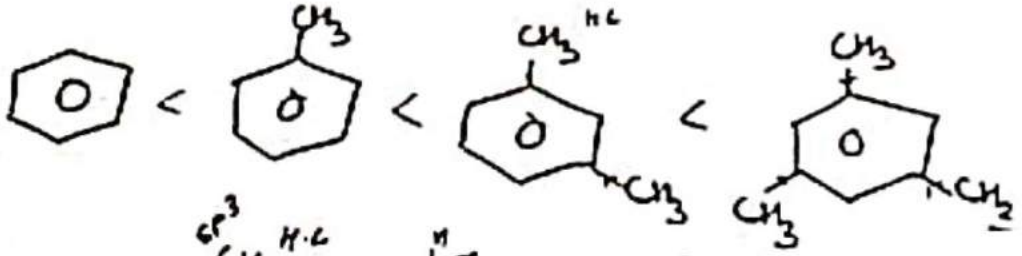
⇒ Reactivity ↑es — in presence of e^- donating group { activating groups: (+M, +H, +I) }

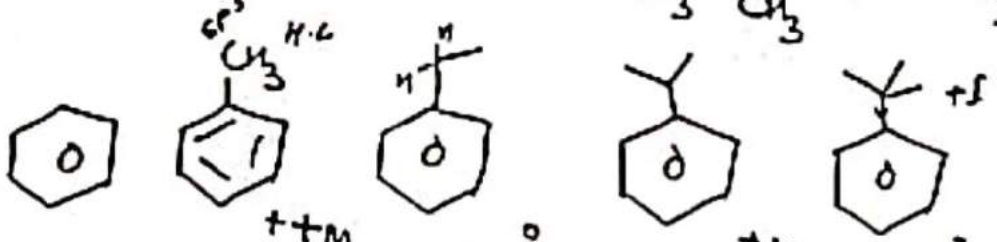
⇒ " " ↓es --- " " " e^- withdrowing groups { deactivating groups: (-M, -I) }

GENERAL ORGANIC CHEMISTRY - I

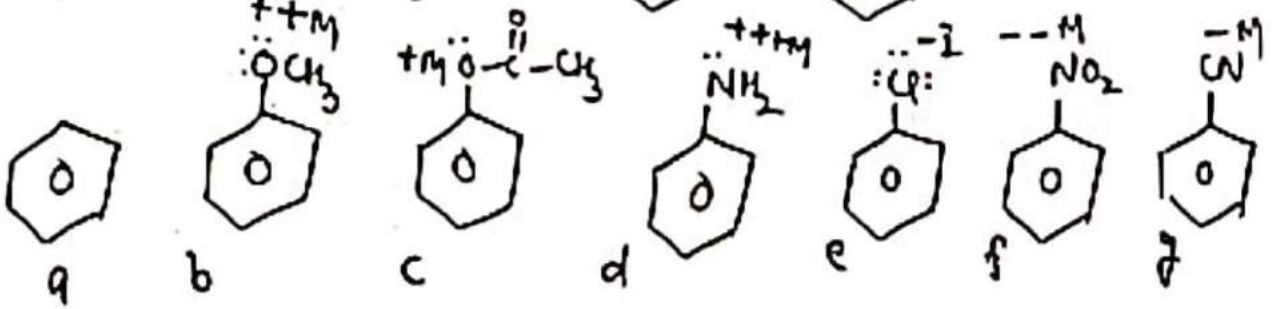
(VIII) Electron density on aromatic rings / Reactivity of aromatic compounds.

Q Identify order of e⁻ density on aro. ring.

(1) 

(2) 

b > c > d > e > a

(3) 

d > b > c > a > e > g > f